

ECON 201A: Economic Theory

Part I: Choice Theory

Lecture Notes

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*Notes taken in the course taught by Shachar Kariv, Department of Economics, UC Berkeley. All errors are mine.

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Notation and conventions

Throughout, X denotes a set of *alternatives* (objects of choice) and Z a set of *prizes* or *consequences*. A binary relation on X is a subset $\succsim \subseteq X \times X$; we write $x \succsim y$ for $(x, y) \in \succsim$ and read it “ x is weakly preferred to y ”. From $\succsim$ we derive

$$x \succ y \iff (x \succsim y \text{ and not } y \succsim x), \quad x \sim y \iff (x \succsim y \text{ and } y \succsim x).$$

$\Delta(Z)$ is the set of (simple) probability distributions on Z ; $\delta_z \in \Delta(Z)$ is the degenerate lottery paying z with certainty. For a finite $Z = \{z_1, \dots, z_n\}$ we identify $\Delta(Z)$ with the simplex

$$\Delta^{n-1} = \left\{ p \in \mathbb{R}_+^n : \sum_{i=1}^n p_i = 1 \right\}.$$

u always denotes a *Bernoulli* (von Neumann–Morgenstern) utility index defined on prizes; U or V denotes a utility function defined on *lotteries*. “Increasing” means weakly increasing unless stated otherwise.

1 Lecture 1. Introduction: preferences, lotteries and expected utility

1.1 Prologue: what is a theory of choice *for*?

A recurrent complaint is that economics treats agents as hyper-rational and its models as “unrealistic”. The premise is wrong: *a theory need not be realistic to be useful*. We do not ask that assumptions photograph the agent’s mind, only that the model’s implications organize and predict behaviour. The relevant question is not “is this assumption true?” but “for what purpose, and how well, does the model work?”. A theory of choice can be useful in three senses:

- **Descriptive:** how people *actually* choose.
- **Prescriptive:** as a practical aid to choice.
- **Normative:** how people *should* choose.

Expected utility theory is strong on the normative and prescriptive dimensions, since its axioms are hard to reject as principles of rational choice, and it is precisely on the descriptive one that it fails. This tension organizes Starmer’s (2000) survey (§A): the predictions of a *descriptive* theory are judged against evidence, not against normative principles.

Anatomy of the model

The “standard” theory of the economic agent (consumer, manager, policy maker) decomposes any choice problem into four inputs:

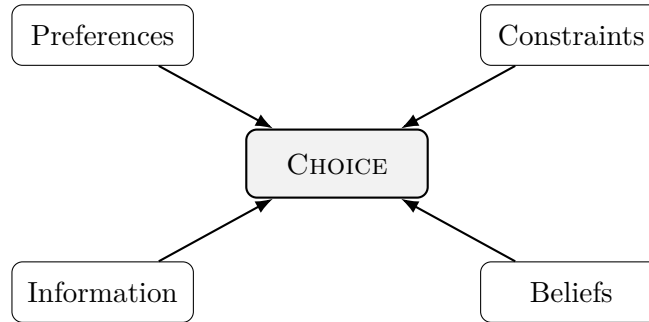


Figure 1: Decision making under certainty and uncertainty. Preferences and constraints are the classical pair; information and beliefs are what uncertainty adds.

Under certainty the bottom half of Figure 1 is inert and the theory reduces to “maximize $\succsim$ over a feasible set $B \subseteq X$ ”. Under uncertainty the agent must also *form beliefs* about what she does not know and *process information*.

The fundamental trade-offs

People’s attitudes towards *risk*, *time*, and *other people* enter every realm of (financial) decision making:

$$\text{risk} \iff \text{return}, \quad \text{today} \iff \text{tomorrow}, \quad \text{self} \iff \text{others}.$$

Four types of questions about preferences

- (1) **Consistency.** Is behaviour consistent with the utility maximization model at all? (*Representation theorems and revealed preference, Afriat.*)
- (2) **Structure.** Is behaviour consistent with a utility function having some special structural properties: monotonicity, convexity, homotheticity, quasi-linearity, expected-utility form?
- (3) **Recoverability.** Can the underlying utility function be recovered from observed choices, and how uniquely? (*Duality, integrability, Afriat.*)
- (4) **Extrapolation.** How can we forecast behaviour in *other* circumstances, new budgets, new lotteries, new institutions?

1.2 Lotteries

Definition 1.1 (Simple lottery). Let Z be a set of *prizes* (consequences). A *simple lottery* on Z is a probability distribution p on Z with finite support. For finite $Z = \{z_1, \dots, z_n\}$, $p = (p_1, \dots, p_n)$ with $p_i \geq 0$ and $\sum_i p_i = 1$, where p_i is the probability of z_i . The set of all such lotteries is $\Delta(Z)$.

Binary lottery: with probability α you get x , with probability $1 - \alpha$ you get y .

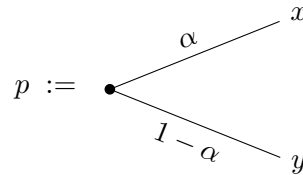


Figure 2: A simple (binary) lottery $p = \alpha\delta_x + (1 - \alpha)\delta_y$.

Risk versus uncertainty (ambiguity)

Figure 3 places the two cases of the Knightian distinction (Knight, 1921) side by side. On the left the probabilities are *known*: this is *risk*. On the right they are not: this is *uncertainty*, or, in the modern terminology, *ambiguity*.



Figure 3: A risky lottery (left) and an ambiguous lottery (right).

Compound lotteries and reduction

A *compound lottery* is a lottery whose prizes are themselves lotteries.

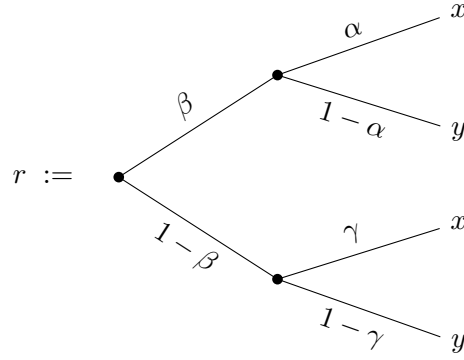


Figure 4: A compound lottery: $r = \beta p_\alpha + (1 - \beta) p_\gamma$ where $p_\alpha = \alpha \delta_x + (1 - \alpha) \delta_y$.

Axiom 1.1 (Reduction of compound lotteries). The decision maker is indifferent between a compound lottery and the simple lottery obtained from it by the calculus of probabilities. For the lottery of Figure 4,

$$r \sim [\beta\alpha + (1 - \beta)\gamma] \delta_x + [\beta(1 - \alpha) + (1 - \beta)(1 - \gamma)] \delta_y.$$

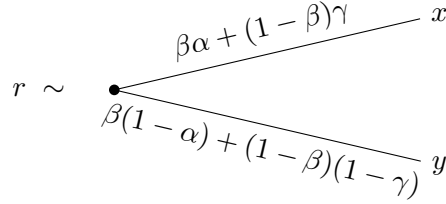


Figure 5: The reduction of the compound lottery of Figure 4.

Reduction says that the agent cares only about the induced distribution over final prizes, not about the *way* the randomization is staged or resolved. [Segal \(1990\)](#) drops it in order to reconcile non-expected-utility preferences with dynamic consistency.

Remark 1.2. For finite Z with $|Z| = n$, $\Delta(Z) = \Delta^{n-1}$ is a compact convex subset of $\mathbb{R}^n$ with n extreme points, the degenerate lotteries δ_{z_i} . Mixture is just convex combination in $\mathbb{R}^n$. For $n = 3$ the simplex is a triangle.

Axiom 1.2 (Weak order). For all lotteries p, q, r : $p \succsim q$ or $q \succsim p$ (completeness), and $p \succsim q$ with $q \succsim r$ gives $p \succsim r$ (transitivity).

Axiom 1.3 (Monotonicity w.r.t. first-order stochastic dominance). For any pair of lotteries p, q with payoff distributions F_p and F_q : if F_p first-order stochastically dominates F_q , then $p \succsim q$.

Completeness and transitivity give a representation $V : \Delta(Z) \rightarrow \mathbb{R}$; monotonicity makes it “well behaved” (increasing).

Definition 1.3 (FOSD). Let $Z \subseteq \mathbb{R}$ and let F, G be the CDFs of p, q . Then p *first-order stochastically dominates* q , written $p \succsim_{\text{FOSD}} q$, if

$$F(t) \leq G(t) \quad \text{for all } t \in \mathbb{R}.$$

Proposition 1.4. F FOSD-dominates G if and only if $\int u dF \geq \int u dG$ for every non-decreasing $u : \mathbb{R} \rightarrow \mathbb{R}$ for which both integrals exist.

Proof sketch. ($\Leftarrow$) Take $u = \mathbf{1}_{[t, \infty)}$, which gives $1 - F(t) \geq 1 - G(t)$. ($\Rightarrow$) For u non-decreasing and, say, differentiable with $u(z)$ bounded, integrate by parts: $\int u dF - \int u dG = \int u'(t) [G(t) - F(t)] dt \geq 0$. The general case follows by approximating u with positive combinations of indicators $\mathbf{1}_{[t, \infty)}$. $\square$

1.3 The expected utility representation theorem

The origin of decision theory, and of game theory, is von Neumann and Morgenstern’s *Theory of Games and Economic Behavior* ([von Neumann and Morgenstern, 1944](#)).

Axiom 1.4 (Independence / substitution). For all lotteries p, q, r and all $\alpha \in (0, 1)$:

$$p \succ q \implies \alpha p + (1 - \alpha)r \succ \alpha q + (1 - \alpha)r.$$

(Equivalently, with $\succsim$ in place of $\succ$ throughout.)

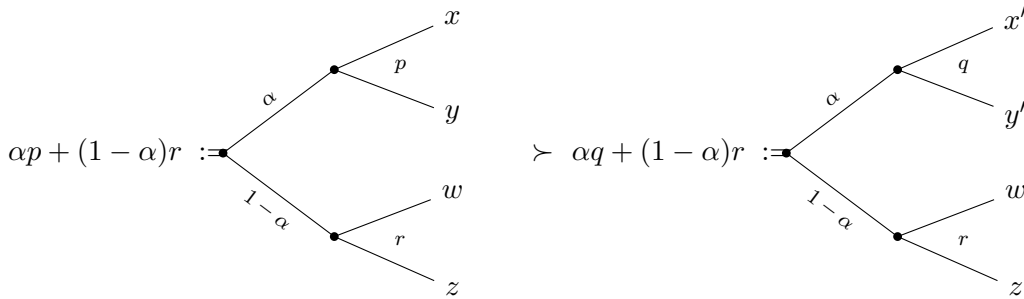


Figure 6: Independence. If $p \succ q$, then mixing *the same* lottery r into both, with *the same* weight $1 - \alpha$, cannot reverse the ranking.

Axiom 1.5 (Archimedean / continuity). For all lotteries p, q, r with $p \succ q \succ r$ there exist $\alpha, \beta \in (0, 1)$ such that

$$\alpha p + (1 - \alpha)r \succ q \succ \beta p + (1 - \beta)r.$$

Equivalently: the sets $\{\alpha \in [0, 1] : \alpha p + (1 - \alpha)r \succeq q\}$ and $\{\alpha \in [0, 1] : q \succeq \alpha p + (1 - \alpha)r\}$ are closed.

Remark 1.5. Continuity is what fails if some prize is “infinitely bad”. If q is “\$1”, p is “\$2” and r is “death”, continuity says there is some probability of death small enough that you would take the gamble to gain a dollar. Most people say they would not.

Definition 1.6 (Expected utility representation). $\succeq$ on $\Delta(Z)$ has an *expected utility* (von Neumann–Morgenstern) representation if there is $u : Z \rightarrow \mathbb{R}$ such that

$$p \succeq q \iff U(p) := \sum_{z \in Z} p(z) u(z) \geq \sum_{z \in Z} q(z) u(z) =: U(q).$$

u is called the *Bernoulli* utility index, U the *von Neumann–Morgenstern* utility function.

Theorem 1.7 (von Neumann–Morgenstern (1944)). *Let Z be finite and $\succeq$ a binary relation on $\Delta(Z)$. Then $\succeq$ satisfies completeness, transitivity, continuity (Axiom 1.5) and independence (Axiom 1.4) if and only if it admits an expected utility representation. Moreover u is unique up to positive affine transformations: $\tilde{u}$ also represents $\succeq$ in this sense if and only if $\tilde{u} = a u + b$ for some $a > 0$, $b \in \mathbb{R}$.*

Proof. Necessity is a direct computation; U is affine, so $U(\alpha p + (1 - \alpha)r) = \alpha U(p) + (1 - \alpha)U(r)$, and independence follows. For sufficiency, assume Z finite so that there are a best and a worst degenerate lottery, $\bar{p} \succeq p \succeq \underline{p}$ for all p . If $\bar{p} \sim \underline{p}$ the result is trivial (u constant), so assume $\bar{p} \succ \underline{p}$.

Step 1. If $p \succ q$ and $\alpha \in (0, 1)$ then $\alpha p + (1 - \alpha)q \succ q$ and $p \succ \alpha p + (1 - \alpha)q$. (Apply independence twice.)

Step 2. (monotonicity in the mixing weight.) If $\alpha > \beta$ then $\alpha \bar{p} + (1 - \alpha)\underline{p} \succ \beta \bar{p} + (1 - \beta)\underline{p}$.

Step 3. (Calibration.) For each p there is a unique $\alpha_p \in [0, 1]$ with $p \sim \alpha_p \bar{p} + (1 - \alpha_p)\underline{p}$. Existence is continuity (Axiom 1.5) plus completeness; uniqueness is Step 2.

Step 4. $U(p) := \alpha_p$ represents $\succeq$: this is Step 2 again.

Step 5. (*U is affine.*) For $\alpha \in [0, 1]$,

$$\begin{aligned} \alpha p + (1 - \alpha)q &\sim \alpha [\alpha_p \bar{p} + (1 - \alpha_p) \underline{p}] + (1 - \alpha)q \\ &\sim \alpha [\alpha_p \bar{p} + (1 - \alpha_p) \underline{p}] + (1 - \alpha) [\alpha_q \bar{p} + (1 - \alpha_q) \underline{p}] \\ &= [\alpha \alpha_p + (1 - \alpha) \alpha_q] \bar{p} + [1 - \alpha \alpha_p - (1 - \alpha) \alpha_q] \underline{p}, \end{aligned}$$

each step by independence and Step 1; hence $U(\alpha p + (1 - \alpha)q) = \alpha U(p) + (1 - \alpha)U(q)$.

Step 6. (*Affine $\Rightarrow$ expected utility.*) Write $p = \sum_i p_i \delta_{z_i}$ and iterate affinity: $U(p) = \sum_i p_i U(\delta_{z_i})$. Set $u(z_i) := U(\delta_{z_i})$.

Uniqueness. If $\tilde{u} = au + b$ with $a > 0$ then $\tilde{U} = aU + b$ represents the same order. Conversely, if $\tilde{U}$ is affine and represents $\succsim$, then $\tilde{U}$ and U agree on the ranking of the calibrating mixtures, which pins down $\tilde{U}$ as an increasing affine function of U . $\square$

Corollary 1.8 (Linearity in probabilities). *U is affine on $\Delta(Z)$: for any finite collection of lotteries p^k and weights $\alpha_k \geq 0$ summing to one, $U(\sum_k \alpha_k p^k) = \sum_k \alpha_k U(p^k)$.*

1.4 The Allais paradox

Maurice Allais (1953) did not believe EU was an adequate description of risk preferences, and he built the following pair of problems as a counterexample.

Allais I

Choose between:

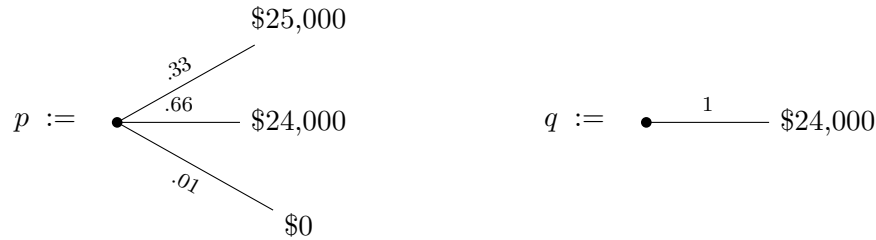


Figure 7: Allais (1953), problem I. Most people choose q : the certainty of \$24,000.

Allais II

Now choose between:



Figure 8: Allais (1953), problem II. Most people choose p' : the odds of winning look similar, but the prizes differ.

Normalize $u(0) = 0$ and $u(25,000) = 1$, and write $m := u(24,000) \in (0, 1)$. Then

$$U(p) = .33 + .66m, \quad U(q) = m, \quad U(p') = .33, \quad U(q') = .34m.$$

Hence

$$q \succsim p \iff m \geq .33 + .66m \iff .34m \geq .33 \iff q' \succsim p'.$$

1.5 The Marschak–Machina probability triangle

Fix three monetary prizes $H > M > L$ (“high”, “medium”, “low”). Any lottery over $\{H, M, L\}$ is a vector (p_L, p_M, p_H) with $p_M = 1 - p_L - p_H$, so it is completely described by the pair (p_L, p_H) . Plot p_L on the horizontal axis and p_H on the vertical axis. The set of lotteries is the triangle $T = \{(p_L, p_H) \in \mathbb{R}_+^2 : p_L + p_H \leq 1\}$, whose vertices are the three degenerate lotteries. This device is due to Marschak (1950) and was popularized by Machina in the 1980s; it is often called the *Machina triangle*.

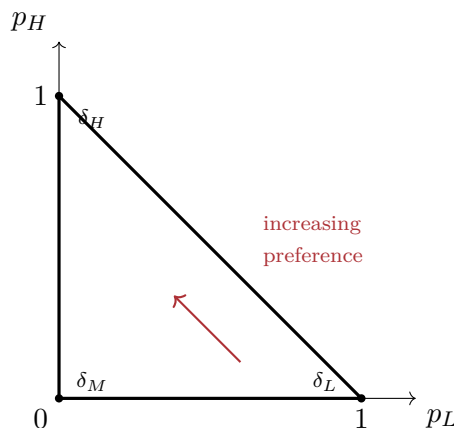


Figure 9: The (Marschak–Machina) probability triangle for three payouts $H > M > L$.

Proposition 1.9 (Direction of FOSD). *Within T , (p'_L, p'_H) FOSD-dominates (p_L, p_H) iff $p'_L \leq p_L$ and $p'_H \geq p_H$ (with one inequality strict). Hence monotonicity is equivalent to “preference increases to the north-west”.*

Proposition 1.10 (EU $\iff$ parallel straight lines). *Let $\succsim$ be a monotone weak order on T . Then $\succsim$ admits an expected utility representation if and only if its indifference curves are parallel straight lines of positive slope, in which case that slope is*

$$s = \frac{u(M) - u(L)}{u(H) - u(M)}$$

and determines the Bernoulli index up to positive affine transformation.

Proof. Normalize $u(L) = 0$ and $u(H) = 1$ and write $m = u(M) \in (0, 1)$, which is without loss because u is unique up to positive affine transformations (Theorem 1.7).

Necessity. Substituting $p_M = 1 - p_L - p_H$,

$$U(p_L, p_H) = p_L u(L) + p_M u(M) + p_H u(H) = m + p_H(1 - m) - p_L m.$$

Since $m < 1$, solving $U = c$ for p_H gives $p_H = \frac{c-m}{1-m} + \frac{m}{1-m} p_L$. The constant is therefore $(c-m)/(1-m)$, the height at which the level- c curve meets the axis $p_L = 0$. The coefficient of p_L does not involve c , so the family is parallel, with slope $m/(1-m) > 0$. That slope equals $(u(M) - u(L))/(u(H) - u(M))$, a ratio of utility differences, hence unchanged if u is replaced by $au + b$ with $a > 0$: unlike m , it does not depend on the normalization.

Sufficiency. If the indifference sets are the lines $p_H - s p_L = k$, then $V := p_H - s p_L$ is constant exactly on them, so V or $-V$ represents $\succsim$, and FOSD picks V . Put $m = s/(1+s) \in (0, 1)$ and $u(L) = 0$, $u(M) = m$, $u(H) = 1$; since $1 - m = 1/(1+s)$, the associated expected utility is

$$U(p_L, p_H) = m + \frac{p_H - s p_L}{1 + s} = m + \frac{1}{1 + s} V(p_L, p_H),$$

an increasing affine transformation of V . As $m \mapsto m/(1-m)$ is a bijection from $(0, 1)$ onto $(0, \infty)$, the slope determines the index. $\square$

Every movement inside the triangle is made at the expense of the middle prize, since $p_M = 1 - p_L - p_H$ is what absorbs it. So s is the rate at which the two extreme prizes trade off: the extra probability of H that just compensates for one more unit of probability of L .

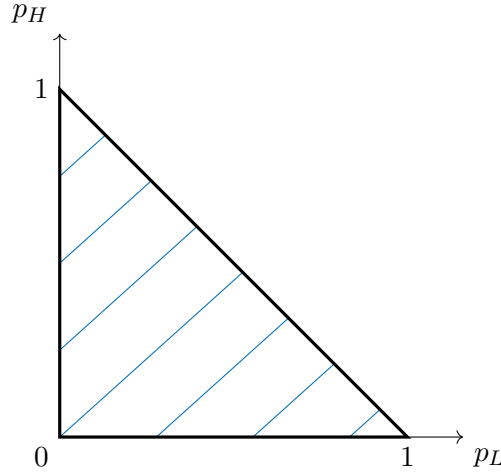


Figure 10: An indifference map of a “loss-neutral” (expected utility) individual. Expected utility theory requires that indifference lines be *parallel* straight lines. The slope measures risk aversion: the steeper the lines, the more risk averse the agent.

Proposition 1.11 (Comparative risk aversion in the triangle). *Let two EU agents A and B have Bernoulli indices u_A, u_B with the same normalization. A is more risk averse than B (in the sense $u_A = h \circ u_B$ with h increasing and concave) if and only if A ’s indifference lines are steeper than B ’s.*

Proof. Normalize both indices by $u_A(L) = u_B(L) = 0$ and $u_A(H) = u_B(H) = 1$. Since $u_A = h \circ u_B$, this forces $h(0) = h(u_B(L)) = u_A(L) = 0$ and $h(1) = h(u_B(H)) = u_A(H) = 1$. A concave h with $h(0) = 0$ and $h(1) = 1$ lies weakly above the diagonal on $[0, 1]$, because $h(t) = h(t \cdot 1 + (1 - t) \cdot 0) \geq t h(1) + (1 - t) h(0) = t$. Applying this at $t = m_B = u_B(M)$ gives $m_A = u_A(M) = h(m_B) \geq m_B$. Finally $s = m/(1 - m)$ is strictly increasing on $(0, 1)$, so $m_A \geq m_B$ gives $s_A \geq s_B$: A ’s lines are steeper. The converse follows by reading the same chain backwards. $\square$

Camerer’s survey (Camerer, 1995) of two decades of experiments reports a *border effect*. The boundary of the triangle is made of the lotteries that give some prize probability zero, that is, gambles over two prizes rather than three; in the interior all three probabilities are strictly positive. Violations of EU are large on the boundary and much smaller, though still statistically significant, in the interior. The likely cause is nonlinear weighting of probabilities near 0 and 1: a comparison on the boundary involves probabilities close to those endpoints, where the weighting function of §A.3 is steep and distorts most, while a comparison in the interior involves moderate probabilities, where that function is nearly linear and behaviour looks like EU. The full set of findings is in §A.4.

Testing EU with a handful of hypothetical pairwise choices does establish that violations exist, but the design has three limitations.

- [1] It says nothing about how *large* the violations are, or how *often* they occur.
- [2] The statistics rest on *ad hoc* assumptions about how subjects make mistakes, with no econometric model behind them.
- [3] The choice problems look nothing like the situations economists apply the theory to.

2 Lecture 2. Preferences

Lecture 1 took the preference relation as given and asked what representations it admits. This lecture goes one step back and asks where the preference relation itself comes from.

Primitive	Object	Interpretation
preference	$\succsim$ on X	what the agent <i>wants</i>
utility	$u : X \rightarrow \mathbb{R}$	a numerical summary of $\succsim$
choice	$C(A) \subseteq A$	what the agent <i>does</i>

The plan is to show that, for finite X , the three are the same object seen from three sides: consistent choice behaviour is equivalent to maximizing a complete and transitive preference, which is equivalent to maximizing a utility function. Only choice is observable, so the content of the theory is the restriction it places on C , and the other two are convenient encodings.

2.1 Two questionnaires

Rubinstein introduces preferences through two *imaginary questionnaires*. The point of doing it twice is that the two routes look different and turn out to carry exactly the same information.

Questionnaire Q . For each unordered pair $x \neq y$ the agent ticks one of three boxes: $x \succ y$, $y \succ x$, or I (indifferent). Answers such as “I cannot compare” or “I have no opinion” are not admissible: ruling them out is exactly the assumption of completeness.

Definition 2.1 (Preferences via Q). A *preference* is a function f assigning to every pair $x, y \in X$ one of the values $x \succ y$, $y \succ x$, I , such that for all $x, y, z \in X$:

- (i) *no order effect*: $f(x, y) = f(y, x)$, i.e. the value assigned to the pair does not depend on the order in which the two options are presented;
- (ii) *transitivity*:

$$\begin{aligned}
 f(x, y) = x \succ y \text{ and } f(y, z) = y \succ z &\implies f(x, z) = x \succ z, \\
 f(x, y) = I \text{ and } f(y, z) = I &\implies f(x, z) = I.
 \end{aligned}$$

The remaining consistency requirement one might want to impose, that $f(x, y) = x \succ y$ and $f(y, z) = I$ should give $f(x, z) = x \succ z$, is not an extra assumption: it follows from (i) and (ii), as the proof of Proposition 2.3 shows.

Questionnaire R . For each *ordered* pair the agent answers one yes-or-no question: is x at least as good as y ? A “yes” to $R(x, y)$ is recorded as $x \succsim y$.

Definition 2.2 (Preferences via R). A *preference* is a binary relation $\succsim$ on X that is *complete* ($x \succsim y$ or $y \succsim x$ for all x, y) and *transitive* ($x \succsim y$ and $y \succsim z$ give $x \succsim z$). Reflexivity, $x \succsim x$, follows from completeness. Such a relation is also called a *weak order*.

Definition 2.2 is the conventional one, and it carries the same information as Definition 2.1.

Proposition 2.3 (Equivalence of the two definitions). *The map sending f to the relation*

$$x \succsim y \iff f(x, y) \in \{x \succ y, I\}$$

is a bijection from the preferences of Definition 2.1 onto the preferences of Definition 2.2. Its inverse sends $\succsim$ to

$$f(x, y) = \begin{cases} x \succ y, & x \succsim y \text{ and not } y \succsim x, \\ y \succ x, & y \succsim x \text{ and not } x \succsim y, \\ I, & x \succsim y \text{ and } y \succsim x. \end{cases}$$

Both maps act one pair at a time, so it is enough to see what they do to a single pair $\{x, y\}$. Figure 11 shows that.

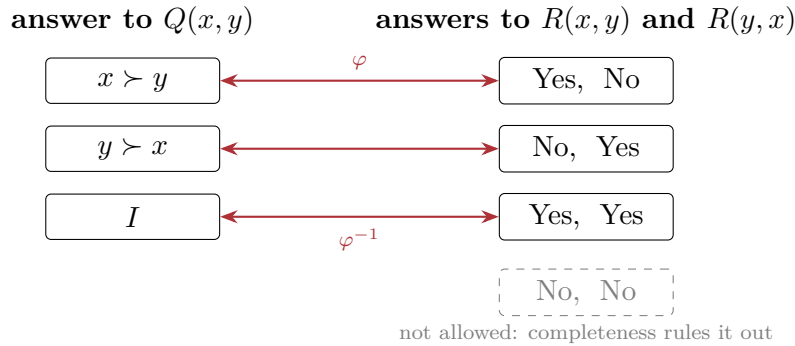


Figure 11: The map φ pair by pair. Reading left to right gives $\succsim$ from f : the answer $x \succ y$ becomes “yes” to $R(x, y)$ and “no” to $R(y, x)$, and I becomes “yes” to both. Reading right to left gives f from $\succsim$. Each side has exactly three possible answers and the arrows pair them off one to one, which is why φ is a bijection.

Proof. The map is well defined. Let f satisfy Definition 2.1 and let $\succsim$ be as displayed. Reflexivity: $f(x, x) = I$, so $x \succsim x$. Completeness: $f(x, y)$ takes one of three values, and in

each case at least one of $x \succsim y$, $y \succsim x$ holds. Transitivity: suppose $x \succsim y$ and $y \succsim z$. Four cases, according to whether each link is strict or indifferent.

- $x \succ y$ and $y \succ z$: then $f(x, z) = x \succ z$ by (ii), so $x \succ z$.
- $f(x, y) = I$ and $f(y, z) = I$: then $f(x, z) = I$ by (ii), so $x \succsim z$.
- $f(x, y) = x \succ y$ and $f(y, z) = I$: we claim $f(x, z) = x \succ z$. Suppose not. Since f assigns exactly one of three values to the pair $\{x, z\}$, either $f(x, z) = z \succ x$ or $f(x, z) = I$. In the first case $z \succ x$ and $x \succ y$ give $f(z, y) = z \succ y$ by (ii), contradicting $f(y, z) = I$. In the second case $f(y, z) = I$ and $f(z, x) = I$ give $f(y, x) = I$ by (ii), contradicting $f(x, y) = x \succ y$. Hence $f(x, z) = x \succ z$ and $x \succ z$.
- $f(x, y) = I$ and $f(y, z) = y \succ z$: symmetric to the previous case, exchanging the roles of x and z .

In every case $x \succsim z$, so $\succsim$ is transitive and Definition 2.2 is satisfied.

The inverse map is well defined. Let $\succsim$ be reflexive, complete and transitive and let f be as displayed. The three cases are exhaustive by completeness and mutually exclusive by construction, and $f(x, y) = f(y, x)$ because the definition treats the pair symmetrically, so (i) holds. For (ii) we use two standard facts about a weak order, both worth recording.

- $\succ$ is transitive. If $x \succ y$ and $y \succ z$ then $x \succ z$ by transitivity of $\succsim$. If also $z \succsim x$, then $z \succsim x \succ z$ gives $z \succ z$, contradicting $y \succ z$. Hence $x \succ z$.
- $\sim$ is transitive. If $x \sim y$ and $y \sim z$ then $x \succsim y \succsim z$ and $z \succsim y \succsim x$, so $x \succsim z$ and $z \succsim x$.

□

2.2 Where transitivity comes from, and where it fails

Completeness is an assumption about the agent's willingness to answer. Transitivity is a genuine restriction, and two natural decision procedures violate it.

Example 2.4 (Aggregation of considerations). Let $X = \{a, b, c\}$ and suppose the agent evaluates options along three *considerations* (criteria, attributes, or the opinions of three advisers), each of which is itself a strict order:

$$a \succ_1 b \succ_1 c, \quad b \succ_2 c \succ_2 a, \quad c \succ_3 a \succ_3 b,$$

and ranks x above y when a majority of considerations does. Comparing the three pairs:

	consideration 1	2	3	majority
a vs b	a	b	a	$a \succ b$
b vs c	b	b	c	$b \succ c$
a vs c	a	c	c	$c \succ a$

so $a \succ b \succ c \succ a$: a cycle. This is Condorcet’s paradox, and the point is that each ingredient is perfectly well behaved. Transitivity is not a property that survives aggregation, whether the things aggregated are voters or the attributes of a car.

Example 2.5 (Similarity and just-noticeable differences). Let $X = \mathbb{R}$ with “larger is better”, but suppose the agent can only tell two numbers apart when they differ by at least one unit:

$$x \succ y \iff x \geq y + 1, \quad x \sim y \iff |x - y| < 1.$$

Then $0 \sim \frac{2}{3}$ and $\frac{2}{3} \sim \frac{4}{3}$, but $\frac{4}{3} \geq 0 + 1$, so $\frac{4}{3} \succ 0$. Indifference is intransitive, so $\succsim$ is not a weak order, and by Proposition 3.2 it has no utility representation.

3 Lecture 3. Utility and choice

3.1 Utility

Definition 3.1 (Utility representation). $u : X \rightarrow \mathbb{R}$ represents $\succsim$ if for all $x, y \in X$,

$$x \succsim y \iff u(x) \geq u(y).$$

Proposition 3.2 (Necessity). *If $\succsim$ admits a utility representation then $\succsim$ is complete and transitive.*

Proof. Let $u : X \rightarrow \mathbb{R}$ represent $\succsim$. Completeness: $u(x), u(y) \in \mathbb{R}$ and $\geq$ is complete on $\mathbb{R}$, so $u(x) \geq u(y)$ or $u(y) \geq u(x)$, hence $x \succsim y$ or $y \succsim x$. Transitivity: $\geq$ is transitive on $\mathbb{R}$, so $u(x) \geq u(y)$ and $u(y) \geq u(z)$ imply $u(x) \geq u(z)$. By the representation, this says that $x \succsim y$ and $y \succsim z$ imply $x \succsim z$. $\square$

Proposition 3.3 (Ordinality). *If u represents $\succsim$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing on the range of u , then $v = g \circ u$ represents $\succsim$. Consequently a representable $\succsim$ has infinitely many representations.*

Proof. $x \succsim y \iff u(x) \geq u(y) \iff g(u(x)) \geq g(u(y)) \iff v(x) \geq v(y)$, where the middle equivalence is exactly strict monotonicity of g . $\square$

What about the converse: are any two representations of the same $\succsim$ related by a strictly increasing transformation? On the range they always are. If u and v both represent $\succsim$, define g on $u(X)$ by $g(u(x)) = v(x)$. This is well defined, because $u(x) = u(x')$ gives $x \sim x'$ and hence $v(x) = v(x')$, and it is strictly increasing, because $u(x) > u(x')$ gives $x \succ x'$ and hence $v(x) > v(x')$. So $v = g \circ u$ with g strictly increasing on $u(X)$. What can fail is extending g to all of $\mathbb{R}$.

Example 3.4 (No strictly increasing g on $\mathbb{R}$). Let $X = (0, 1)$ with the usual order. Both $u(x) = x$ and $v(x) = \tan\left(\pi\left(x - \frac{1}{2}\right)\right)$ represent it, and $v = g \circ u$ with $g = \tan\left(\pi\left(\cdot - \frac{1}{2}\right)\right)$, strictly increasing on the common range $(0, 1)$. No strictly increasing $\tilde{g} : \mathbb{R} \rightarrow \mathbb{R}$ agrees with g there: it would need $\tilde{g}(0) < v(x)$ for every $x \in (0, 1)$, and v takes every real value. This is why Proposition 3.3 asks for g to be strictly increasing on the range of u and not on $\mathbb{R}$.

Lemma 3.5 (Existence of minimal and maximal elements). *Let $\succsim$ be complete and transitive on X and let $A \subseteq X$ be finite and nonempty. Then A has a minimal element, i.e. some $a \in A$ with $x \succsim a$ for every $x \in A$, and a maximal element, i.e. some $b \in A$ with $b \succsim x$ for every $x \in A$.*

Proof. By induction on $|A|$. If $A = \{a\}$ then a is minimal by reflexivity. Suppose the claim holds for sets of size n and let $|A| = n + 1$. Fix $x \in A$ and let y be minimal in $A \setminus \{x\}$, so $z \succsim y$ for every $z \in A \setminus \{x\}$. By completeness, either $x \succsim y$ or $y \succ x$.

- If $x \succsim y$, then $z \succsim y$ for every $z \in A$, so y is minimal in A .
- If $y \succ x$, then for every $z \in A \setminus \{x\}$ transitivity gives $z \succsim y \succ x$, hence $z \succ x$; together with $x \succ x$ this makes x minimal in A .

□

Theorem 3.6 (X finite). *Let $\succsim$ be complete and transitive on a finite X and put*

$$L(x) = \{z \in X : x \succsim z\}, \quad u(x) = |L(x)|.$$

Then u represents $\succsim$, with values in $\{1, 2, \dots, |X|\}$.

Proof. Step 1. Better bundles have larger lower sets. If $x \succ z$ then $L(z) \subseteq L(x)$: any w with $z \succsim w$ also has $x \succsim w$ by transitivity. Hence $x \succ y$ implies $u(x) \geq u(y)$.

Step 2. The converse. Suppose $u(x) \geq u(y)$ and $x \succsim y$ fails, so $y \succ x$ by completeness. Then $L(x) \subseteq L(y)$ by Step 1, and the inclusion is strict: $y \in L(y)$ by reflexivity while $y \notin L(x)$. Since X is finite, a strict inclusion is a strict inequality of sizes, so $u(x) < u(y)$, a contradiction.

Finally $x \in L(x) \subseteq X$, so $1 \leq u(x) \leq |X|$. Finiteness is used only in Step 2, to turn a strict inclusion into a strict inequality of cardinalities. □

Theorem 3.7 (Representation, X countable). *If $\succsim$ is complete and transitive on a countable X , then $\succsim$ has a utility representation with values in $(0, 1]$.*

Proof. Write $L(x) = \{y \in X : x \succsim y\}$ for the weak lower contour set of x . The argument rests on one observation.

Claim. $x \succ z$ implies $L(z) \subseteq L(x)$, and $x \succ z$ implies $L(z) \subsetneq L(x)$.

For the first part, let $y \in L(z)$, i.e. $z \succsim y$. Then $x \succ z \succsim y$, so $x \succ y$ by transitivity and $y \in L(x)$. For the second, $x \succ z$ gives $L(z) \subseteq L(x)$ by the first part, and $x \in L(x)$ by reflexivity, while $x \notin L(z)$: if it were, $z \succsim x$ would contradict $x \succ z$. So the inclusion is strict.

Enumerate $X = \{x_1, x_2, \dots\}$ and put

$$u(x) = \sum_{i: x \succsim x_i} 2^{-i} = \sum_{i: x_i \in L(x)} 2^{-i} \in (0, 1].$$

The series has nonnegative terms dominated by $\sum_{i \geq 1} 2^{-i} = 1$, so it converges absolutely and u is well defined; it is strictly positive because the index i with $x_i = x$ always contributes. If $x \succsim z$ then $L(z) \subseteq L(x)$, so the index set summed over for z is contained in the one for x and, all terms being positive, $u(x) \geq u(z)$. If $x \succ z$, let i be the index with $x_i = x$. By the claim $L(z) \subseteq L(x)$ and $x \in L(x) \setminus L(z)$, so

$$u(x) - u(z) = \sum_{j: x_j \in L(x) \setminus L(z)} 2^{-j} \geq 2^{-i} > 0.$$

Conversely suppose $u(x) \geq u(z)$ and that $x \succsim z$ fails. Completeness gives $z \succ x$, and the previous paragraph applied with the roles exchanged gives $u(z) > u(x)$, a contradiction. Hence $x \succsim z \iff u(x) \geq u(z)$. $\square$

Example 3.8 (Lexicographic preferences). Let $X = [0, 1] \times [0, 1]$ and let $\succsim_L$ rank first by the first coordinate, breaking ties by the second:

$$(a_1, a_2) \succsim_L (b_1, b_2) \iff a_1 > b_1 \text{ or } (a_1 = b_1 \text{ and } a_2 \geq b_2).$$

This relation is complete and transitive, and it is exactly the “look at the most important criterion first” procedure, so it is not exotic. It has no utility representation.

Proof. Suppose u represents $\succsim_L$. For each $a \in [0, 1]$ we have $(a, 1) \succ_L (a, 0)$ and therefore $u(a, 1) > u(a, 0)$, so the open interval

$$I_a = (u(a, 0), u(a, 1)) \subset \mathbb{R}$$

is nonempty. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, choose a rational $q(a) \in I_a$. This defines a map $q : [0, 1] \rightarrow \mathbb{Q}$, and the codomain is the whole point of the argument. The map is injective. Let $b > a$. Then $(b, 0) \succ_L (a, 1)$, because the first coordinate already decides, so $u(b, 0) > u(a, 1)$. Hence every point of I_b lies above every point of I_a ; in particular $I_a \cap I_b = \emptyset$ and $q(a) \neq q(b)$. Therefore, q is an injection from $[0, 1]$ into $\mathbb{Q}$. But $[0, 1]$ has the cardinality of the continuum and $\mathbb{Q}$ is countable, and no injection from an uncountable set into a countable one exists. $\square$

Remark 3.9 (Could we do without utility?). Yes, in principle: every statement about maximizing u over a budget set can be restated as a statement about $\succsim$ -maximal elements,

and existence results such as Lemma 3.5 are proved directly on $\succsim$. What would be lost is calculus. Marginal rates of substitution, first-order conditions, comparative statics by the implicit function theorem, and econometric estimation all operate on a function, not on a binary relation.

3.2 Continuity

Definition 3.10 (Continuity, two forms). Let $X \subseteq \mathbb{R}^n$ and let $\succsim$ be a weak order on X .

- (I) $\succsim$ is continuous if whenever $a \succ b$ there are open balls $B_a \ni a$ and $B_b \ni b$ such that $x \succ y$ for all $x \in B_a$ and all $y \in B_b$.
- (II) $\succsim$ is continuous if whenever $a_n \succsim b_n$ for all n , $a_n \rightarrow a$ and $b_n \rightarrow b$, we have $a \succsim b$; equivalently, the graph $\{(x, y) \in X \times X : x \succsim y\}$ is closed in the product topology.

Proposition 3.11. (I) and (II) are equivalent, and both are equivalent to the contour set form: $U(x) = \{y : y \succsim x\}$ and $L(x) = \{y : x \succsim y\}$ are closed for every x .

Proof. It is enough to close the cycle (I) $\Rightarrow$ (II) $\Rightarrow$ closed contour sets $\Rightarrow$ (I).

(I) $\Rightarrow$ (II). Let $a_n \succsim b_n$, $a_n \rightarrow a$, $b_n \rightarrow b$ and suppose $a \succ b$ fails, so $b \succ a$ by completeness. Then (I) gives balls $B_b \ni b$ and $B_a \ni a$ with $y \succ x$ throughout, and for n large $b_n \succ a_n$, contradicting $a_n \succsim b_n$.

(II) $\Rightarrow$ closed contour sets. If $y_n \succsim x$ and $y_n \rightarrow y$, apply (II) to the pairs (y_n, x) to get $y \succsim x$; if $x \succsim y_n$ and $y_n \rightarrow y$, apply it to the pairs (x, y_n) .

Closed contour sets $\Rightarrow$ (I). Let $a \succ b$. If $a \succ w \succ b$ for some w , take $B_a = \{z : z \succ w\}$ and $B_b = \{z : w \succ z\}$, open as complements of $L(w)$ and $U(w)$; then $x \succ w \succ y$ throughout. If no such w exists, call the pair a *jump*; then $\{z : z \succ b\} = \{z : z \succsim a\}$, since $z \succ b$ together with $a \succ z$ would put z between b and a . Take $B_a = \{z : z \succ b\}$ and $B_b = \{z : a \succ z\}$, and $x \succ a \succ y$ throughout. $\square$

Proposition 3.12. If $\succsim$ is represented by a continuous $u : X \rightarrow \mathbb{R}$, then $\succsim$ is continuous.

Proof. If $a_n \succsim b_n$, $a_n \rightarrow a$ and $b_n \rightarrow b$, then $u(a_n) \geq u(b_n)$, and continuity of u carries the weak inequality to the limit: $u(a) \geq u(b)$. $\square$

Example 3.13 ($\succsim_L$ is not continuous). On $X = [0, 1]^2$ let $a_n = (1 - \frac{1}{n}, 1) \rightarrow a = (1, 1)$ and $b = (1, 0)$. The first coordinate decides, so $b \succ_L a_n$ for every n , while in the limit the first coordinates tie and $a \succ_L b$: form (II) fails.

3.3 Choice

Preferences are not observable; choices are. This section takes choice behaviour as the primitive and asks what has to be true of it for the preference-and-utility story to be a correct description.

Definition 3.14 (Choice function). Let $\mathcal{A}$ be the collection of nonempty subsets of X . A *choice function* assigns to each $A \in \mathcal{A}$ a subset $C(A) \subseteq A$, read as the set of elements of A the agent would be content to take.

Given $\succsim$, the induced choice is

$$C_{\succsim}(A) = \{x \in A : x \succsim y \text{ for all } y \in A\}, \quad (1)$$

and given u , the induced choice is

$$C_u(A) = \{x \in A : u(x) \geq u(y) \text{ for all } y \in A\}. \quad (2)$$

Definition 3.15. A choice function C satisfies

C1. *finite nonemptiness* if $C(A) \neq \emptyset$ for every $A \in \mathcal{A}$ that is finite;

C2. *choice coherence* if for all $x, y \in X$ and all $A, B \in \mathcal{A}$,

$$x, y \in A \cap B, \quad x \in C(A), \quad y \notin C(A) \quad \implies \quad y \notin C(B).$$

C2 says: if y was available and rejected in a problem where x was chosen, then y cannot be chosen in any other problem where x is also available: so no complementarities.

Definition 3.16 (Sen 1971). A choice function or correspondence C satisfies

$$\begin{aligned} (\alpha) \quad & x \in A \subseteq B \text{ and } x \in C(B) \implies x \in C(A); \\ (\beta) \quad & x, y \in C(A), \quad A \subseteq B, \quad y \in C(B) \implies x \in C(B). \end{aligned}$$

α is *contraction consistency*: shrinking the menu, without removing the chosen element, cannot change the fact that it is chosen. It fails exactly when the menu itself carries information.

β is *expansion consistency*: if two options are tied on a small menu and one of them survives on a larger menu, so must the other. It fails exactly when ties are fragile, i.e. when indifference is not transitive.

The two guard against opposite pathologies, which is why neither implies the other. See Example 3.28.

Lemma 3.17. *C2 implies α and β . Conversely, if C satisfies C1 and both α and β on a collection closed under intersection, then C satisfies C2.*

Proof. Step by step:

Step 1. $C2 \Rightarrow \alpha$. Let $x \in A \subseteq B$ with $x \in C(B)$ and suppose $x \notin C(A)$. By C1 there is $y \in C(A)$, and $x, y \in A \subseteq B$. Applying C2 to the pair (y, x) and the problems (A, B) gives $x \notin C(B)$, a contradiction.

Step 2. $C2 \Rightarrow \beta$. Let $x, y \in C(A)$, $A \subseteq B$, $y \in C(B)$, and suppose $x \notin C(B)$. Then $x, y \in A \cap B$, so C2 with the roles of A and B exchanged gives $x \notin C(A)$, a contradiction.

Step 3. α and $\beta \Rightarrow C2$. Let $x, y \in A \cap B$, $x \in C(A)$, $y \notin C(A)$, and suppose $y \in C(B)$. Put $E = A \cap B$. By α , $x \in C(E)$ and $y \in C(E)$. Now $x, y \in C(E)$, $E \subseteq A$ and $x \in C(A)$, so β gives $y \in C(A)$, a contradiction.

□

Theorem 3.18 (X finite). *Let X be finite. The following three constructions produce exactly the same objects.*

- (i) *If C satisfies C1 and C2, there is a complete and transitive $\succsim$ with $C = C_{\succsim}$, and a $u : X \rightarrow \mathbb{R}$ with $C = C_u$.*
- (ii) *If $\succsim$ is complete and transitive, then $C_{\succsim}$ satisfies C1 and C2, and some u represents $\succsim$.*
- (iii) *For any $u : X \rightarrow \mathbb{R}$, C_u satisfies C1 and C2, and the relation it represents is complete and transitive. Moreover the choice function generated by that relation through (1) is exactly C_u .*

Proof. Step 1. Part (ii). $C_{\succsim}(A)$ is the set of maximal elements of A , nonempty by Lemma 3.5, so C1 holds. For C2, let $x, y \in A \cap B$ with $x \in C_{\succsim}(A)$ and $y \notin C_{\succsim}(A)$. Some $w \in A$ has $w \succ y$, and $x \succsim w$, so $x \succ y$ by transitivity; since $x \in B$, y is not maximal in B . The existence of u is Theorem 3.6.

Step 2. Part (iii). For $x \succsim y \iff u(x) \geq u(y)$ the relation is complete and transitive because $\geq$ is, and $C_u = C_{\succsim}$; now apply Step 1.

Step 3. Part (i): the candidate relation. Define $x \succsim y : \iff x \in C(\{x, y\})$. Reflexivity is $C(\{x\}) = \{x\}$ and completeness is $C(\{x, y\}) \neq \emptyset$, both by C1.

Step 4. Part (i): transitivity. Let $x \in C(\{x, y\})$ and $y \in C(\{y, z\})$, and put $T = \{x, y, z\}$, so $C(T) \neq \emptyset$ by C1. Exactly one of three cases occurs. If $x \in C(T)$ there is nothing to prove. If $y \in C(T)$, then α gives $y \in C(\{x, y\})$, and β applied to $x, y \in C(\{x, y\})$ with $y \in C(T)$ gives $x \in C(T)$. If $z \in C(T)$, the same argument on $\{y, z\}$ gives $y \in C(T)$ and the previous case applies. So $x \in C(T)$, and α gives $x \in C(\{x, z\})$, that is $x \succsim z$.

Step 5. Part (i): $C = C_{\succsim}$. If $x \in C(A)$ then α gives $x \in C(\{x, y\})$ for every $y \in A$, so $x \in C_{\succsim}(A)$. Conversely let $x \in C_{\succsim}(A)$ and pick $y \in C(A)$, which exists by C1. Then $x \in C(\{x, y\})$, and α applied to $y \in C(A)$ gives $y \in C(\{x, y\})$; now β with $\{x, y\} \subseteq A$ and $y \in C(A)$ gives $x \in C(A)$. Finally u exists by Theorem 3.6, and $C_u = C_{\succsim} = C$.

□

Requiring C to be defined on *every* nonempty subset of X is demanding: nobody observes choices from arbitrary menus.

Definition 3.19 (Domain, context). A *domain* is a collection D of nonempty subsets of X , the menus on which the agent's behaviour is observed. The pair (X, D) is a *context*, and from now on C is defined on D only.

Definition 3.20 (Rationalizability). A choice function C on D is *rationalizable* if there is a preference $\succsim$ on X with $C(A) = C_{\succsim}(A)$ for every $A \in D$.

Definition 3.21 (Condition α , single-valued form). A choice function C on D satisfies *condition α* if for all $A, B \in D$ with $A \subseteq B$ and $C(B) \in A$ we have $C(A) = C(B)$.

Example 3.22 (The second-best rule violates α). Fix a strict order $a \succ b \succ c$ on $X = \{a, b, c\}$ and let $C(A)$ be the best element of A among the non-maximal ones, i.e. the second best, with $C(A)$ the unique element when $|A| = 1$. Then $C(\{a, b, c\}) = b$ and $b \in \{b, c\} \subseteq \{a, b, c\}$, so α requires $C(\{b, c\}) = b$; but the second best of $\{b, c\}$ is c . The menu matters: dropping a promotes c .

Theorem 3.23. Let D be closed under unions, i.e. $A, B \in D \Rightarrow A \cup B \in D$. If C satisfies condition α , and is a choice function, then C is rationalizable.

Proof. Step 1. The revealed relation. Define a binary relation R on X by

$$x R y \quad :\Longleftrightarrow \quad x \neq y \text{ and } \exists A \in D \text{ with } y \in A \text{ and } C(A) = x,$$

read as “ x was chosen over y at least once”.

Step 2. R is transitive. Let $x R y$ and $y R z$, witnessed by $A \ni y$ with $C(A) = x$ and $B \ni z$ with $C(B) = y$. Then $A \cup B \in D$ and $C(A \cup B)$ lies in A or in B , so α makes it equal to $C(A)$ or to $C(B)$, that is $C(A \cup B) \in \{x, y\}$. If $C(A \cup B) = y$ then, since $y \in A$, α gives $C(A) = y \neq x$, a contradiction. So $C(A \cup B) = x$, and $z \in B \subseteq A \cup B$ gives $x R z$.

Step 3. R is a strict partial order. If $x R y$ and $y R x$, Step 2 gives $x R x$, which the definition excludes. So R is transitive and asymmetric, in particular acyclic.

Step 4. Extension to a preference. By Szpilrajn's theorem ([Szpilrajn, 1930](#)) there is a complete and transitive $\succsim$ whose strict part contains R : extend R to a linear order on X and read $\succsim$ off it.

Step 5. $C = C_{\succsim}$. For $A \in D$ and $x \in A$ with $x \neq C(A)$, the definition of R gives $C(A) R x$, hence $C(A) \succ x$. So $C(A)$ is $\succsim$ -maximal in A , and since C is single-valued, $C(A) = C_{\succsim}(A)$.

□

Remark 3.24. Condition α alone suffices here only because C is single-valued: β is what disciplines ties.

3.4 Choice correspondences and revealed preference

Definition 3.25 (Choice correspondence). A *choice correspondence* assigns to every menu A in the domain D of Definition [3.19](#) a nonempty subset $\emptyset \neq C(A) \subseteq A$.

Why would $C(A)$ have more than one element? There are two answers. The **first** is indifference: the agent is equally happy with all of them. The **second** is that the menu A does not describe the whole problem. The same menu can be put to the agent in different ways, and the way it is put can change what she takes. Call that presentation the *frame* f : the order in which the options are listed, which one she gets if she does nothing, the words used to describe them, who is asking. So (A, f) is a menu together with its presentation, and $C(A, f)$ is the single option she takes when menu A comes with frame f :

$$C(A) = \{x : x = C(A, f) \text{ for some frame } f\}.$$

Definition 3.26 (Weak axiom of revealed preference). C satisfies the *WARP* if for all $A, B \in D$ and all $x, y \in A \cap B$,

$$x \in C(A) \text{ and } y \in C(B) \implies x \in C(B).$$

In words: if x is ever revealed at least as good as y , then y is never revealed strictly better than x .

Proposition 3.27. *On a collection closed under intersection, the WARP is equivalent to α and β together, and to choice coherence C2.*

Proof. Step 1. WARP $\Rightarrow \alpha$. Let $x \in A \subseteq B$ with $x \in C(B)$ and take $y \in C(A)$. Then $x, y \in A \cap B = A$, and the WARP with the problems (B, A) and the pair (x, y) gives $x \in C(A)$.

Step 2. WARP $\Rightarrow \beta$. Let $x, y \in C(A)$, $A \subseteq B$ and $y \in C(B)$. Then $x, y \in A \cap B = A$, and the WARP gives $x \in C(B)$.

Step 3. α and $\beta \Rightarrow$ WARP. This is Step 3 of Lemma 3.17, since C2 is the contrapositive of the WARP.

□

Example 3.28 (α but not β : intransitive indifference). The agent can tell x from z and prefers x , but cannot tell x from y nor y from z , as in Example 2.5:

A	$\{x, y\}$	$\{y, z\}$	$\{x, z\}$	$\{x, y, z\}$
$C(A)$	$\{x, y\}$	$\{y, z\}$	$\{x\}$	$\{x, y\}$

Condition α holds: every element chosen from a menu is still chosen from each submenu containing it. Condition β fails: $y, z \in C(\{y, z\})$ and $y \in C(\{x, y, z\})$, but $z \notin C(\{x, y, z\})$. The induced relation $x \succsim y \iff x \in C(\{x, y\})$ is complete but not transitive: $z \sim y$ and $y \sim x$ while $x \succ z$. That is precisely what β excludes.

Example 3.29 (β but not α : the menu carries information). Sen’s polite guest. Three slices of cake $s < m < \ell$, and the rule is “take the largest slice that is not the largest available”.

A	$\{s, m\}$	$\{m, \ell\}$	$\{s, \ell\}$	$\{s, m, \ell\}$
$C(A)$	$\{s\}$	$\{m\}$	$\{s\}$	$\{m\}$

Condition α fails: $m \in C(\{s, m, \ell\})$ and $m \in \{s, m\} \subseteq \{s, m, \ell\}$, yet $m \notin C(\{s, m\})$. Condition β holds vacuously, because C is single-valued and β requires two distinct elements in $C(A)$.

Procedural rationality

Theorem 3.18 says that consistent choice looks like utility maximization. It says nothing about how the agent decides, and Simon's *satisficing* procedure (Simon, 1955) shows how far apart the two can be. The agent has a level she considers good enough, goes through the options in whatever order they reach her, and stops at the first one that clears it; if none does, she is left with the last one she looked at.

Definition 3.30 (Satisficing). Fix a valuation $v : X \rightarrow \mathbb{R}$, an *aspiration level* v^* , and a linear order O giving the sequence in which the agent examines alternatives. For $A \subseteq X$ let $L(A, O)$ be the list of A written in O -order. Then $C(A)$ is the first element of $L(A, O)$ with $v \geq v^*$, and the last element of $L(A, O)$ if there is no such element.

Proposition 3.31. *The choice function induced by satisficing satisfies condition α , hence is rationalizable on any union-closed domain.*

Proof. Let $A \subseteq B$ and $x = C(B) \in A$.

If $v(x) \geq v^*$, then x is the O -first element of B meeting the threshold, so no element of B preceding x meets it; a fortiori no element of A preceding x meets it. Since $x \in A$ meets it, x is the O -first element of A meeting the threshold, so $C(A) = x$.

If $v(x) < v^*$, then no element of B meets the threshold, for otherwise the first such element, not x , would have been chosen; and x is the O -last element of B . Since $A \subseteq B$, no element of A meets the threshold either, and $x \in A$ is O -last in B hence O -last in A . So $C(A) = x$. $\square$

4 Lecture 4. Monotonicity and continuity

From here on the set of alternatives carries the structure of a commodity space, $X = \mathbb{R}_+^K$, and a bundle $x = (x_1, \dots, x_K)$ records the quantity $x_k \geq 0$ of each commodity. The question changes: every restriction placed on $\succsim$ shows up as a restriction on the shape of u . On top of completeness and transitivity, the three classical restrictions are monotonicity, continuity and convexity, and a preference relation satisfying them is called *well behaved*.

Property of $\succsim$	Property of u
monotone	nondecreasing
strongly monotone	strictly increasing
continuous	continuous
convex	quasi-concave
strictly convex	strictly quasi-concave
homothetic and continuous	homogeneous of degree one
quasi-linear in x_1	$u(x) = x_1 + v(x_{-1})$
separable	$u(x) = U(v_1(x_1), \dots, v_K(x_K))$
strongly separable	$u(x) = v_1(x_1) + \dots + v_K(x_K)$
of class C^r	C^r with $\nabla u \neq 0$

Table 1: The roadmap. Each line is proved below.

How much of this can be tested depends on the number of goods. With $K = 2$ the pairwise condition already contains the whole of the acyclicity condition: a data set satisfies GARP if and only if it satisfies WARP (Theorem 10.5). WARP compares two observations at a time, so with two goods no data set can display an intransitive chain that is not a pairwise inconsistency. Transitivity is testable only from three goods on.

Example 4.1 (A cycle that no pair reveals). Let $K = 3$ and take the three observations

$$\begin{aligned} x^1 &= (3, 2, 1), & x^2 &= (1, 3, 2), & x^3 &= (2, 1, 3), \\ p^1 &= (2, 1, 2), & p^2 &= (2, 2, 1), & p^3 &= (1, 2, 2). \end{aligned}$$

Each bundle costs 10 at its own prices, the next one in the list costs 9 and the previous one costs 11. So x^1 is revealed strictly preferred to x^2 , x^2 to x^3 and x^3 to x^1 , a cycle, while no reverse comparison is ever available, so WARP holds. Every pair of observations is consistent with utility maximization and the three together are not.

4.1 Monotonicity

Write $x \geq y$ if $x_k \geq y_k$ for every k , and $x \gg y$ if $x_k > y_k$ for every k .

Definition 4.2 (Monotonicity and local nonsatiation).

- (i) $\succsim$ is *monotone* if $x \geq y$ implies $x \succsim y$, and $x \gg y$ implies $x \succ y$.
- (ii) $\succsim$ is *strongly monotone* if $x \geq y$ and $x \neq y$ imply $x \succ y$.
- (iii) $\succsim$ satisfies *local nonsatiation* if for every $y \in X$ and every $\varepsilon > 0$ there is $x \in X$ with $\|x - y\| \leq \varepsilon$ and $x \succ y$.

Example 4.3 (Leontief). Let $x \succsim y$ if and only if $\min_k x_k \geq \min_k y_k$. This is monotone but not strongly monotone: $(1, 2) \geq (1, 1)$ and the two bundles differ, yet both have minimum 1, so $(1, 2) \sim (1, 1)$.

Example 4.4 (Three benchmarks). Perfect substitutes $u(x) = \sum_k a_k x_k$ with every $a_k > 0$ are strongly monotone. Cobb-Douglas $u(x) = \prod_k x_k^{a_k}$ with every $a_k > 0$ is monotone but not strongly monotone on $\mathbb{R}_+^K$, since $(1, 0) \sim (0, 0)$ although $(1, 0) \geq (0, 0)$ and the two bundles differ; on the interior $\mathbb{R}_{++}^K$ it is strongly monotone.

Proposition 4.5. *Strong monotonicity implies monotonicity, and monotonicity implies local nonsatiation.*

Proof. Let $\succsim$ be strongly monotone. If $x \geq y$ then either $x = y$, and $x \succsim y$ by reflexivity, or $x \neq y$ and $x \succ y$; if $x \gg y$ then $x \neq y$, so $x \succ y$.

Let $\succsim$ be monotone, $y \in X$ and $\varepsilon > 0$. The bundle $x = y + \frac{\varepsilon}{2\sqrt{K}} \mathbf{1}$ lies in X and satisfies $x \gg y$, hence $x \succ y$, and $\|x - y\| = \varepsilon/2 \leq \varepsilon$. $\square$

Example 4.6 (Where local nonsatiation fails). $u(x) = -\|x - x^*\|^2$ has a bliss point and fails local nonsatiation at x^* and nowhere else. The relation $x \succsim y$ if $\lfloor x_1 + x_2 \rfloor \geq \lfloor y_1 + y_2 \rfloor$ fails it at every bundle whose coordinates do not add up to an integer, since its indifference sets are bands with nonempty interior.

Proposition 4.7. *Let u represent $\succsim$. Then $\succsim$ is monotone if and only if u is nondecreasing and $u(x) > u(y)$ whenever $x \gg y$, and $\succsim$ is strongly monotone if and only if u is strictly increasing.*

Proof. Both are the definitions read through $x \succsim y \iff u(x) \geq u(y)$ and $x \succ y \iff u(x) > u(y)$. $\square$

4.2 Continuity and the diagonal index

Continuity was defined in §3.2. On $X = \mathbb{R}_+^K$ the convenient form is the sequential one: if $x^n \succsim y^n$ for every n , $x^n \rightarrow x$ and $y^n \rightarrow y$, then $x \succsim y$.

Lemma 4.8 (Diagonal index). *Let $\succsim$ be a continuous and monotone weak order on $X = \mathbb{R}_+^K$. For every $x \in X$ there is exactly one $t(x) \geq 0$ with $x \sim t(x)\mathbf{1}$, where $\mathbf{1} = (1, \dots, 1)$, and $u = t$ is a continuous representation of $\succsim$.*

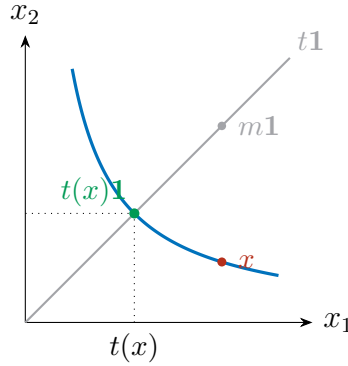


Figure 12: The diagonal index.

Proof. Fix x and let $m = \max_k x_k$. Then $m\mathbf{1} \geq x \geq 0$, so $m\mathbf{1} \succsim x \succsim 0$ by monotonicity. The sets

$$A = \{t \geq 0 : t\mathbf{1} \succsim x\}, \quad B = \{t \geq 0 : x \succsim t\mathbf{1}\}$$

are closed, being preimages of the closed contour sets $U(x)$ and $L(x)$ under the continuous map $t \mapsto t\mathbf{1}$; they are nonempty, since $m \in A$ and $0 \in B$; and they cover $[0, \infty)$ by completeness. Since $[0, \infty)$ is connected they intersect, and any t in the intersection satisfies $x \sim t\mathbf{1}$. It is unique: if $t > t'$ then $t\mathbf{1} \gg t'\mathbf{1}$, so $t\mathbf{1} \succ t'\mathbf{1}$ and both cannot be indifferent to x . Monotonicity also gives $t\mathbf{1} \succsim t'\mathbf{1}$ if and only if $t \geq t'$, so

$$x \succsim y \iff t(x)\mathbf{1} \succsim t(y)\mathbf{1} \iff t(x) \geq t(y),$$

and $u = t$ represents $\succsim$. For continuity, note that for $a \geq 0$

$$\{u > a\} = \{x : x \succ a\mathbf{1}\}, \quad \{u < a\} = \{x : a\mathbf{1} \succ x\},$$

which are open as complements of closed contour sets, while for $a < 0$ the first set is X and the second is empty. Preimages of open rays are open, so u is continuous. $\square$

5 Lecture 5. Convexity, homotheticity, quasi-linearity and separability

5.1 Convexity

Definition 5.1 (Convexity). $\succsim$ is *convex* if the upper contour set $U(x) = \{y \in X : y \succsim x\}$ is convex for every x . It is *strictly convex* if $y \succsim x$, $z \succsim x$ and $y \neq z$ imply $\alpha y + (1 - \alpha)z \succ x$ for every $\alpha \in (0, 1)$.

Convexity says that averages are at least as good as the extremes they average (Figure 13).

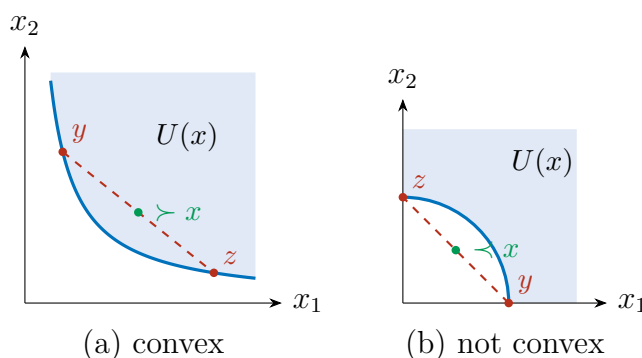


Figure 13: Definition 5.1. In (a) the upper contour set is convex: the segment between two bundles at least as good as x stays inside it, and here it stays strictly inside, which is strict convexity. In (b) the preference is for extremes, $u(x) = x_1^2 + x_2^2$: the upper contour set is the region outside the arc, and the midpoint of the segment joining two indifferent bundles falls out of it, so the average is strictly worse than either extreme.

Definition 5.2 (Concavity and quasi-concavity). u is *concave* if $u(\lambda x + (1 - \lambda)y) \geq \lambda u(x) + (1 - \lambda)u(y)$ for all x, y and $\lambda \in [0, 1]$, and *quasi-concave* if $\{x : u(x) \geq u(y)\}$ is convex for every y . Concavity implies quasi-concavity.

Proposition 5.3. If u represents $\succsim$, then $\succsim$ is convex if and only if u is quasi-concave.

Example 5.4 (Convex, strictly convex, neither). Cobb-Douglas on $\mathbb{R}_{++}^K$ is strictly convex. Leontief and perfect substitutes are convex but not strictly, because their indifference sets contain segments. A preference for extremes fails convexity: $u(x) = x_1^2 + x_2^2$ gives $u(1, 0) = u(0, 1) = 1$ and $u(\frac{1}{2}, \frac{1}{2}) = \frac{1}{2}$, so the upper contour set of $(1, 0)$ is not convex, as in panel (b) of Figure 13.

Example 5.5 (Convex preferences with no concave representation). On $X = \mathbb{R}$ let $u(x) = \max\{0, x\}$ and let $\succsim$ be the preference it represents: all $x \leq 0$ are indifferent to one another and $[0, \infty)$ carries the usual order. The upper level sets are intervals, so $\succsim$ is convex. No representation is concave: any v representing $\succsim$ is constant at some c on $(-\infty, 0]$ and has $v(1) > c$, so concavity at $-\frac{1}{2} = \frac{1}{2}(-2) + \frac{1}{2}(1)$ would force $c = v(-\frac{1}{2}) \geq \frac{1}{2}(c + v(1)) > c$ (Figure 14).

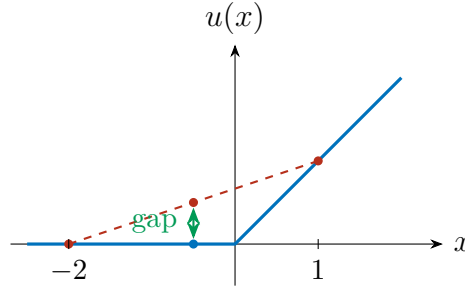


Figure 14: Example 5.5. The graph of $u(x) = \max\{0, x\}$ is flat on $(-\infty, 0]$ and rises afterwards. The chord from $(-2, 0)$ to $(1, 1)$ lies strictly above the graph at $x = -\frac{1}{2}$, so no representation of $\succsim$ can be concave, although $\succsim$ is convex.

5.2 Homotheticity

Definition 5.6 (Homotheticity). $\succsim$ is *homothetic* if $x \succsim y$ implies $\alpha x \succsim \alpha y$ for every $\alpha \geq 0$.

Proposition 5.7. If $\succsim$ admits a representation homogeneous of degree $\lambda > 0$, that is $u(\alpha x) = \alpha^\lambda u(x)$ for every $\alpha > 0$, then $\succsim$ is homothetic.

Proof. For $\alpha > 0$, multiplying by $\alpha^\lambda > 0$ preserves the inequality, so $u(x) \geq u(y) \iff u(\alpha x) \geq u(\alpha y)$. For $\alpha = 0$ the two bundles coincide. $\square$

Theorem 5.8. A continuous, monotone and homothetic $\succsim$ on $\mathbb{R}_+^K$ has a continuous representation homogeneous of degree one.

Proof. Take $u = t$ from Lemma 4.8, so $x \sim t(x)\mathbf{1}$. Homotheticity applied to both directions of that indifference gives $\alpha x \sim \alpha t(x)\mathbf{1} = (\alpha t(x))\mathbf{1}$ for $\alpha \geq 0$, and uniqueness of the diagonal index gives $t(\alpha x) = \alpha t(x)$. $\square$

Example 5.9. Perfect substitutes, Leontief, Cobb-Douglas and CES $u(x) = \left(\sum_k a_k x_k^\rho\right)^{1/\rho}$ with $\rho < 1$ and $\rho \neq 0$ are homothetic, and all of them are already homogeneous of degree

one, so t reproduces them up to a positive factor. Stone-Geary $u(x) = \prod_k (x_k - \gamma_k)^{a_k}$ with some $\gamma_k \neq 0$ is not homothetic.

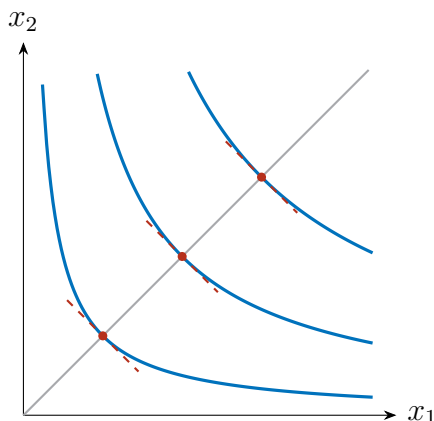


Figure 15: Homothetic preferences. Each indifference curve is a radial blow-up of the others, so along any ray from the origin the marginal rate of substitution is the same and the tangents are parallel. Drawn for $u(x) = x_1 x_2$, whose curves through $\alpha \mathbf{1}$ are the images of the curve through $\mathbf{1}$ under scaling by α .

5.3 Quasi-linearity

Now let $X = \mathbb{R} \times \mathbb{R}_+^{K-1}$, so the first commodity, the *numeraire*, is unbounded in both directions. Write $e_1 = (1, 0, \dots, 0)$.

Definition 5.10 (Quasi-linearity). $\succsim$ is *quasi-linear in x_1* if $x \succsim y$ implies $x + \varepsilon e_1 \succsim y + \varepsilon e_1$ for every $\varepsilon > 0$.

Indifference curves are then horizontal translates of one another along the numeraire axis (Figure 16), so the marginal rate of substitution between any two of the other goods does not depend on how much numeraire the agent holds.

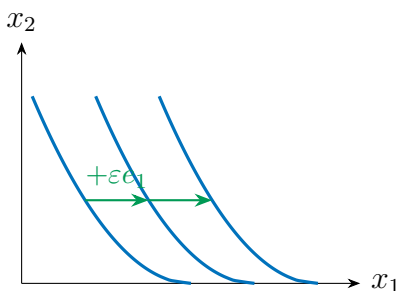


Figure 16: Quasi-linearity in x_1 . Drawn for $v(x_2) = 2\sqrt{x_2}$.

Lemma 5.11 (Translation invariance). *Let $\succsim$ on $X = \mathbb{R} \times \mathbb{R}_+^{K-1}$ be continuous, strongly monotone and quasi-linear in x_1 . Then for all $x, y \in X$ and every $t \in \mathbb{R}$,*

$$x \succsim y \iff x + te_1 \succsim y + te_1, \quad x \sim y \iff x + te_1 \sim y + te_1.$$

Proof. Let $t > 0$; $t = 0$ is trivial. The forward implication is Definition 5.10. For the converse, let $x + te_1 \succsim y + te_1$ and suppose $y \succ x$. The set $\{z : y \succ z\}$ is open and contains x , and the first coordinate is unrestricted, so $y \succ x + \delta e_1$ for some $\delta > 0$. Shifting by t and using strong monotonicity, $y + te_1 \succ x + (\delta + t)e_1 \succ x + te_1$, a contradiction. For $t < 0$ apply the case $-t > 0$ to $x + te_1$ and $y + te_1$. The statement for $\sim$ follows by applying the first equivalence in both directions. $\square$

Theorem 5.12. *Let $\succsim$ on $X = \mathbb{R} \times \mathbb{R}_+^{K-1}$ be continuous, strongly monotone and quasi-linear in x_1 . For every $x_{-1} \in \mathbb{R}_+^{K-1}$ there is a unique $v(x_{-1}) \geq 0$ with $(v(x_{-1}), 0, \dots, 0) \sim (0, x_{-1})$, and*

$$u(x) = x_1 + v(x_{-1})$$

represents $\succsim$.

Proof. Throughout, $(s, \mathbf{0})$ abbreviates the bundle $(s, 0, \dots, 0)$ on the numeraire axis, and $e_1 = (1, 0, \dots, 0)$.

Step 1. On the numeraire axis the order is the order of $\mathbb{R}$:

$$(s, \mathbf{0}) \succsim (s', \mathbf{0}) \iff s \geq s'.$$

If $s > s'$ then $(s, \mathbf{0}) \geq (s', \mathbf{0})$ coordinatewise and the two bundles differ, so strong monotonicity gives $(s, \mathbf{0}) \succ (s', \mathbf{0})$. If $s = s'$ the two bundles are equal, and $\succsim$ is reflexive. Conversely, if $s < s'$ the first case applied the other way round gives $(s', \mathbf{0}) \succ (s, \mathbf{0})$, which is incompatible with $(s, \mathbf{0}) \succsim (s', \mathbf{0})$.

Step 2. v is a well defined function of x_{-1} . Fix $y = x_{-1} \in \mathbb{R}_+^{K-1}$ and define

$$A = \{t \geq 0 : (t, \mathbf{0}) \succsim (0, y)\}.$$

We first show that A is nonempty. If $y = \mathbf{0}$, then $0 \in A$. Suppose $y \neq \mathbf{0}$, and consider the compact segment $C = \{\lambda y : 0 \leq \lambda \leq 1\}$. For $K = 2$, this is just the interval $[0, y]$. Strong monotonicity gives

$$(1, z) \succ (0, z) \quad \text{for every } z \in C.$$

By continuity and compactness, there is a single $\delta > 0$ such that

$$z, w \in C, \quad \|z - w\| < \delta \implies (1, z) \succ (0, w).$$

Indeed, otherwise there would be sequences $z_n, w_n \in C$ with $\|z_n - w_n\| \rightarrow 0$ and $(1, z_n) \not\succ (0, w_n)$. By completeness, $(0, w_n) \succeq (1, z_n)$. Compactness gives a subsequence along which $z_n \rightarrow z \in C$ and $w_n \rightarrow z$. Continuity would then imply $(0, z) \succeq (1, z)$, contradicting strong monotonicity.

Choose an integer $N \geq 1$ with $\|y\|/N < \delta$. For $k = 0, \dots, N-1$,

$$\left(1, \frac{k}{N}y\right) \succ \left(0, \frac{k+1}{N}y\right).$$

Lemma 5.11 also preserves strict preference, since it preserves both weak comparisons. Adding $N - k - 1$ to both numeraire coordinates gives

$$\left(N - k, \frac{k}{N}y\right) \succ \left(N - k - 1, \frac{k+1}{N}y\right).$$

By transitivity,

$$(N, \mathbf{0}) \succ \left(N - 1, \frac{1}{N}y\right) \succ \dots \succ (0, y).$$

Thus $N \in A$, proving nonemptiness.

To obtain indifference, let $s = \inf A \geq 0$. The set A is closed by continuity, so $s \in A$ and $(s, \mathbf{0}) \succeq (0, y)$. If $s = 0$, monotonicity gives $(0, y) \succeq (0, \mathbf{0})$. If $s > 0$, choose $t_n \in [0, s)$ with $t_n \rightarrow s$. Since $t_n \notin A$, completeness gives $(0, y) \succ (t_n, \mathbf{0})$, and continuity implies $(0, y) \succeq (s, \mathbf{0})$. In either case, $(s, \mathbf{0}) \sim (0, y)$.

For uniqueness, suppose $(s, \mathbf{0}) \sim (0, x_{-1})$ and $(s', \mathbf{0}) \sim (0, x_{-1})$. Transitivity gives $(s, \mathbf{0}) \sim (s', \mathbf{0})$, so both $(s, \mathbf{0}) \succeq (s', \mathbf{0})$ and $(s', \mathbf{0}) \succeq (s, \mathbf{0})$ hold, and Step 1 gives $s \geq s'$ and $s' \geq s$, that is $s = s'$. Taking $x_{-1} = 0$ and using reflexivity gives $v(0) = 0$.

Step 3. For every $t \in \mathbb{R}$,

$$x \sim y \iff x + te_1 \sim y + te_1.$$

This is Lemma 5.11; both $x + te_1$ and $y + te_1$ lie in X for every real t .

Step 4. Every bundle is indifferent to a point of the axis:

$$x = (x_1, x_{-1}) \sim (x_1 + v(x_{-1}), \mathbf{0}) \quad \text{for every } x \in X.$$

By the definition of v , $(v(x_{-1}), \mathbf{0}) \sim (0, x_{-1})$. Apply Step 3 with the shift $t = x_1$, which may be negative: the left side becomes $(v(x_{-1}) + x_1, \mathbf{0})$ and the right side becomes $(0 + x_1, x_{-1}) = x$.

Step 5. u represents $\succsim$. Fix $x, y \in X$ and write $a = x_1 + v(x_{-1}) = u(x)$ and $b = y_1 + v(y_{-1}) = u(y)$, so that $x \sim (a, \mathbf{0})$ and $y \sim (b, \mathbf{0})$ by Step 4. Recall that $x \sim z$ means both $x \succsim z$ and $z \succsim x$.

Suppose $x \succsim y$. Chaining $(a, \mathbf{0}) \succsim x$, $x \succsim y$ and $y \succsim (b, \mathbf{0})$ by transitivity gives $(a, \mathbf{0}) \succsim (b, \mathbf{0})$, and Step 1 gives $a \geq b$, that is $u(x) \geq u(y)$.

Suppose $u(x) \geq u(y)$, that is $a \geq b$. Step 1 gives $(a, \mathbf{0}) \succsim (b, \mathbf{0})$, and chaining $x \succsim (a, \mathbf{0})$, $(a, \mathbf{0}) \succsim (b, \mathbf{0})$ and $(b, \mathbf{0}) \succsim y$ by transitivity gives $x \succsim y$.

So $x \succsim y$ if and only if $u(x) \geq u(y)$, as desired.

□

The compensation argument also works on $X = \mathbb{R}^K$, where the proof of Lemma 5.11 applies unchanged. For any $y \in \mathbb{R}^{K-1}$, use the same compact segment from $\mathbf{0}$ to y . The forward chain gives $(N, \mathbf{0}) \succ (0, y)$; the reversed chain gives $(N, y) \succ (0, \mathbf{0})$, which becomes $(0, y) \succ (-N, \mathbf{0})$ after translation. Thus

$$(N, \mathbf{0}) \succ (0, y) \succ (-N, \mathbf{0}).$$

The sets of $t \in [-N, N]$ satisfying $(t, \mathbf{0}) \succsim (0, y)$ and $(0, y) \succsim (t, \mathbf{0})$ are closed, contain opposite endpoints and cover the interval by completeness. Connectedness implies that they intersect, yielding compensation for signed bundles as well.

Theorem 5.13. *If $\succsim$ on $\mathbb{R}^K$ is continuous, strongly monotone and quasi-linear in every coordinate, then it has a linear representation $u(x) = \alpha_1 x_1 + \dots + \alpha_K x_K$ with every $\alpha_k > 0$.*

Proposition 5.14 (Quasi-linear and homothetic means linear). *Let $\succsim$ on $\mathbb{R} \times \mathbb{R}_+$ be quasi-linear in x_1 with representation $u(x) = x_1 + v(x_2)$ as in Theorem 5.12. If $\succsim$ is also homothetic, then $v(x_2) = cx_2$ with $c > 0$, so $\succsim$ are perfect substitutes.*

Proof. Here $v(0) = 0$. For $r, s \geq 0$ the bundles $(v(r), s)$ and $(v(s), r)$ have the same value of u , hence are indifferent, so homotheticity gives $(\alpha v(r), \alpha s) \sim (\alpha v(s), \alpha r)$ for $\alpha \geq 0$, that is $\alpha v(r) + v(\alpha s) = \alpha v(s) + v(\alpha r)$. Setting $r = 0$ leaves $v(\alpha s) = \alpha v(s)$, and $s = 1$ gives $v(\alpha) = c\alpha$ with $c = v(1) > 0$. □

5.4 Separability

Definition 5.15 (Separability). Let $I \subseteq \{1, \dots, K\}$ and write a bundle as $x = (x_I, x_{-I})$, the levels inside the group I and the levels outside it. Then $\succsim$ is *separable in I* if for all x_I, x'_I and all x_{-I}, x'_{-I} ,

$$(x_I, x_{-I}) \succsim (x'_I, x_{-I}) \iff (x_I, x'_{-I}) \succsim (x'_I, x'_{-I}).$$

It is *separable* if it is separable in every single coordinate, and *strongly separable* if it is separable in every group I of $\{1, \dots, K\}$.

Example 5.16. Take $K = 3$ and $I = \{1\}$. Separability in coordinate 1 says that whether two units of good 1 beat one unit does not depend on how much of goods 2 and 3 the bundle carries. For perfect substitutes $u(x) = x_1 + x_2 + x_3$ it holds, since $(2, x_2, x_3) \succ (1, x_2, x_3)$ whatever x_2 and x_3 are. For Leontief $u(x) = \min_k x_k$ it fails: $(2, 1, 1) \sim (1, 1, 1)$, because the extra unit is worth nothing when the other goods are scarce, while $(2, 3, 3) \succ (1, 3, 3)$.

Theorem 5.17 (Separable groups can be summarized). *Let $\succsim$ on $X = \mathbb{R}_+^K$ be continuous and strongly monotone, let $I_1, \dots, I_m$ partition $\{1, \dots, K\}$, and let $\succsim$ be separable in every group I_j . Then:*

- (i) *the within-group relation $\succsim_j$ on $\mathbb{R}_+^{I_j}$, defined by $x_{I_j} \succsim_j x'_{I_j}$ if $(x_{I_j}, \bar{x}_{-I_j}) \succsim (x'_{I_j}, \bar{x}_{-I_j})$ for some context $\bar{x}_{-I_j}$, does not depend on the context and is itself a continuous and strongly monotone preference relation;*
- (ii) *there are continuous functions v_j representing $\succsim_j$ and a continuous U , strictly increasing in each argument, such that*

$$u(x) = U(v_1(x_{I_1}), \dots, v_m(x_{I_m}))$$

represents $\succsim$.

Proof. Write $I = I_j$ and $x = (x_I, x_{-I})$; a *context* is a value of x_{-I} .

Step 1. Separability $\Rightarrow \succsim_j$ is well defined. Its definition asks for *some* context; the display in Definition 5.15 makes the comparison the same at every context. Hence $\succ_j$ and $\sim_j$ are well defined too.

Step 2. $\succsim$ complete and transitive $\Rightarrow$ so is $\succsim_j$. Read the comparisons at $\bar{x}_{-I}$, on the bundles $(\cdot, \bar{x}_{-I})$.

Step 3. $\succsim$ continuous $\Rightarrow \succsim_j$ continuous. The contour sets of $\succsim_j$ are preimages of contour sets of $\succsim$ under the continuous map $b \mapsto (b, \bar{x}_{-I})$, hence closed.

Step 4. $\succsim$ strongly monotone $\Rightarrow \succsim_j$ strongly monotone. $x_I \geq x'_I$ with $x_I \neq x'_I$ gives $(x_I, \bar{x}_{-I}) \geq (x'_I, \bar{x}_{-I})$ with the bundles distinct. Steps 1 to 4 are part (i).

Step 5. Steps 2 to 4 $\Rightarrow$ each group has an index. Lemma 4.8 applies to $\succsim_j$ on $\mathbb{R}_+^{I_j}$ and gives a continuous v_j representing $\succsim_j$ with $x_{I_j} \sim_j v_j(x_{I_j})\mathbf{1}$.

Step 6. The same lemma for $\succsim$ itself. It applies on $\mathbb{R}_+^K$ because $\succsim$ is continuous and strongly monotone, and gives a continuous w representing $\succsim$, with $x \sim w(x)\mathbf{1}$.

Step 7. From m group levels back to a bundle. For $s = (s_1, \dots, s_m) \in \mathbb{R}_+^m$ let $\delta(s) \in X$ be the bundle that gives every commodity of group I_j the level s_j :

$$\delta(s)_k = s_j \quad \text{for the } j \text{ with } k \in I_j, \quad \text{that is} \quad \delta(s)_{I_j} = s_j \mathbf{1}.$$

The map δ is linear, hence continuous, and so is $U := w \circ \delta$.

Step 8. Step 1 $\Rightarrow$ swapping one group preserves indifference. If $z_I \sim_j y_I$, the two weak comparisons hold at every context, so at y_{-I} : $(y_I, y_{-I}) \sim (z_I, y_{-I})$.

Step 9. Steps 5 and 8 $\Rightarrow$ swap all groups. Put $y^0 = x$ and let y^j be y^{j-1} with the block I_j replaced by $v_j(x_{I_j})\mathbf{1}$. Each swap gives $y^{j-1} \sim y^j$, so by transitivity

$$x \sim y^m = \delta(v_1(x_{I_1}), \dots, v_m(x_{I_m})).$$

Step 10. Steps 6 and 9 $\Rightarrow$ part (ii). Apply w to that indifference:

$$w(x) = U(v_1(x_{I_1}), \dots, v_m(x_{I_m})) \quad \text{for every } x.$$

So $U \circ (v_1, \dots, v_m)$ is w itself, and w represents $\succsim$.

Step 11. Strong monotonicity $\Rightarrow U$ strictly increasing. If s and s' differ only in the j -th coordinate, with $s_j > s'_j$, then $\delta(s) \geq \delta(s')$ and the bundles differ, so $\delta(s) \succ \delta(s')$ and $U(s) > U(s')$.

□

Example 5.18 (Strongly separable). Cobb-Douglas is strongly separable, since $\log u(x) = \sum_k a_k \log x_k$ is additive and additivity delivers the display for every group I at once. CES is strongly separable for the same reason, through $\sum_k a_k x_k^\rho$.

Example 5.19 (Separable but not strongly separable). On $\mathbb{R}_{++}^3$ let $u(x) = x_1x_2 + x_3$. It is separable in each single coordinate: for coordinate 1 the comparison reduces to $(x_1 - x'_1)x_2 \geq 0$, that is to $x_1 \geq x'_1$, whatever the rest of the bundle is, and coordinates 2 and 3 work the same way. It is not separable in the group $I = \{1, 3\}$: comparing $(x_1, x_3) = (2, \frac{1}{2})$ with $(x'_1, x'_3) = (1, \frac{3}{2})$ amounts to $x_2 \geq 1$, so the first wins when $x_2 = 2$ and the second when $x_2 = 1/2$. Separability in every single coordinate therefore does not give the additive form.

5.5 Differentiable preferences

Being differentiable is a property of a *function*, and it does not survive an arbitrary strictly increasing change of representation. So asking for “differentiable preferences” means asking for something about $\succsim$ itself, not about one of its representations.

Definition 5.20 (Differentiable preferences). $\succsim$ on an open $X \subseteq \mathbb{R}^K$ is *differentiable at* x if there is $v(x) \in \mathbb{R}^K$, $v(x) \neq 0$, such that for every direction d and all small enough $\varepsilon > 0$,

$$v(x) \cdot d > 0 \Rightarrow x + \varepsilon d \succ x, \quad v(x) \cdot d < 0 \Rightarrow x \succ x + \varepsilon d.$$

It is *differentiable* if this holds at every x .

The vector $v(x)$ is a local normal, determined up to a positive scalar: the indifference set through x has a supporting hyperplane there, improving directions on one side and worsening directions on the other (Figure 17). It does not follow that the indifference set is a C^1 hypersurface, nor that $x \mapsto v(x)/\|v(x)\|$ is continuous. This is differentiability of the relation at one point, not a differentiable structure on X .

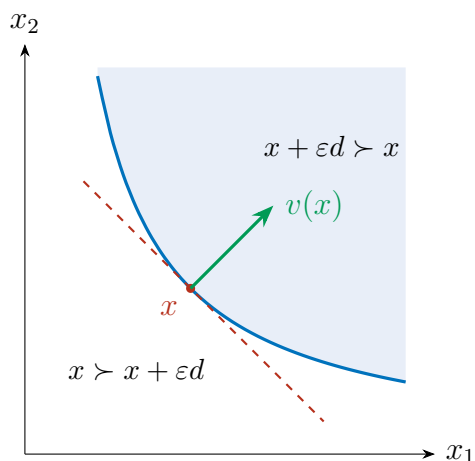


Figure 17: Definition 5.20. The dashed line is the supporting hyperplane $\{y : v(x) \cdot (y - x) = 0\}$ at x ; directions d with $v(x) \cdot d > 0$ point into the shaded better set.

Example 5.21 (Leontief is not differentiable). At $x = (1, 1)$ the bundle $x + \varepsilon d$ is strictly better than x only when $d \gg 0$, so Definition 5.20 would need $\{d : v(x) \cdot d > 0\} \subseteq \{d : d \gg 0\}$. An open halfspace is never contained in the open positive orthant, so no $v(x)$ works. The reason is visible in the picture: the upper contour set is the quadrant $\{y : y \geq x\}$, and a hyperplane through x supports it exactly when its normal q satisfies $q \geq 0$ and $q \neq 0$. The supporting normals therefore fill a whole quarter turn of directions, from $(1, 0)$ to $(0, 1)$, and no single one of them is the normal (Figure 18).

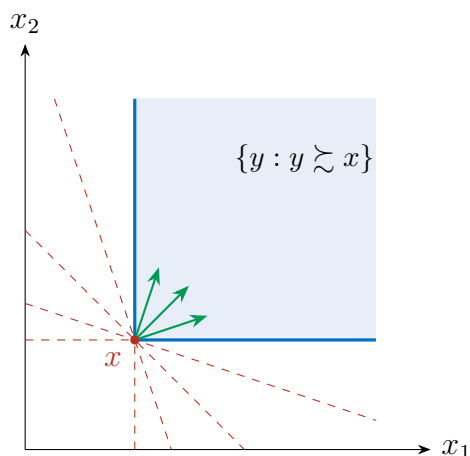


Figure 18: Example 5.21. Every line through x with non-positive slope supports the upper contour set there; the dashed lines are a sample, the horizontal and the vertical ones being the two extremes, and the arrows are the corresponding normals. Whichever normal $v(x)$ is picked, the halfspace $\{d : v(x) \cdot d > 0\}$ contains directions along the indifference set that do not improve x .

Two things are worth keeping in mind. First, the condition is local and says nothing about how $v(x)$ moves with x : it can jump from one bundle to the next. Second, it is what a first order condition needs. If x maximizes $\succsim$ on the budget set $\{y : p \cdot y \leq p \cdot x\}$ and $\succsim$ is differentiable at x , then $v(x)$ is proportional to p , which is the usual tangency condition. Appendix C takes this further: what it means for $\succsim$ to be of class C^r , stated through the indifference set as a manifold, why that version is ordinal after all, what it buys for demand, and a definition of differentiability that does not mention derivatives at all.

6 Lecture 6. The consumer's problem

6.1 Improvement directions

Definition 6.1 (Improvements and improvement directions). Let $x \in X$. A vector $d \in \mathbb{R}^K$ with $x + d \in X$ is an *improvement* at x if $x + d \succ x$. It is an *improvement direction* at x if there is $\lambda^* > 0$ such that

$$x + \lambda d \succ x \quad \text{for every } 0 < \lambda < \lambda^*.$$

Write $D_{\succsim}(x)$ for the set of improvement directions at x , and, for a map $v : X \rightarrow \mathbb{R}_+^K$,

$$D_v(x) = \{d \in \mathbb{R}^K : d \cdot v(x) > 0\}$$

for the set of directions whose value is positive at the prices $v(x)$.

An improvement is a single bundle that beats x . An improvement direction is a heading such that *every* short enough step along it beats x .

Lemma 6.2 (Every improvement is an improvement direction). *Let $x \in X$ and let d be an improvement at x . Then $d \in D_{\succsim}(x)$ in either of the following cases:*

- (i) $\succsim$ is strictly convex;
- (ii) $\succsim$ is convex, strongly monotone and continuous.

Proof. There are 2 cases:

Case (i). For $\lambda \in (0, 1)$ write $x + \lambda d = \lambda(x + d) + (1 - \lambda)x$. The two bundles $x + d$ and x are both $\succsim x$, the first strictly, and they are distinct because $d \neq 0$. Strict convexity gives $\lambda(x + d) + (1 - \lambda)x \succ x$. So $\lambda^* = 1$ works.

Case (ii). Since $\{y : y \succ x\}$ is open and contains $x + d$, and since $x + d \neq 0$, we may pick $\eta > 0$ small and set

$$z = x + d - \eta e, \quad e_k = \begin{cases} 1, & (x + d)_k > 0, \\ 0, & (x + d)_k = 0, \end{cases}$$

so that $z \in X$, $z \succ x$, $z \leq x + d$, and $z_k < (x + d)_k$ at every k with $(x + d)_k > 0$. There is at least one such k : otherwise $x + d = 0$ and $x + d \succ x \succsim 0 = x + d$, which is impossible.

Fix $\lambda \in (0, 1)$. Convexity applied to $z \succ x$ gives $\lambda z + (1 - \lambda)x \succsim x$. Also

$$x + \lambda d - (\lambda z + (1 - \lambda)x) = \lambda[(x + d) - z] = \lambda \eta e \geq 0,$$

and the two bundles differ, so strong monotonicity gives $x + \lambda d \succ \lambda z + (1 - \lambda)x$.

Transitivity yields $x + \lambda d \succ x$ for every $\lambda \in (0, 1)$, so again $\lambda^* = 1$ works.

□

Lemma 6.3 ($D_{\succsim}(x)$ is an open cone). *Let x be interior to X and let $\succsim$ be continuous, convex and strongly monotone. Then $D_{\succsim}(x)$ is a cone, and it is open.*

Proof. Recall that $C \subseteq \mathbb{R}^K$ is a *cone* if $d \in C$ and $\alpha > 0$ imply $\alpha d \in C$: the set is closed under rescaling by positive numbers.

Step 1. $D_{\succsim}(x)$ is a cone. Let $d \in D_{\succsim}(x)$, with constant λ^* , and let $\alpha > 0$. Then $x + \lambda(\alpha d) = x + (\lambda\alpha)d \succ x$ whenever $\lambda\alpha < \lambda^*$, that is for every $0 < \lambda < \lambda^*/\alpha$. So $\alpha d \in D_{\succsim}(x)$, with constant λ^*/α .

Step 2. $D_{\succsim}(x)$ is open. Let $d \in D_{\succsim}(x)$ and fix $\lambda_0 \in (0, \lambda^*)$, so that $x + \lambda_0 d \succ x$. The set $\{y : y \succ x\}$ is open, and x is interior, so $x + \lambda_0 d' \succ x$ for every d' in some neighbourhood of d . For such a d' the vector $\lambda_0 d'$ is an improvement at x , hence an improvement direction by Lemma 6.2(ii), and then d' itself is one by Step 1.

□

Definition 6.4 (Differentiable preferences, direction form). $\succsim$ is *differentiable* if there is a map $v : X \rightarrow \mathbb{R}_+^K$ with

$$D_{\succsim}(x) = D_v(x) \quad \text{for every } x \in X.$$

Example 6.5 (Two standard cases). For $u(x) = \alpha x_1 + \beta x_2$ with $\alpha, \beta > 0$ the improvement directions at any x are those with $\alpha d_1 + \beta d_2 > 0$, so $\succsim$ is differentiable with the constant $v(x) = (\alpha, \beta)$. For $u(x) = \min\{\alpha x_1, \beta x_2\}$ the relation is differentiable at every x with $\alpha x_1 \neq \beta x_2$, with

$$v(x) = \begin{cases} (1, 0), & \alpha x_1 < \beta x_2, \\ (0, 1), & \alpha x_1 > \beta x_2, \end{cases}$$

since only the binding good can be improved. On the diagonal $\alpha x_1 = \beta x_2$ it fails, which is Example 5.21.

Theorem 6.6 (Smooth utility gives differentiable preferences). *Let $\succsim$ on the interior of X be represented by a u that is differentiable, strictly increasing in each coordinate with $\partial u / \partial x_k > 0$, and quasi-concave. Then $\succsim$ is differentiable, with $v(x) = \nabla u(x)$.*

Proof. First note that $\succsim$ inherits from u everything the lemmas need: continuity, since u is differentiable hence continuous; convexity, since u is quasi-concave (Proposition 5.3); and strong monotonicity, since u has positive partial derivatives.

Step 1. $D_{\nabla u}(x) \subseteq D_{\succsim}(x)$. Let $d \cdot \nabla u(x) > 0$. Differentiability of u gives

$$u(x + \lambda d) = u(x) + \lambda d \cdot \nabla u(x) + o(\lambda),$$

so $u(x + \lambda d) > u(x)$, that is $x + \lambda d \succ x$, for every small enough $\lambda > 0$. Hence $d \in D_{\succsim}(x)$.

Step 2. Let $d \in D_{\succsim}(x)$. Then $u(x + \lambda d) > u(x)$ for all small $\lambda > 0$, so

$$\frac{u(x + \lambda d) - u(x)}{\lambda} > 0,$$

and letting $\lambda \downarrow 0$ gives $d \cdot \nabla u(x) \geq 0$.

Step 3. Suppose $d \in D_{\succsim}(x)$ with $d \cdot \nabla u(x) = 0$. By Lemma 6.3 the set $D_{\succsim}(x)$ is open, so $d' = d - \eta \mathbf{1} \in D_{\succsim}(x)$ for every small $\eta > 0$. But

$$d' \cdot \nabla u(x) = d \cdot \nabla u(x) - \eta \mathbf{1} \cdot \nabla u(x) = -\eta \mathbf{1} \cdot \nabla u(x) < 0,$$

because every partial derivative is positive. The expansion of Step 1, read with a negative directional derivative, then gives $u(x + \lambda d') < u(x)$ for all small $\lambda > 0$, contradicting $d' \in D_{\succsim}(x)$.

Step 4. Conclusion. Steps 2 and 3 give $D_{\succsim}(x) \subseteq D_{\nabla u}(x)$, and with Step 1 the two sets are equal.

□

6.2 Budget sets and consumer's problem

Definition 6.7 (Budget set). For a price vector $p \in \mathbb{R}_{++}^K$ and wealth $\omega > 0$,

$$B(p, \omega) = \{x \in \mathbb{R}_+^K : p \cdot x \leq \omega\}.$$

Since $B(\lambda p, \lambda \omega) = B(p, \omega)$ for every $\lambda > 0$, wealth is normalized to $\omega = 1$ and we write $B(p)$.

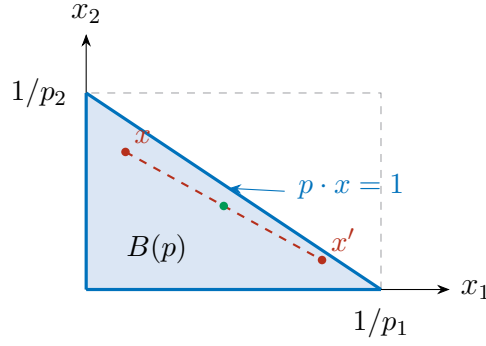


Figure 19: The budget set with $K = 2$. It is bounded, since it sits inside the dashed box $[0, 1/p_1] \times [0, 1/p_2]$, and closed, since it is cut out by the weak inequalities $p \cdot x \leq 1$ and $x \geq 0$, so it is compact. It is convex: the segment between any two of its points, such as the dashed one, does not leave it. In general it is the simplex with vertices 0 and e_k/p_k , $k = 1, \dots, K$.

Proposition 6.8. $B(p)$ is nonempty, compact and convex (Figure 19).

Proof. It contains 0. It is closed, being the intersection of the closed sets $\{x : p \cdot x \leq 1\}$ and $\mathbb{R}_+^K$, and bounded, since $x \in B(p)$ forces $0 \leq x_k \leq 1/p_k$ for every k ; by Heine-Borel it is compact. For convexity, let $x, x' \in B(p)$ and $\alpha \in [0, 1]$. Then $\alpha x + (1 - \alpha)x' \geq 0$ and

$$p \cdot (\alpha x + (1 - \alpha)x') = \alpha p \cdot x + (1 - \alpha)p \cdot x' \leq \alpha + (1 - \alpha) = 1.$$

□

Definition 6.9 (The consumer's problem). Given $p \in \mathbb{R}_{++}^K$, the *consumer's problem* is to find a $\succsim$ -best bundle in $B(p)$, that is some $x^* \in B(p)$ with $x^* \succsim y$ for every $y \in B(p)$.

Theorem 6.10 (Existence). *If $\succsim$ is continuous, the consumer's problem has a solution.*

Proof. Debreu's theorem (Debreu, 1954), proved in Appendix B (Theorem B.21), says that a continuous $\succsim$ on the connected and separable set $\mathbb{R}_+^K$ admits a continuous representation u . If $\succsim$ is also monotone, Lemma 4.8 gives one directly. Either way the consumer's problem becomes $\max\{u(x) : x \in B(p)\}$ where $B(p)$ is compact by Proposition 6.8 and u is continuous, so a maximizer exists by the Weierstrass theorem, and a maximizer of u on $B(p)$ is a $\succsim$ -best bundle.

If we only work with the preference:

Step 1. For $x \in X$ let $x_{\prec} = \{y \in X : x \succ y\}$ be its strict lower contour set, which is open because $\succsim$ is continuous.

Step 2. Suppose, towards a contradiction, that no $x \in B(p)$ is $\succsim$ -best. Then every $z \in B(p)$ is beaten by some $x \in B(p)$, that is $z \in x_{\prec}$, so the family $\{x_{\prec} : x \in B(p)\}$ is an open cover of $B(p)$.

Step 3. $B(p)$ is compact, so finitely many of them cover it: there are $x^1, \dots, x^n \in B(p)$ with $B(p) \subseteq x^1_{\prec} \cup \dots \cup x^n_{\prec}$.

Step 4. The finite set $\{x^1, \dots, x^n\}$ has a $\succsim$ -maximal element x^i by Lemma 3.5. Since $x^i \in B(p)$, it lies in some $x^j_{\prec}$, that is $x^j \succ x^i$ with $j \leq n$, contradicting the maximality of x^i .

□

Proposition 6.11 (Shape of the solution set). *Let $S(p)$ be the set of $\succsim$ -best bundles in $B(p)$.*

(i) *If $\succsim$ is convex, then $S(p)$ is convex.*

(ii) *If $\succsim$ is strictly convex, then $S(p)$ has at most one element.*

Proof. (i) Let $x, x' \in S(p)$ and $\alpha \in (0, 1)$. Then $\alpha x + (1 - \alpha)x' \in B(p)$ because $B(p)$ is convex. Both x and x' lie in the upper contour set $U(x)$, the first by reflexivity and the second because two best bundles are indifferent, and $U(x)$ is convex by Definition 5.1, so $\alpha x + (1 - \alpha)x' \succsim x$. Since $x \succsim z$ for every $z \in B(p)$, transitivity makes $\alpha x + (1 - \alpha)x'$ best as well.

(ii) Let $x, x' \in S(p)$ with $x \neq x'$; being both best they satisfy $x \sim x'$. The midpoint $\frac{1}{2}x + \frac{1}{2}x'$ lies in $B(p)$, and strict convexity applied to $x \succsim x$ and $x' \succsim x$ gives $\frac{1}{2}x + \frac{1}{2}x' \succ x$, contradicting that x is best. □

7 Lecture 7. Demand

Lecture 6 and this one are organized by a single principle: each property of $\succsim$ buys exactly one conclusion about the consumer's problem, and uses nothing else (Table 2).

Property of $\succsim$	Conclusion	Where
continuous	a solution exists	Theorem 6.10
convex	the set of solutions is convex	Proposition 6.11(i)
strictly convex	at most one solution	Proposition 6.11(ii)
monotone	every solution exhausts the budget	Proposition 7.12
differentiable	no budget-neutral improvement at a solution: (3) is necessary	Theorem 7.3
differentiable, continuous, convex, strongly monotone	$p \cdot x^* = 1$ and (3) are sufficient	Theorem 7.6

Table 2: The roadmap of Lectures 6–7.

7.1 First order conditions

Throughout, $\succsim$ is differentiable in the sense of Definition 6.4, with *subjective values* $v(x) = (v_1(x), \dots, v_K(x))$.

Remark 7.1. $v_k(x)$ is what a marginal unit of commodity k is worth at x , and only the direction of $v(x)$ matters. When $\succsim$ has a smooth representation, Theorem 6.6 settles what it is:

$$v(x) = \nabla u(x), \quad v_k(x) = \frac{\partial u}{\partial x_k}(x),$$

the marginal utility of commodity k .

Figure 20 recalls what Definition 6.4 asks. At a bundle where $\succsim$ is differentiable the set of improvement directions is an *open half-space* through the origin, and $v(x)$ is its normal. At the Leontief kink it is an open quadrant, and no single vector describes it.

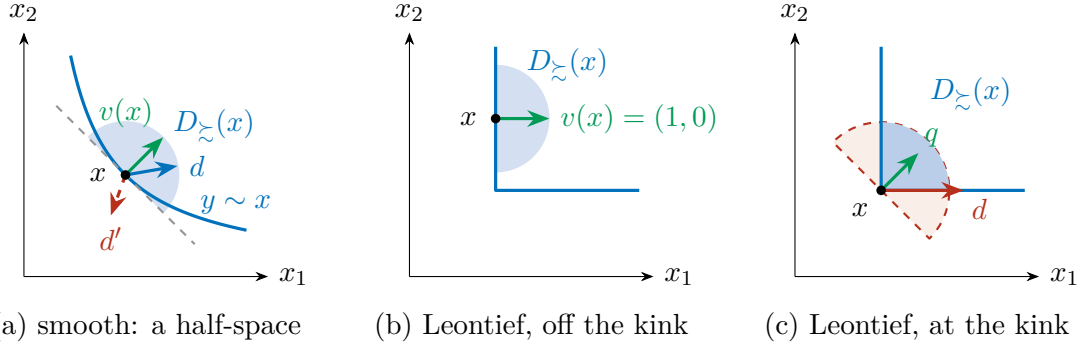


Figure 20: Improvement directions (shaded). (a) Smooth: an open half-space with normal $v(x)$. (b) Leontief off the kink: $v(x) = (1, 0)$. (c) At the kink: an open quadrant, strictly inside any half-space $\{d : d \cdot q > 0\}$.

Remark 7.2 (Differentiability is a half-space condition). $\succsim$ is differentiable at x if and only if $D_{\succsim}(x)$ is an open half-space $\{d : d \cdot q > 0\}$ through the origin: $D_v(x)$ is one, and conversely such a half-space is $D_q(x)$. If $\succsim$ is monotone then $D_{\succsim}(x) \supseteq \mathbb{R}_{++}^K$, which forces $q \geq 0$, so the normal is a legitimate vector of subjective values. This is the whole content of Definition 6.4.

Theorem 7.3 (Necessary conditions). *Let x^* be a $\succsim$ -best bundle in $B(p)$ and let k be a commodity with $x_k^* > 0$. Then*

$$\frac{v_k(x^*)}{p_k} \geq \frac{v_j(x^*)}{p_j} \quad \text{for every } j \neq k. \quad (3)$$

In particular, if $x_k^ > 0$ and $x_j^* > 0$ then $v_k(x^*)/v_j(x^*) = p_k/p_j$.*

Proof. If we assumed differentiability, this would be straightforward using $du(\mathbf{x}) = 0$ along indifference sets.

Step 1. The trade. Suppose (3) fails, so $v_j(x^*)/p_j > v_k(x^*)/p_k$ for some j . Consider the direction that sells good k and spends the proceeds on good j ,

$$d = -\frac{1}{p_k} e_k + \frac{1}{p_j} e_j,$$

where e_k is the k -th unit vector.

Step 2. It is feasible. $p \cdot d = -1 + 1 = 0$, so $p \cdot (x^* + \lambda d) = p \cdot x^* \leq 1$ for every λ ; and $x_k^* > 0$ gives $x_k^* - \lambda/p_k \geq 0$ for every small $\lambda > 0$. So $x^* + \lambda d \in B(p)$ for all small $\lambda > 0$.

Step 3. It is an improvement direction.

$$d \cdot v(x^*) = -\frac{v_k(x^*)}{p_k} + \frac{v_j(x^*)}{p_j} > 0,$$

so $d \in D_v(x^*) = D_{\succsim}(x^*)$ and $x^* + \lambda d \succ x^*$ for all small $\lambda > 0$. Together with Step 2 this contradicts the optimality of x^* .

Step 4. Equality among consumed goods. If $x_k^* > 0$ and $x_j^* > 0$, apply (3) once with the roles of k and j as written and once with them exchanged.

□

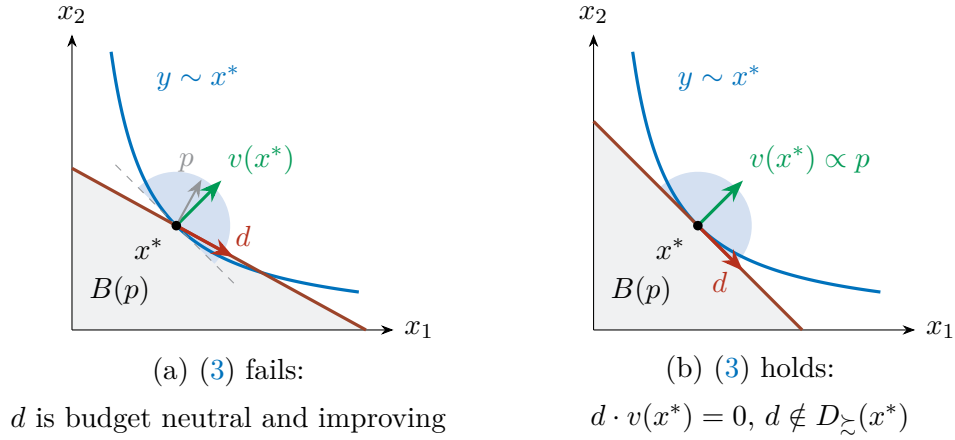


Figure 21: Theorem 7.3. (a) The budget-neutral trade d lies in $D_{\succsim}(x^*)$, so x^* is not optimal. (b) At a tangency $v(x^*) \propto p$ and no budget-neutral direction improves.

Corollary 7.4 (The same conditions in classical form). Let $\succsim$ be represented by a differentiable quasi-concave u with positive partial derivatives, so that $v = \nabla u$, let x^* solve the consumer's problem, and set

$$\mu := \frac{1}{p_k} \frac{\partial u}{\partial x_k}(x^*) \quad \text{for any } k \text{ with } x_k^* > 0,$$

which by Theorem 7.3 does not depend on which such k is used. Then

$$\frac{\partial u(x^*)/\partial x_k}{\partial u(x^*)/\partial x_j} = \frac{p_k}{p_j} \quad \text{whenever } x_k^*, x_j^* > 0, \quad \text{and} \quad \frac{\partial u}{\partial x_k}(x^*) \begin{cases} = \mu p_k, & x_k^* > 0, \\ \leq \mu p_k, & x_k^* = 0. \end{cases}$$

The first display is the familiar tangency condition: the marginal rate of substitution between two consumed goods equals their price ratio, so the indifference curve through

x^* is tangent to the budget line. The second is the Karush-Kuhn-Tucker condition for $\max u(x)$ subject to $p \cdot x \leq 1$ and $x \geq 0$, with μ the Lagrange multiplier, the marginal utility of income. A commodity left unbought is one whose marginal utility per dollar is below μ even at zero consumption.

Remark 7.5 (Quasi-concavity is doing real work). Corollary 7.4 rests on Theorem 6.6, and there quasi-concavity is not decoration. Let $u(x) = x_1 + x_2 + (x_1 - x_2)^2$ near $x = (1, 1)$, where both partial derivatives are positive. Then $\nabla u(1, 1) = (1, 1)$ and $d = (1, -1)$ has $d \cdot \nabla u(x) = 0$, yet

$$u(x + \lambda d) = 2 + 4\lambda^2 > u(x) \quad \text{for every } \lambda \neq 0.$$

So $d \in D_{\succsim}(x)$ while $d \notin D_{\nabla u}(x)$. At $p = (1, 1)$ the bundle $(1, 1)$ satisfies the tangency condition and is not optimal: moving along the budget line raises utility. Differentiability of u pins down first-order improvements; quasi-concavity is what rules out second-order rescues, and this u is not quasi-concave, since $u(1.1, 0.9) = u(0.9, 1.1) = 2.04 > u(1, 1)$. It is the same role convexity played in Lemma 6.2.

Theorem 7.6 (Sufficient conditions). *Let $\succsim$ be strongly monotone, convex, continuous and differentiable, and let $x^* \in B(p)$ satisfy budget balancedness $p \cdot x^* = 1$ together with (3) at every k with $x_k^* > 0$. Then x^* is a $\succsim$ -best bundle in $B(p)$.*

Proof. Condition (3) says that $v(x^*)$ and p are proportional on the commodities actually bought, and that $v(x^*)$ falls short of that proportion on the rest: writing μ for the common ratio,

$$v_k(x^*) \leq \mu p_k \quad \text{for every } k, \quad \text{with equality whenever } x_k^* > 0.$$

Step 1. The multiplier. This is the μ of Corollary 7.4, the marginal utility of income. Since $p \cdot x^* = 1$, some $x_k^* > 0$. Strong monotonicity makes $\mathbf{1}$ an improvement direction, so $\mathbf{1} \cdot v(x^*) > 0$ and $v(x^*) \neq 0$. By (3) the ratio

$$\mu := \frac{v_k(x^*)}{p_k}$$

is the same for every k with $x_k^* > 0$, it is positive, and $v_j(x^*) \leq \mu p_j$ for every j .

Step 2. Suppose x^ is not best.* Then there is $y \in B(p)$ with $y \succ x^*$. The vector $d = y - x^*$ is an improvement at x^* , hence an improvement direction by Lemma 6.2(ii), hence

$$v(x^*) \cdot (y - x^*) > 0.$$

Step 3. But the budget says the opposite. Term by term,

$$\left(p_k - \frac{v_k(x^*)}{\mu}\right)(y_k - x_k^*) \geq 0 \quad \text{for every } k :$$

if $x_k^* > 0$ the first factor is zero by Step 1; if $x_k^* = 0$ the first factor is ≥ 0 by $v_k(x^*) \leq \mu p_k$ and the second is $y_k \geq 0$. Summing over k and using $p \cdot y \leq 1 = p \cdot x^*$,

$$0 \geq p \cdot (y - x^*) \geq \frac{1}{\mu} v(x^*) \cdot (y - x^*),$$

so $v(x^*) \cdot (y - x^*) \leq 0$, contradicting Step 2.

□

Remark 7.7 (Two features of the proof). (i) No utility function and no Lagrangian appear: μ plays the role of the multiplier, but it is defined directly from (3). (ii) The direction of the logic matters. The term-by-term inequality of Step 3 holds at an unconsumed commodity only because $y_k - x_k^* = y_k \geq 0$ there. That is exactly the corner at which (3) can hold strictly, and it is why the conditions are inequalities rather than the equalities remembered from intermediate micro (Figure 22).

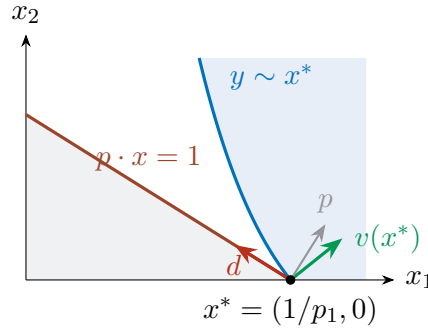


Figure 22: A corner solution: (3) holds strictly, and the only feasible budget-neutral direction d worsens.

7.2 The demand function

Definition 7.8 (Demand function). A *demand function* is a map $x : \mathbb{R}_{++}^K \rightarrow \mathbb{R}_+^K$ assigning to every price vector p a bundle $x(p) \in B(p)$. It is the demand *induced* by $\succsim$, the *Marshallian* demand, if $x(p)$ is the unique $\succsim$ -best bundle in $B(p)$ for every p .

Proposition 7.9 (Homogeneity of degree zero). *Reinstating wealth*, $x(\lambda p, \lambda \omega) = x(p, \omega)$ for every $\lambda > 0$.

Proof. $B(\lambda p, \lambda \omega) = B(p, \omega)$, and the problem depends on p and ω only through the budget set. $\square$

Theorem 7.10 (Continuity of demand). *If $\succsim$ is continuous and the induced demand is single valued, then $x(\cdot)$ is continuous.*

The result also follows from the maximum theorem (Berge's theorem) applied to a continuous representation.

Proof. Let $p^n \rightarrow p$ in $\mathbb{R}_{++}^K$ and write $x^n = x(p^n)$.

Step 1. The sequence is bounded. For n large, $p_k^n \geq p_k/2$ for every k , so $0 \leq x_k^n \leq 1/p_k^n \leq 2/p_k$.

Step 2. Limit points are feasible. Let $x^{n_m} \rightarrow \bar{x}$ along a subsequence. Then $p^{n_m} \cdot x^{n_m} \leq 1$ passes to the limit and gives $p \cdot \bar{x} \leq 1$, so $\bar{x} \in B(p)$.

Step 3. Approximating a competitor. Fix $y \in B(p)$ and set

$$t_n = \min \left\{ 1, \frac{1}{p^n \cdot y} \right\}, \quad y^n = t_n y$$

(with $y^n = 0$ if $y = 0$). Then $y^n \in B(p^n)$, and $p^n \cdot y \rightarrow p \cdot y \leq 1$ gives $t_n \rightarrow 1$, so $y^n \rightarrow y$.

Step 4. Limit points are optimal. Since x^{n_m} is best in $B(p^{n_m})$ and $y^{n_m} \in B(p^{n_m})$, we have $x^{n_m} \succsim y^{n_m}$ for every m . Continuity of $\succsim$ applied to $x^{n_m} \rightarrow \bar{x}$ and $y^{n_m} \rightarrow y$ gives $\bar{x} \succsim y$. As $y \in B(p)$ was arbitrary, $\bar{x}$ is best in $B(p)$, so $\bar{x} = x(p)$ by uniqueness.

Step 5. Conclusion. The sequence (x^n) is bounded and every convergent subsequence has the same limit $x(p)$, so $x^n \rightarrow x(p)$. $\square$

Remark 7.11 (Without strict convexity). If $\succsim$ is continuous but not strictly convex, demand is the correspondence $p \mapsto S(p)$ of Proposition 6.11. Steps 1–4 of the proof still show that every limit of $x^n \in S(p^n)$ lies in $S(p)$ and that the x^n stay bounded, which is upper hemicontinuity. Lower hemicontinuity can fail: with perfect substitutes $u = x_1 + x_2$, $S(1, 1)$ is the whole budget line, while $S(1 - \varepsilon, 1) = \{(1/(1 - \varepsilon), 0)\}$ for every $\varepsilon > 0$.

Proposition 7.12 (Walras' law). *If $\succsim$ is monotone, then every solution of the consumer's problem satisfies $p \cdot x(p) = 1$.*

Proof. Suppose $p \cdot x(p) < 1$. For $\varepsilon > 0$ small, $p \cdot (x(p) + \varepsilon \mathbf{1}) < 1$, so $x(p) + \varepsilon \mathbf{1} \in B(p)$, and $x(p) + \varepsilon \mathbf{1} \gg x(p)$ gives $x(p) + \varepsilon \mathbf{1} \succ x(p)$ by monotonicity, contradicting optimality. $\square$

7.3 Demand functions from preferences

Definition 7.13 (Three notions). Let $x(\cdot)$ be a demand function. Then $\succsim$ *fully rationalizes* x if $x(p)$ is the *unique* $\succsim$ -best bundle in $B(p)$ for every p ; x is *rationalizable* if some $\succsim$ makes $x(p)$ a $\succsim$ -best bundle in $B(p)$ for every p ; and a *monotone* $\succsim$ *rationalizes* x if it does so with $\succsim$ monotone.

Remark 7.14. The middle notion is empty: total indifference, $x \sim y$ for all x, y , makes every bundle best in every budget set, so it rationalizes every demand function. The content is in full rationalization and in rationalization by a monotone $\succsim$.

Example 7.15 (A demand function that is not rationalizable). With $K = 2$ let $x(p)$ spend everything on the (weakly) more expensive good, ties going to good 1:

$$x(p) = \begin{cases} (1/p_1, 0), & p_1 \geq p_2, \\ (0, 1/p_2), & p_1 < p_2. \end{cases}$$

At $p = (1, 1)$ she takes $(1, 0)$ with $(0, 1)$ affordable, and at $q = (1 - \varepsilon, 1)$ she takes $(0, 1)$ with $(1, 0)$ affordable and strictly inside the budget. The first says $(1, 0) \succsim (0, 1)$; the second says $(0, 1) \succsim (1/(1 - \varepsilon), 0) \succ (1, 0)$ for a monotone $\succsim$, a cycle (Figure 23). By total indifference it is of course rationalizable in the empty sense of Remark 7.14.

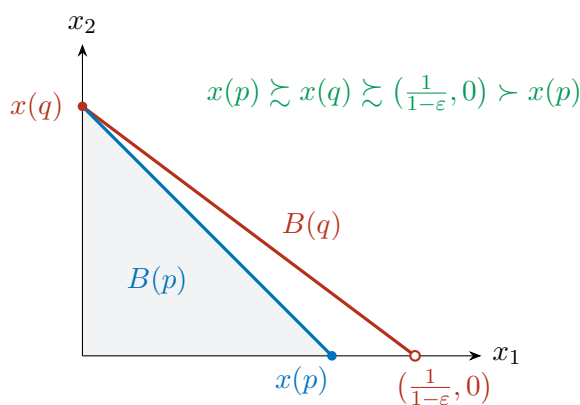


Figure 23: Example 7.15 with $\varepsilon = \frac{1}{4}$: a strict revealed preference cycle.

Remark 7.16 (Why not condition α). Lecture 3 characterized choice from a preference by condition α on domains closed under unions (Theorem 3.23), and by the WARP on domains closed under intersections (Proposition 3.27). Neither applies here: $B(p)$ is infinite, and neither $B(p) \cup B(q)$ nor $B(p) \cap B(q)$ is a budget set unless one contains the

other. So instead of a consistency condition on the domain we use the revealed preference that the demand function itself induces.

Definition 7.17 (Weak and strong axioms). Write $x(p) R^D x(q)$, read “ $x(p)$ is directly revealed preferred to $x(q)$ ”, if $p \cdot x(q) \leq 1$. A demand function satisfies

- the *weak axiom* (WA) if no two distinct chosen bundles are directly revealed preferred to each other;
- the *strong axiom* (SA) if no chain $x(p^1) R^D x(p^2) R^D \dots R^D x(p^n)$ of distinct bundles closes up, that is $p^n \cdot x(p^1) > 1$ for every such chain.

The WA forbids a reversal between two observations, the SA along a chain of any length (Figure 24). So the SA implies the WA, and the question is the converse.

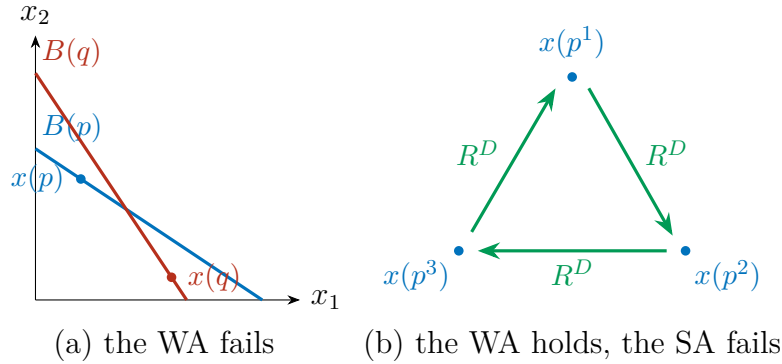


Figure 24: (a) Each chosen bundle is affordable at the other’s prices: the WA fails. (b) No two-way comparison, yet a three-cycle: the WA holds, the SA fails ($K \geq 3$).

Proposition 7.18 (The WA is necessary). *If some $\succsim$ fully rationalizes x , then x satisfies the WA.*

Proof. Let $x(p) \neq x(q)$ with $x(q) \in B(p)$. Uniqueness of the best bundle gives $x(p) \succ x(q)$. If also $x(p) \in B(q)$, then $x(q)$ best in $B(q)$ gives $x(q) \succsim x(p)$, a contradiction. So $q \cdot x(p) > 1$. $\square$

Theorem 7.19 (Houthakker 1950). *A demand function satisfying Walras’ law and homogeneity of degree zero is rationalizable if and only if it satisfies the SA.*

8 Lecture 8. Classic demand theory

Wealth is reinstated: $B(p, \omega) = \{x \in \mathbb{R}_+^K : p \cdot x \leq \omega\}$ with $p \gg 0$ and $\omega > 0$. Unless stated otherwise, $\succsim$ is continuous and locally nonsatiated (Definition 4.2) and represented by a continuous u ; $\bar{u}$ denotes a utility level with $\bar{u} > u(0)$ in the range of u .

8.1 The utility maximization problem

Definition 8.1 (UMP). The *indirect utility function* and the *Marshallian demand* are

$$v(p, \omega) = \max_{x \in B(p, \omega)} u(x), \quad x(p, \omega) \in \arg \max_{x \in B(p, \omega)} u(x).$$

Proposition 8.2 (Properties of v).

- (i) v is nonincreasing in p and strictly increasing in ω .
- (ii) v is homogeneous of degree zero in (p, ω) .
- (iii) v is quasi-convex in (p, ω) : $\{(p, \omega) : v(p, \omega) \leq \bar{u}\}$ is convex for every $\bar{u}$.¹
- (iv) v is continuous on $\mathbb{R}_{++}^K \times \mathbb{R}_{++}$.

Proof. (i) If $p' \geq p$ then $B(p', \omega) \subseteq B(p, \omega)$, and a maximum over a smaller set is no larger. If $\omega' > \omega$, let $x = x(p, \omega)$; local nonsatiation gives $y \succ x$ with $\|y - x\| \|p\| \leq \omega' - \omega$, so $p \cdot y \leq \omega'$ and $v(p, \omega') \geq u(y) > u(x) = v(p, \omega)$.

(ii) $B(\lambda p, \lambda \omega) = B(p, \omega)$.

(iii) Let $v(p, \omega) \leq \bar{u}$, $v(p', \omega') \leq \bar{u}$, and $(p'', \omega'') = \lambda(p, \omega) + (1 - \lambda)(p', \omega')$. If $x \in B(p'', \omega'')$ then $\lambda p \cdot x + (1 - \lambda)p' \cdot x \leq \lambda \omega + (1 - \lambda)\omega'$, so $p \cdot x \leq \omega$ or $p' \cdot x \leq \omega'$. Hence $B(p'', \omega'') \subseteq B(p, \omega) \cup B(p', \omega')$ and $v(p'', \omega'') \leq \bar{u}$.

(iv) By (ii), $v(p, \omega) = v(p/\omega, 1)$, and $p \mapsto v(p, 1)$ is continuous by Berge's theorem. $\square$

Quasi-convexity reads: the worse of two budgets is no better than their average. §8.4 gives the reading behind it.

Remark 8.3. Local nonsatiation is what makes v strictly increasing in ω in (i). With locally satiated preferences it may not be: if $\succsim$ has a bliss point x^* , then as soon as $p \cdot x^* \leq \omega$ extra wealth buys nothing better, and $v(p, \cdot)$ is flat from there on.

¹A function f is *quasi-convex* if every lower level set $\{z : f(z) \leq c\}$ is convex, equivalently if $f(\lambda z + (1 - \lambda)z') \leq \max\{f(z), f(z')\}$ for all z, z' and $\lambda \in [0, 1]$. It is the mirror image of quasi-concavity (Definition 5.2): f is quasi-convex exactly when $-f$ is quasi-concave.

8.2 The expenditure minimization problem

Definition 8.4 (EMP). The *expenditure function* and the *Hicksian* (compensated) demand are

$$e(p, \bar{u}) = \min_{u(x) \geq \bar{u}} p \cdot x, \quad h(p, \bar{u}) \in \arg \min_{u(x) \geq \bar{u}} p \cdot x.$$

The minimum exists: pick $\bar{x}$ with $u(\bar{x}) \geq \bar{u}$; the set $\{x \geq 0 : u(x) \geq \bar{u}, p \cdot x \leq p \cdot \bar{x}\}$ is closed because u is continuous and bounded because $p \gg 0$, and $p \cdot x$ is continuous on it.

Proposition 8.5 (Properties of e and h).

- (i) No excess utility: $u(h(p, \bar{u})) = \bar{u}$.
- (ii) e is nondecreasing in p and strictly increasing in $\bar{u}$.
- (iii) e is homogeneous of degree one in p , and h of degree zero.
- (iv) e is concave in p .
- (v) e is continuous in p on $\mathbb{R}_{++}^K$.
- (vi) Compensated law of demand: $(p' - p) \cdot (h(p', \bar{u}) - h(p, \bar{u})) \leq 0$.

Proof. Write $h = h(p, \bar{u})$ and $h' = h(p', \bar{u})$.

(i) Suppose $u(h) > \bar{u}$. Then $h \neq 0$, since $\bar{u} > u(0)$. By continuity $u(\alpha h) > \bar{u}$ for $\alpha < 1$ close to 1, and $p \cdot (\alpha h) < p \cdot h$, contradicting minimality.

(ii) If $p' \geq p$ then $e(p', \bar{u}) = p' \cdot h' \geq p \cdot h' \geq e(p, \bar{u})$, since h' is feasible at p . If $\bar{u}' > \bar{u}$ and $x' = h(p, \bar{u}')$, the argument of (i) gives $\alpha < 1$ with $u(\alpha x') > \bar{u}$, so $e(p, \bar{u}) \leq \alpha p \cdot x' < e(p, \bar{u}')$.

(iii) The constraint does not involve p , and minimizing $\lambda p \cdot x$ has the same minimizers as minimizing $p \cdot x$, with value multiplied by λ .

(iv) Let $p'' = \lambda p + (1 - \lambda)p'$ and $h'' = h(p'', \bar{u})$. Since h'' is feasible at p and at p' ,

$$e(p'', \bar{u}) = \lambda p \cdot h'' + (1 - \lambda)p' \cdot h'' \geq \lambda e(p, \bar{u}) + (1 - \lambda)e(p', \bar{u}).$$

(v) A concave function on an open convex set is continuous.

(vi) Minimality gives $p' \cdot h' \leq p' \cdot h$ and $p \cdot h \leq p \cdot h'$. Adding, $p' \cdot h' + p \cdot h \leq p' \cdot h + p \cdot h'$, which rearranges to the claim. $\square$

In (iv) the whole content is that e is a *minimum of linear functions* of p , one for each bundle that reaches $\bar{u}$. That is also the picture behind Shephard's lemma (Figure 26). Taking $p' = p + te_k$ in (vi) gives $h_k(p', \bar{u}) \leq h_k(p, \bar{u})$: compensated demand never slopes up.

8.3 Duality

Proposition 8.6 (UMP and EMP are dual).

- (i) If x^* solves the UMP at (p, ω) , it also solves the EMP at utility $v(p, \omega)$, and $e(p, v(p, \omega)) = \omega$.
- (ii) If x^* solves the EMP at $(p, \bar{u})$, it also solves the UMP at wealth $e(p, \bar{u})$, and $v(p, e(p, \bar{u})) = \bar{u}$.

Hence, when the solutions are unique (Figure 25),

$$\begin{aligned} (1A) \quad v(p, e(p, \bar{u})) &= \bar{u}, & (1B) \quad e(p, v(p, \omega)) &= \omega, \\ (2A) \quad x(p, \omega) &= h(p, v(p, \omega)), & (2B) \quad h(p, \bar{u}) &= x(p, e(p, \bar{u})). \end{aligned}$$

Proof. (i) Local nonsatiation gives $p \cdot x^* = \omega$, the argument of Proposition 7.12. Suppose $u(y) \geq u(x^*)$ and $p \cdot y < \omega$. Local nonsatiation gives y' close to y with $u(y') > u(y) \geq u(x^*)$ and still $p \cdot y' < \omega$, so $y' \in B(p, \omega)$ beats x^* . Hence no bundle reaching $u(x^*)$ is cheaper than x^* .

(ii) By Proposition 8.5(i), $u(x^*) = \bar{u} > u(0)$, so $x^* \neq 0$ and $e = p \cdot x^* > 0$. Suppose $y \in B(p, e)$ and $u(y) > \bar{u}$. Then $y \neq 0$, and by continuity $u(\alpha y) > \bar{u}$ for some $\alpha < 1$, while $p \cdot (\alpha y) < e$: a cheaper bundle reaching $\bar{u}$. So x^* is best in $B(p, e)$ and $v(p, e) = u(x^*) = \bar{u}$. $\square$

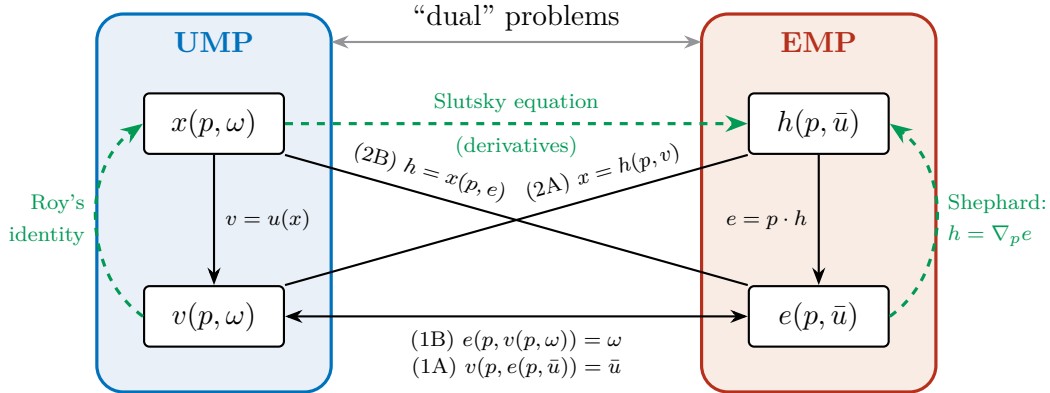


Figure 25: The dual consumer (after MWG Fig. 3.G.3). Solid: identities by construction. Dashed: identities that need a proof.

8.4 Shephard's lemma and Roy's identity

Assume from here on that $\succsim$ is strictly convex, so that x and h are single valued, and that x, h, v, e are continuously differentiable, with e twice so in p . Both identities below are the envelope theorem applied twice, in the form of the next lemma; see Appendix D.

Lemma 8.7. *Let ϕ, ψ be defined on an open set $\Theta \subseteq \mathbb{R}^n$, with $\phi \geq \psi$ on Θ and $\phi(\theta^*) = \psi(\theta^*)$. If both are differentiable at θ^* , then $\nabla\phi(\theta^*) = \nabla\psi(\theta^*)$.*

Proof. $\phi - \psi \geq 0$ attains its minimum 0 at the interior point θ^* , so its gradient vanishes there. $\square$

Theorem 8.8 (Shephard's lemma). $\frac{\partial e}{\partial p_k}(p, \bar{u}) = h_k(p, \bar{u})$ for every k .

Proof. Fix p^* and $h^* = h(p^*, \bar{u})$. The bundle h^* reaches $\bar{u}$, so it is feasible in the EMP at every p :

$$\phi(p) := p \cdot h^* \geq e(p, \bar{u}) =: \psi(p), \quad \text{with equality at } p = p^*.$$

By Lemma 8.7, $\nabla_p e(p^*, \bar{u}) = \nabla\phi(p^*) = h^*$. $\square$

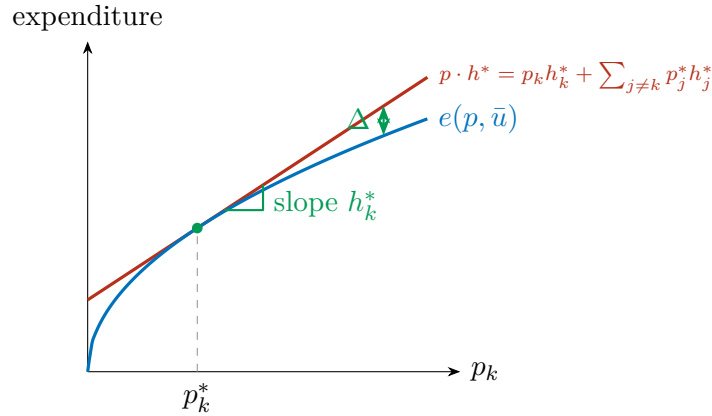


Figure 26: Shephard's lemma, all prices but p_k held at p^* . Red: the cost $p \cdot h^*$ of the old bundle, linear in p_k . Blue: the least cost of reaching $\bar{u}$. The two touch at p_k^* and therefore share the slope h_k^* ; Δ is the saving from substituting away from h^* .

Figure 26 fixes every price except p_k . The red line is what the old bundle h^* costs as p_k moves; h^* is held fixed, so the cost is linear in p_k with slope h_k^* . The blue curve is the least cost of reaching $\bar{u}$, which lets the consumer re-optimize as p_k moves. Re-optimizing can only save money, so the curve lies below the line; at p_k^* the old bundle is already optimal, so the two meet.

Definition 8.9 (Indirect preference). Let D be a family of nonempty subsets of X on which $C_{\succsim}$ is single valued. The *indirect preference* $\succsim^*$ on D is

$$A \succsim^* B \iff C_{\succsim}(A) \succsim C_{\succsim}(B).$$

Three facts. First, $\succsim^*$ is a preference relation, and if u represents $\succsim$ then

$$v(A) = u(C_{\succsim}(A))$$

represents $\succsim^*$: that is what an indirect utility function is. Second, the choice can be read back off $\succsim^*$: if $a \in A$ and $A \succ^* A \setminus \{a\}$, then $C_{\succsim}(A) = a$, so on a rich enough D nothing is lost by working with $\succsim^*$. Third, a standard decision maker never dislikes having more options,

$$B \subseteq A \implies A \succsim^* B,$$

and a violation is a *preference for commitment*. [Gul and Pesendorfer \(2001\)](#) replace that monotonicity by *set betweenness*,

$$A \succsim^* B \implies A \succsim^* A \cup B \succsim^* B,$$

which is compatible with temptation and self-control.

For the consumer D is the family of budget sets, indexed by the $K + 1$ parameters (p, ω) , and

$$(p, \omega) \succsim^* (p', \omega') \iff x(p, \omega) \succsim x(p', \omega'),$$

a relation represented by $v(p, \omega)$. Read this way, Proposition 8.2(iii) is a statement about $\succsim^*$: an average of two budget parameters is never better than the better of the two.

Theorem 8.10 (Roy's identity). *If $\partial v / \partial \omega(p, \omega) \neq 0$, then*

$$x_k(p, \omega) = - \frac{\partial v / \partial p_k(p, \omega)}{\partial v / \partial \omega(p, \omega)} \quad \text{for every } k.$$

Proof. Fix (p^*, ω^*) and $x^* = x(p^*, \omega^*)$, so $p^* \cdot x^* = \omega^*$. At any prices p , wealth $p \cdot x^*$ still buys x^* , so

$$\phi(p) := v(p, p \cdot x^*) \geq u(x^*) = v(p^*, \omega^*) =: \psi(p), \quad \text{with equality at } p = p^*.$$

ψ is constant, so Lemma 8.7 and the chain rule give $\partial v / \partial p_k + x_k^* \partial v / \partial \omega = 0$ at (p^*, ω^*) . $\square$

A supporting hyperplane argument. Work in (p, ω) -space with the indirect preference of Definition 8.9, and put $x^* = x(p^*, \omega^*)$. Let

$$H = \{(p, \omega) : p \cdot x^* = \omega\} = \{(p, \omega) : (x^*, -1) \cdot (p - p^*, \omega - \omega^*) = 0\},$$

the budget parameters at which x^* is exactly affordable; the second form uses $p^* \cdot x^* = \omega^*$. For every $(p, \omega) \in H$ the bundle x^* lies in $B(p, \omega)$, so $x(p, \omega) \succeq x^*$ and therefore

$$(p, \omega) \succeq^* (p^*, \omega^*) \quad \text{for every } (p, \omega) \in H.$$

So H passes through (p^*, ω^*) and lies inside its upper contour set: H supports the indirect indifference surface there. If v is differentiable, that surface has exactly one supporting hyperplane at the point, the one orthogonal to $\nabla v(p^*, \omega^*)$. Hence the gradient is proportional to the normal $(x^*, -1)$ of H ,

$$\frac{\partial v}{\partial p_k}(p^*, \omega^*) = c x_k^*, \quad \frac{\partial v}{\partial \omega}(p^*, \omega^*) = -c,$$

and eliminating c gives Roy's identity; $c < 0$, since v increases in ω .

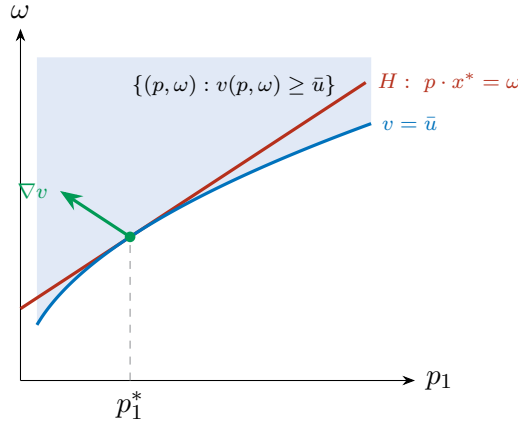


Figure 27: Roy's identity as a supporting hyperplane, with all prices but p_1 held at p^* . The shaded region is the upper contour set of $\succeq^*$ at (p^*, ω^*) , bounded by the indirect indifference curve $\omega = e(p, \bar{u})$. The line H lies inside it and touches it at (p^*, ω^*) , so it supports that set and separates it from the budgets that are worse. Its normal is $(x^*, -1)$, and therefore so is ∇v up to a factor, which is Roy's identity.

9 Lecture 9. Slutsky and integrability

Fix (p, ω^0) , write $x = x(p, \omega^0)$ and $\bar{u} = v(p, \omega^0)$, and let prices move to p' . To compensate the consumer is to choose a new wealth $\omega^c(p')$; the *compensation* is the transfer $\omega^c(p') - \omega^0$. There are two standard choices of ω^c : $\omega^S(p') = p' \cdot x$, under which the old bundle stays affordable, and $\omega^H(p') = e(p', \bar{u})$, under which the old utility stays attainable. Both equal ω^0 at $p' = p$, and $\omega^H \leq \omega^S$ everywhere, since x already reaches $\bar{u}$. They also have the same gradient there: $\partial \omega^S / \partial p'_j = x_j$ and $\partial \omega^H / \partial p'_j|_{p'=p} = h_j(p, \bar{u}) = x_j$, by Theorem 8.8 and (2A). Only ω^H depends on the unobservable $\succsim$, and h is exactly its record.

9.1 The Slutsky equation

Theorem 9.1 (Slutsky equation). *For $\bar{u} = v(p, \omega)$ and all k, j ,*

$$\frac{\partial x_k}{\partial p_j}(p, \omega) = \underbrace{\frac{\partial h_k}{\partial p_j}(p, \bar{u})}_{\text{substitution}} - \underbrace{x_j(p, \omega) \frac{\partial x_k}{\partial \omega}(p, \omega)}_{\text{income}}.$$

Proof. Differentiate (2B), $h_k(p, \bar{u}) = x_k(p, e(p, \bar{u}))$, in p_j :

$$\frac{\partial h_k}{\partial p_j} = \frac{\partial x_k}{\partial p_j} + \frac{\partial x_k}{\partial \omega} \frac{\partial e}{\partial p_j} = \frac{\partial x_k}{\partial p_j} + \frac{\partial x_k}{\partial \omega} h_j(p, \bar{u}),$$

by Shephard's lemma, and at $\bar{u} = v(p, \omega)$ we have $e = \omega$ and $h_j = x_j$ by (1B) and (2A). $\square$

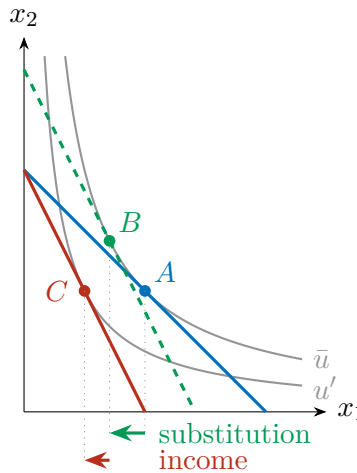


Figure 28: Hicks' decomposition when p_1 rises: $A \rightarrow B$ substitution, $B \rightarrow C$ income ($u = \sqrt{x_1 x_2}$, $\omega = 4$, $p_1 : 1 \rightarrow 2$).

9.2 The Slutsky matrix: symmetry and negative semidefiniteness

Definition 9.2 (Slutsky matrix). $S(p, \omega)$ is the $K \times K$ matrix with entries

$$S_{kj}(p, \omega) = \frac{\partial x_k}{\partial p_j}(p, \omega) + x_j(p, \omega) \frac{\partial x_k}{\partial \omega}(p, \omega).$$

It is built from the observable Marshallian demand alone.

Theorem 9.3. *If x is the demand of a consumer satisfying the standing assumptions, then at every (p, ω) , with $\bar{u} = v(p, \omega)$:*

- (i) $S(p, \omega) = D_p^2 e(p, \bar{u})$;
- (ii) S is symmetric;
- (iii) S is negative semidefinite, and in particular $S_{kk} \leq 0$;
- (iv) $S p = 0$ and $p^\top S = 0$.

Proof. (i) By Theorem 9.1, $S_{kj} = \partial h_k / \partial p_j$, and by Theorem 8.8, $h_k = \partial e / \partial p_k$, so $S_{kj} = \partial^2 e / \partial p_j \partial p_k$.

(ii) Mixed partials of a C^2 function commute (Schwarz).

(iii) e is concave in p (Proposition 8.5), and a C^2 function is concave exactly when its Hessian is negative semidefinite. Then $S_{kk} = e_k^\top S e_k \leq 0$.

(iv) Both follow from the demand function alone. Differentiating Walras' law $p \cdot x(p, \omega) = \omega$ in p_j and in ω gives $x_j + \sum_k p_k \partial x_k / \partial p_j = 0$ and $\sum_k p_k \partial x_k / \partial \omega = 1$, so

$$\sum_k p_k S_{kj} = \sum_k p_k \frac{\partial x_k}{\partial p_j} + x_j \sum_k p_k \frac{\partial x_k}{\partial \omega} = -x_j + x_j = 0.$$

Differentiating $x_k(\lambda p, \lambda \omega) = x_k(p, \omega)$ in λ at $\lambda = 1$ gives Euler's equation $\sum_j p_j \partial x_k / \partial p_j + \omega \partial x_k / \partial \omega = 0$, so, using $\sum_j p_j x_j = \omega$,

$$\sum_j S_{kj} p_j = \sum_j p_j \frac{\partial x_k}{\partial p_j} + \frac{\partial x_k}{\partial \omega} \sum_j p_j x_j = -\omega \frac{\partial x_k}{\partial \omega} + \omega \frac{\partial x_k}{\partial \omega} = 0.$$

□

Part (iv) puts p in the kernel of S and of $S^\top$, so S is singular, with rank at most $K - 1$. The two halves say different things. $S p = 0$ reads the direction $dp = p dt$, a proportional rise of all prices: compensated demand does not move, because scaling every price and the required expenditure by the same factor changes nothing real. $p^\top S = 0$ reads any direction dp : the compensated change $S dp$ in the bundle costs nothing at current prices, $p^\top S dp = 0$, because compensation holds expenditure fixed, so the move is a pure rearrangement.

Remark 9.4 (What symmetry says). Symmetry is $\partial h_k / \partial p_j = \partial h_j / \partial p_k$: once the consumer is compensated, the effect of the price of coffee on the demand for tea equals the effect of the price of tea on the demand for coffee.

Proposition 9.5 (Symmetry is free when $K = 2$). *Let $K = 2$ and let x be C^1 , homogeneous of degree zero and satisfy Walras' law. Then S is symmetric.*

Proof. By Theorem 9.3(iv), whose proof used only these two properties, $Sp = 0$ and $p^\top S = 0$. The first row of $Sp = 0$ gives $S_{12} = -S_{11}p_1/p_2$, and the first column of $p^\top S = 0$ gives $S_{21} = -S_{11}p_1/p_2$. $\square$

9.3 Integrability

Integrability recovers preferences from demand in two moves: $x \rightarrow e$, by reading Shephard's lemma backwards, which Theorem 9.6 makes possible; and $e \rightarrow \succsim$, by duality. Let $x(p, \omega)$ be C^1 , homogeneous of degree zero, and satisfy Walras' law. The remaining conditions are symmetry and negative semidefiniteness of its Slutsky matrix.

Step 1. Integrating demand

Shephard's lemma and duality require

$$\frac{\partial e}{\partial p_k}(p, \bar{u}) = x_k(p, e(p, \bar{u})), \quad k = 1, \dots, K, \quad e(p^0, \bar{u}) = \omega^0. \quad (4)$$

Demand gives the compensated slopes; integrating them reconstructs e . They must be compatible: differentiating in p_j gives

$$\frac{\partial^2 e}{\partial p_j \partial p_k} = \frac{\partial x_k}{\partial p_j} + \frac{\partial x_k}{\partial \omega} x_j = S_{kj}(p, e(p, \bar{u})). \quad (5)$$

Mixed partials commute, so $S_{kj} = S_{jk}$ is necessary.

Theorem 9.6 (Frobenius, applied to demand). *For C^1 demand on an open domain, (4) has a unique local solution through every (p^0, ω^0) if and only if S is symmetric everywhere.*

Thus symmetry makes compensation locally independent of the price path. At fixed p^0 , different initial wealths generate solutions that never cross; their ordered constants of integration serve as utility labels $\bar{u}$.

Step 2. The solution is an expenditure function

Since $D_p^2 e = S$, negative semidefiniteness makes e concave in prices. Demand is nonnegative, so e is nondecreasing. Walras' law gives

$$\frac{d}{d\lambda} e(\lambda p, \bar{u}) = \frac{e(\lambda p, \bar{u})}{\lambda}, \quad \text{hence} \quad e(\lambda p, \bar{u}) = \lambda e(p, \bar{u}).$$

Step 3. From e back to $\succsim$

This step is duality: a closed convex set that is closed upward is determined by the cheapest way of reaching it at each price. Suppose for a moment that a preference $\succsim$ is behind e , with upper contour sets $U(\bar{u}) = \{z : u(z) \geq \bar{u}\}$. The EMP is the statement

$$e(p, \bar{u}) = \min_{z \in U(\bar{u})} p \cdot z,$$

so every $z \in U(\bar{u})$ costs at least $e(p, \bar{u})$ at every price, that is $U(\bar{u}) \subseteq V(\bar{u})$, where

$$V(\bar{u}) = \bigcap_{p \gg 0} \left\{ z \in \mathbb{R}_+^K : p \cdot z \geq e(p, \bar{u}) \right\}.$$

The inclusion is an equality when $U(\bar{u})$ is closed, convex and upward closed: if $z \notin U(\bar{u})$, a separating hyperplane gives $p \neq 0$ and c with $p \cdot z < c \leq p \cdot y$ for every $y \in U(\bar{u})$; upward closedness forces $p \geq 0$, and then $p \cdot z < e(p, \bar{u})$, so $z \notin V(\bar{u})$.

Each $V(\bar{u})$ is an intersection of closed half-spaces with the closed convex set $\mathbb{R}_+^K$, so it is closed and convex. The sets are nested: if $\bar{u}' > \bar{u}$ and $z \in V(\bar{u}')$, then, since $\bar{u} \mapsto e(p, \bar{u})$ is strictly increasing, $p \cdot z \geq e(p, \bar{u}') > e(p, \bar{u})$ for every $p \gg 0$, so $V(\bar{u}') \subseteq V(\bar{u})$ (Figure 29). Hence

$$u(z) = \sup \{ \bar{u} : z \in V(\bar{u}) \}$$

is a utility function; write $z \succsim y$ for $u(z) \geq u(y)$. For $\omega = e(p, \bar{u})$, concavity and Walras' law give, at every p' ,

$$e(p', \bar{u}) \leq e(p, \bar{u}) + x(p, \omega) \cdot (p' - p) = p' \cdot x(p, \omega).$$

Hence $x(p, \omega) \in V(\bar{u})$. No affordable bundle reaches a higher level: $y \in V(\bar{u}')$ with $\bar{u}' > \bar{u}$ would imply $p \cdot y \geq e(p, \bar{u}') > e(p, \bar{u}) = \omega$. Thus x is optimal.

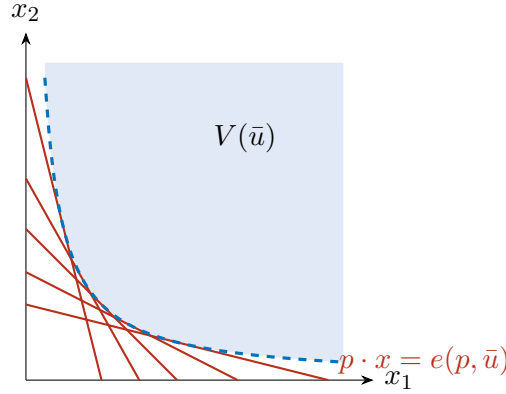


Figure 29: Recovering preferences: $V(\bar{u})$ lies on or above every line $p \cdot x = e(p, \bar{u})$; their dashed envelope is the indifference curve.

Theorem 9.7 (Hurwicz and Uzawa 1971). *Let $x : \mathbb{R}_{++}^K \times \mathbb{R}_{++} \rightarrow \mathbb{R}_+^K$ be C^1 , homogeneous of degree zero, satisfy Walras' law, and have a symmetric negative semidefinite Slutsky matrix everywhere; assume in addition a bound on the income derivatives $\partial x / \partial \omega$. Then there is a monotone, convex, continuous $\succsim$ whose demand is x . The homogeneity, Walras and Slutsky conditions are also necessary under the standing assumptions.*

Proof. Necessity is Proposition 7.9, Proposition 7.12 and Theorem 9.3. For sufficiency:

Step 1. Fix $\bar{u} > 0$, $p \in \mathbb{R}_{++}^K$ and a C^1 path $p(\cdot) : [0, 1] \rightarrow \mathbb{R}_{++}^K$ with $p(0) = p^0$, $p(1) = p$. Label each solution of (4) by its initial wealth at p^0 , $\omega^0 = \bar{u}$, so that $e(p^0, \bar{u}) = \bar{u}$. Along the path, (4) reads

$$\frac{d}{dt} e(p(t), \bar{u}) = x(p(t), e(p(t), \bar{u})) \cdot \dot{p}(t).$$

Walras' law and $x \geq 0$ give $x_k(p(t), \omega) \leq \omega / p_k(t)$ for every $\omega > 0$ and $k = 1, \dots, K$, so

$$\left| \frac{d}{dt} e(p(t), \bar{u}) \right| \leq \sum_{k=1}^K x_k(p(t), e(p(t), \bar{u})) |\dot{p}_k(t)| \leq e(p(t), \bar{u}) \ell(t),$$

where $\ell(t) = \sum_{k=1}^K |\dot{p}_k(t)| / p_k(t)$. While $e(p(t), \bar{u}) > 0$, dividing by it gives

$$-\ell(t) \leq \frac{d}{dt} \ln e(p(t), \bar{u}) \leq \ell(t).$$

Integrating from 0 to t and using $e(p(0), \bar{u}) = e(p^0, \bar{u}) = \bar{u}$,

$$-L(t) \leq \ln e(p(t), \bar{u}) - \ln \bar{u} \leq L(t), \quad L(t) = \int_0^t \sum_{k=1}^K \frac{|\dot{p}_k(s)|}{p_k(s)} ds,$$

and exponentiating,

$$\bar{u} \exp(-L(t)) \leq e(p(t), \bar{u}) \leq \bar{u} \exp(L(t)).$$

In particular $e(p(t), \bar{u})$ stays positive and bounded. Let M bound $|\partial x(q, \omega)/\partial \omega|$ on the compact set $\{(q, \omega) : q = p(s), s \in [0, 1], \omega \in [\bar{u} \exp(-L(1)), \bar{u} \exp(L(1))]\}$. By the mean value theorem, for ω, ω' in that interval,

$$|x(p(s), \omega) \cdot \dot{p}(s) - x(p(s), \omega') \cdot \dot{p}(s)| \leq M |\dot{p}(s)| |\omega - \omega'|,$$

so the right side is Lipschitz in ω where the solution lives, and the solution exists and is unique on $[0, 1]$. By Theorem 9.6 and simple connectedness of $\mathbb{R}_{++}^K$, $e(p(1), \bar{u})$ does not depend on the path. This defines $e(p, \bar{u})$ on $\mathbb{R}_{++}^K \times \mathbb{R}_{++}$ with

$$e(p^0, \bar{u}) = \bar{u}, \quad \nabla_p e(p, \bar{u}) = x(p, e(p, \bar{u})).$$

Step 2. For $\bar{u} < \bar{u}'$ the solutions $e(\cdot, \bar{u})$ and $e(\cdot, \bar{u}')$ never cross, so $\bar{u} \mapsto e(p, \bar{u})$ is continuous and strictly increasing, and by the bounds of Step 1 it maps $\mathbb{R}_{++}$ onto $\mathbb{R}_{++}$: for every (p, ω) there is a unique $\bar{u}$ with $e(p, \bar{u}) = \omega$.

Step 3. By (5), $D_p^2 e(p, \bar{u}) = S(p, e(p, \bar{u}))$ is negative semidefinite, so $p \mapsto e(p, \bar{u})$ is concave. For $\omega = e(p, \bar{u})$ and every $p' \gg 0$,

$$e(p', \bar{u}) \leq e(p, \bar{u}) + \nabla_p e(p, \bar{u}) \cdot (p' - p) = \omega + x(p, \omega) \cdot (p' - p) = p' \cdot x(p, \omega),$$

using Walras' law $p \cdot x(p, \omega) = \omega$.

Step 4. Set

$$V(\bar{u}) = \bigcap_{p \gg 0} \{z \in \mathbb{R}_+^K : p \cdot z \geq e(p, \bar{u})\}, \quad u(z) = \sup\{\bar{u} > 0 : z \in V(\bar{u})\}.$$

$V(\bar{u})$ is closed, convex and upward closed, and $V(\bar{u}') \subseteq V(\bar{u})$ for $\bar{u}' > \bar{u}$. Hence $\{z : u(z) \geq c\} = \bigcap_{\bar{u} < c} V(\bar{u})$ is closed and convex, and u is nondecreasing. By continuity of $\bar{u} \mapsto e(p, \bar{u})$,

$$u(z) \geq \bar{u} \iff z \in V(\bar{u}).$$

Step 5. Let $e(p, \bar{u}) = \omega$. Step 3 gives $p' \cdot x(p, \omega) \geq e(p', \bar{u})$ for all $p' \gg 0$, so $x(p, \omega) \in V(\bar{u})$ and $u(x(p, \omega)) \geq \bar{u}$.

Step 6. If $p \cdot y \leq \omega$ and $u(y) > \bar{u}$, then $y \in V(\bar{u}')$ for some $\bar{u}' > \bar{u}$ and

$$\omega \geq p \cdot y \geq e(p, \bar{u}') > e(p, \bar{u}) = \omega,$$

a contradiction. With Step 5, $u(x(p, \omega)) = \bar{u}$ and $x(p, \omega)$ solves the UMP at (p, ω) .

Step 7. If $p \cdot y \leq \omega$ and $u(y) \geq \bar{u}$, then $y \in V(\bar{u})$ and

$$\varphi(p') = p' \cdot y - e(p', \bar{u}) \geq 0 \quad \forall p' \gg 0, \quad \varphi(p) = p \cdot y - e(p, \bar{u}) \leq \omega - \omega = 0.$$

So p is an interior minimum of φ and $0 = \nabla \varphi(p) = y - \nabla_p e(p, \bar{u}) = y - x(p, \omega)$.

The demand of $\succsim$ at (p, ω) is $x(p, \omega)$. $\square$

Example 9.8 (Integrating Cobb-Douglas demand). For $x_1 = \alpha\omega/p_1$ and $x_2 = (1 - \alpha)\omega/p_2$, $0 < \alpha < 1$, the system gives $d \ln e = \alpha d \ln p_1 + (1 - \alpha) d \ln p_2$. Integrating,

$$e(p, \bar{u}) = \bar{u} p_1^\alpha p_2^{1-\alpha},$$

with $\bar{u}$ the integration constant, so $e(p^0, \bar{u}) = \bar{u}$ at $p^0 = (1, 1)$. By Step 4, $x \in V(\bar{u})$ iff $p \cdot x \geq \bar{u} p_1^\alpha p_2^{1-\alpha}$ for all $p \gg 0$, so

$$u(x) = \inf_{p \gg 0} \frac{p \cdot x}{p_1^\alpha p_2^{1-\alpha}} = \inf_{p_1 > 0} \frac{p_1 x_1 + x_2}{p_1^\alpha},$$

setting $p_2 = 1$ since the ratio is homogeneous of degree zero in p . The first order condition $x_1 p_1^{-\alpha} - \alpha(p_1 x_1 + x_2) p_1^{-\alpha-1} = 0$ gives $p_1 = \alpha x_2 / ((1 - \alpha) x_1)$ and $p_1 x_1 + x_2 = x_2 / (1 - \alpha)$, hence

$$u(x) = \frac{x_2}{1 - \alpha} \left(\frac{(1 - \alpha) x_1}{\alpha x_2} \right)^\alpha = \frac{x_1^\alpha x_2^{1-\alpha}}{\alpha^\alpha (1 - \alpha)^{1-\alpha}}.$$

10 Lecture 10. Revealed preference and Afriat's theorem

Lecture 9 recovered preferences from a differentiable demand function. Here the observations are a finite list of choices. Prices and expenditures reveal comparisons directly. The question is whether *one* locally nonsatiated utility function can explain all the observations.

10.1 Finite data and rationalization

Let

$$D = \{(\mathbf{p}^t, \mathbf{x}^t)\}_{t=1}^T, \quad \mathbf{p}^t \gg 0, \quad \mathbf{x}^t \in \mathbb{R}_+^K, \quad w^t = \mathbf{p}^t \cdot \mathbf{x}^t > 0.$$

Observation t means that the consumer chose $\mathbf{x}^t$ from $B^t = \{\mathbf{z} \geq 0 : \mathbf{p}^t \cdot \mathbf{z} \leq w^t\}$. Wealth is inferred from expenditure: this uses budget exhaustion. If actual wealth were observed to exceed $\mathbf{p}^t \cdot \mathbf{x}^t$, local nonsatiation would already reject the choice.

Definition 10.1 (Rationalizing finite data). A utility function $u : \mathbb{R}_+^K \rightarrow \mathbb{R}$ *rationalizes* D if

$$u(\mathbf{x}^t) \geq u(\mathbf{z}) \quad \text{for every } \mathbf{z} \in B^t \text{ and every } t.$$

The chosen bundle need only be a maximizer; other bundles may tie.

A constant utility rationalizes every data set, so maximization alone has no observable content. We require *local nonsatiation*: every neighbourhood of every bundle contains a strictly preferred bundle (Definition 4.2).

Lemma 10.2 (What affordability reveals). *If a locally nonsatiated u rationalizes D , then*

$$\mathbf{p}^t \cdot \mathbf{z} \leq w^t \implies u(\mathbf{x}^t) \geq u(\mathbf{z}), \quad \mathbf{p}^t \cdot \mathbf{z} < w^t \implies u(\mathbf{x}^t) > u(\mathbf{z}).$$

Proof. The first implication is rationalization. For the second, strict budget slack leaves a neighbourhood of $\mathbf{z}$ affordable. Local nonsatiation supplies a bundle $\mathbf{z}'$ in that neighbourhood with $u(\mathbf{z}') > u(\mathbf{z})$; optimality gives $u(\mathbf{x}^t) \geq u(\mathbf{z}') > u(\mathbf{z})$. $\square$

10.2 Revealed comparisons and GARP

When $\mathbf{x}^t$ was chosen, $\mathbf{x}^s$ was affordable if $\mathbf{p}^t \cdot \mathbf{x}^s \leq w^t = \mathbf{p}^t \cdot \mathbf{x}^t$. Then the consumer picked $\mathbf{x}^t$ over $\mathbf{x}^s$, and we say $\mathbf{x}^t$ is *directly revealed preferred* to $\mathbf{x}^s$; if $\mathbf{x}^s$ was strictly cheaper, *directly strictly revealed preferred*:

$$\mathbf{x}^t R^D \mathbf{x}^s \iff \mathbf{p}^t \cdot \mathbf{x}^s \leq \mathbf{p}^t \cdot \mathbf{x}^t, \quad \mathbf{x}^t P^D \mathbf{x}^s \iff \mathbf{p}^t \cdot \mathbf{x}^s < \mathbf{p}^t \cdot \mathbf{x}^t.$$

The *revealed preference* relation R is the transitive closure of R^D : $\mathbf{x}^t R \mathbf{x}^s$ if a chain $\mathbf{x}^t R^D \mathbf{x}^{t_1} R^D \dots R^D \mathbf{x}^s$ connects them. Lemma 10.2 gives a weak utility inequality at each link, so every rationalizer satisfies

$$\mathbf{x}^t R \mathbf{x}^s \implies u(\mathbf{x}^t) \geq u(\mathbf{x}^s).$$

Axiom 10.1 (Generalized axiom of revealed preference). The data satisfy *GARP* if, for every t, s ,

$$\mathbf{x}^t R \mathbf{x}^s \implies \text{not } \mathbf{x}^s P^D \mathbf{x}^t, \quad \text{equivalently} \quad \mathbf{x}^t R \mathbf{x}^s \implies \mathbf{p}^s \cdot \mathbf{x}^t \geq \mathbf{p}^s \cdot \mathbf{x}^s.$$

GARP rules out a cycle of affordable comparisons with at least one strict link.

Proposition 10.3. *Rationalization by a locally nonsatiated utility implies GARP.*

Proof. A violation gives $\mathbf{x}^t R \mathbf{x}^s$ and $\mathbf{x}^s P^D \mathbf{x}^t$. The weak chain implies $u(\mathbf{x}^t) \geq u(\mathbf{x}^s)$, while strict budget slack and local nonsatiation imply $u(\mathbf{x}^s) > u(\mathbf{x}^t)$ by Lemma 10.2, a contradiction. $\square$

The related axioms differ in which cycles they exclude:

$$\begin{aligned} \text{GARP} : \quad & \mathbf{x}^t R \mathbf{x}^s \implies \text{not } \mathbf{x}^s P^D \mathbf{x}^t, \\ \text{WARP} : \quad & \mathbf{x}^t R^D \mathbf{x}^s \implies \text{not } \mathbf{x}^s P^D \mathbf{x}^t, \\ \text{WARP}_{\text{sv}} : \quad & \mathbf{x}^t R^D \mathbf{x}^s, \mathbf{x}^t \neq \mathbf{x}^s \implies \text{not } \mathbf{x}^s R^D \mathbf{x}^t, \\ \text{SARP} : \quad & \mathbf{x}^t R \mathbf{x}^s, \mathbf{x}^t \neq \mathbf{x}^s \implies \text{not } \mathbf{x}^s R \mathbf{x}^t. \end{aligned}$$

Here *WARP* denotes the pairwise weak–strict condition, also called *WGARP*: it excludes two-observation cycles with a strict link. GARP therefore implies WARP in every dimension. The subscript “sv” stands for *single-valued*. WARP_{sv} excludes pairwise weak reversals; SARP excludes weak cycles of any length among distinct bundles, including equality cycles. GARP excludes only cycles with a strict link, so it allows ties. Consequently SARP implies both GARP and WARP_{sv} , whereas GARP need not imply WARP_{sv} .

Example 10.4 (Two different selections from the same budget). Let $\mathbf{p}^1 = \mathbf{p}^2 = (1, 1)$, $\mathbf{x}^1 = (1, 0)$ and $\mathbf{x}^2 = (0, 1)$, so $w^1 = w^2 = 1$ and both observations face the same budget line $z_1 + z_2 = 1$. Then $\mathbf{p}^1 \cdot \mathbf{x}^2 = 1 \leq 1 = \mathbf{p}^1 \cdot \mathbf{x}^1$ and $\mathbf{p}^2 \cdot \mathbf{x}^1 = 1 \leq 1 = \mathbf{p}^2 \cdot \mathbf{x}^2$, so $\mathbf{x}^1 R^D \mathbf{x}^2$ and $\mathbf{x}^2 R^D \mathbf{x}^1$, both with equality, hence neither P^D link holds.

- GARP holds: the only cycle is $\mathbf{x}^1 R^D \mathbf{x}^2 R^D \mathbf{x}^1$, and it has no strict link.

- WARP_{sv} fails: $\mathbf{x}^1 R^D \mathbf{x}^2$ with $\mathbf{x}^1 \neq \mathbf{x}^2$, yet $\mathbf{x}^2 R^D \mathbf{x}^1$. SARP fails for the same reason.

Yet $u(\mathbf{z}) = z_1 + z_2$ rationalizes the data: every bundle on $z_1 + z_2 = 1$, including $\mathbf{x}^1$ and $\mathbf{x}^2$, maximizes it. WARP_{sv} and SARP fail here because the observations select different tied maximizers.

10.3 Two goods: WARP and GARP are equivalent

With two goods, checking pairs already rules out every revealed cycle with a strict link. The geometry is special: budget boundaries are lines in the plane, and a shortest violating cycle would have to keep moving right without ever returning to its starting bundle.

Theorem 10.5 (Rose 1958; Chambers and Echenique 2016, Theorem 3.4). *For finite data with $K = 2$ and $\mathbf{p}^t \gg 0$, WARP holds if and only if GARP holds. Here WARP is the pairwise weak-strict condition above.*

Proof. GARP implies WARP because a two-observation cycle is a cycle. For the converse, suppose GARP fails and choose a shortest cycle

$$\mathbf{x}^0 R^D \mathbf{x}^1 R^D \dots R^D \mathbf{x}^{L-1} R^D \mathbf{x}^0$$

with at least one strict link. If $L = 2$, WARP fails. Suppose $L \geq 3$; indices below are read modulo L .

1. *Minimality restricts the budgets.* The bundles are distinct: a repeated bundle splits the cycle into two shorter closed chains, one containing a strict link. For any $j \notin \{i, i+1\}$,

$$\mathbf{p}^i \cdot \mathbf{x}^j \geq \mathbf{p}^i \cdot \mathbf{x}^i. \quad (6)$$

Otherwise the strict shortcut $i \rightarrow j$, followed by the original path from j to i , gives a shorter violating cycle. Moreover, the predecessor is *strictly* outside budget i :

$$\mathbf{p}^i \cdot \mathbf{x}^{i-1} > \mathbf{p}^i \cdot \mathbf{x}^i. \quad (7)$$

If equality held, the link $i-1 \rightarrow i$ would also be an equality, to avoid a strict two-cycle. Both budget lines would then pass through the same two distinct bundles, so the budgets would coincide. Deleting observation i would preserve affordability and any strict link, contradicting minimality.

2. *No cycle bundle dominates another.* If $\mathbf{x}^k \geq \mathbf{x}^\ell$ for $k \neq \ell$, positive prices give

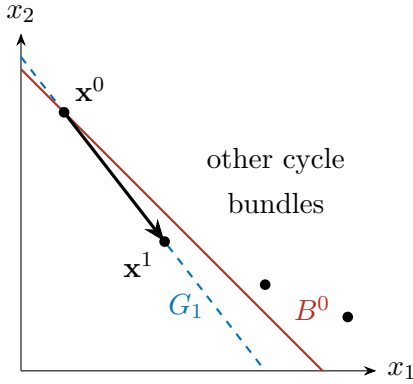
$$\mathbf{p}^{k-1} \cdot \mathbf{x}^\ell < \mathbf{p}^{k-1} \cdot \mathbf{x}^k \leq \mathbf{p}^{k-1} \cdot \mathbf{x}^{k-1}.$$

This contradicts (6) when $\ell \neq k - 1$, and is impossible when $\ell = k - 1$. Thus, with two goods, all first coordinates are distinct, and a bundle further right is strictly lower.

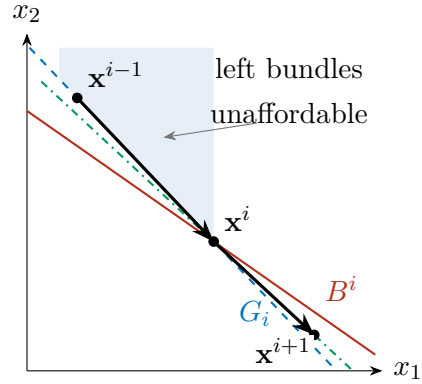
3. *The cycle must always move right.* Start at the leftmost bundle $\mathbf{x}^0$. Its successor $\mathbf{x}^1$ lies to its right. Every other cycle bundle lies on or above budget line 0, while $\mathbf{x}^1$ lies on or below it. Consequently every cycle bundle lies on or above the line G_1 through $\mathbf{x}^0, \mathbf{x}^1$; see Figure 30(a).

Inductively, suppose $\mathbf{x}^{i-1}$ lies left of $\mathbf{x}^i$ and all cycle bundles lie on or above their connecting line G_i . By (7), budget line i lies strictly below the predecessor, so it is flatter than G_i . To the left of $\mathbf{x}^i$ it lies strictly below G_i : no cycle bundle there is affordable. Hence the successor must lie to the right (Figure 30(b)).

The new connecting line G_{i+1} is no steeper than G_i and no flatter than budget line i . Left of $\mathbf{x}^i$, all cycle bundles remain weakly above G_{i+1} because it lies weakly below G_i . Right of $\mathbf{x}^i$, every bundle other than the successor lies weakly above budget line i by (6), hence weakly above G_{i+1} ; the successor lies on it. This proves the induction. The last successor cannot return to the leftmost bundle, a contradiction. Therefore $L = 2$, so WARP fails. $\square$



(a) Start at the leftmost bundle.



(b) An affordable successor must lie right.

Figure 30: The geometry of the two-good proof. Solid red lines are budget boundaries; arrows follow the cycle. (a) All cycle bundles lie weakly above G_1 . (b) Budget line i is flatter than G_i because the predecessor is strictly unaffordable. Every cycle bundle to the left lies above that budget; the next link must move right and preserves the supporting-line property. The green line is G_{i+1} .

Remark 10.6. With three or more goods, WARP need not imply GARP: Example 4.1 has a strict three-cycle and no pairwise violation. More generally, violating cycles can have

arbitrarily large minimum length, so checking cycles up to any fixed length cannot replace GARP (Chambers and Echenique, 2016, Remark 3.5).

10.4 Afriat's theorem

Suppose the data come from a concave, differentiable, strictly increasing u with interior choices. The first order condition at observation t is $\nabla u(\mathbf{x}^t) = \lambda_t \mathbf{p}^t$, with $\lambda_t > 0$ the marginal utility of wealth, and concavity puts the tangent plane above the graph:

$$u(\mathbf{x}^s) \leq u(\mathbf{x}^t) + \nabla u(\mathbf{x}^t) \cdot (\mathbf{x}^s - \mathbf{x}^t) = u(\mathbf{x}^t) + \lambda_t \mathbf{p}^t \cdot (\mathbf{x}^s - \mathbf{x}^t).$$

Write $d_{ts} = \mathbf{p}^t \cdot (\mathbf{x}^s - \mathbf{x}^t)$. With $U_t = u(\mathbf{x}^t)$, these are finitely many linear inequalities in the unknowns U_t, λ_t . Afriat's theorem gives the converse: whenever such numbers exist, the lower envelope of the planes is a utility that rationalizes the data, and they exist exactly when GARP holds.

Theorem 10.7 (Afriat 1967). *For finite data $D = \{(\mathbf{p}^t, \mathbf{x}^t)\}_{t=1}^T$, the following are equivalent:*

- (i) D satisfies GARP.
- (ii) A locally nonsatiated utility function rationalizes D .
- (iii) A continuous, strictly increasing, concave utility function rationalizes D .
- (iv) There are $U_t \in \mathbb{R}$ and $\lambda_t > 0$, $t = 1, \dots, T$, satisfying the Afriat inequalities

$$U_s \leq U_t + \lambda_t \mathbf{p}^t \cdot (\mathbf{x}^s - \mathbf{x}^t) \quad \text{for every } t, s. \quad (8)$$

Proof. The implication chain is

$$(ii) \Rightarrow (i) \Rightarrow (iv) \Rightarrow (iii) \Rightarrow (ii).$$

(ii) $\Rightarrow$ (i) is Proposition 10.3. The hard step (i) $\Rightarrow$ (iv) uses a theorem of the alternative for linear inequalities; its proof is omitted (Chambers and Echenique, 2016, Section 3.1.1 and Lemma 1.12).

For (iv) $\Rightarrow$ (iii), define the supporting affine functions $\phi_t(\mathbf{z}) = U_t + \lambda_t \mathbf{p}^t \cdot (\mathbf{z} - \mathbf{x}^t)$ and their lower envelope

$$u(\mathbf{z}) = \min_{1 \leq t \leq T} \phi_t(\mathbf{z}). \quad (9)$$

At observation s , Afriat's inequalities give $\phi_t(\mathbf{x}^s) \geq U_s$ for all t , with equality at $t = s$. Hence $u(\mathbf{x}^s) = U_s$. If $\mathbf{z}$ is affordable in budget s , then

$$u(\mathbf{z}) \leq \phi_s(\mathbf{z}) \leq U_s = u(\mathbf{x}^s),$$

which proves rationalization, including possible ties.

A finite minimum of continuous functions is continuous. Affinity gives

$$u(\theta\mathbf{z} + (1 - \theta)\mathbf{y}) = \min_t \{\theta\phi_t(\mathbf{z}) + (1 - \theta)\phi_t(\mathbf{y})\} \geq \theta u(\mathbf{z}) + (1 - \theta)u(\mathbf{y}),$$

so u is concave. For $\mathbf{z} \geq \mathbf{y}$, $\mathbf{z} \neq \mathbf{y}$, positivity and finiteness give $\varepsilon = \min_t \lambda_t \mathbf{p}^t \cdot (\mathbf{z} - \mathbf{y}) > 0$. Thus $\phi_t(\mathbf{z}) \geq \phi_t(\mathbf{y}) + \varepsilon$ for every t , and $u(\mathbf{z}) \geq u(\mathbf{y}) + \varepsilon$: strict increase. Finally, strict increase implies local nonsatiation, proving (iii) $\Rightarrow$ (ii). $\square$

Example 10.8. Take $\mathbf{p}^1 = (1, 2)$, $\mathbf{x}^1 = (2, 1)$, $\mathbf{p}^2 = (2, 1)$, $\mathbf{x}^2 = (1, 2)$. Own expenditure is 4 and each other bundle costs 5, so $d_{12} = d_{21} = 1$. The numbers $U_1 = U_2 = 4$ and $\lambda_1 = \lambda_2 = 1$ satisfy all the Afriat inequalities. The two affine functions are

$$\begin{aligned}\phi_1(\mathbf{z}) &= 4 + (1, 2) \cdot (\mathbf{z} - (2, 1)) = z_1 + 2z_2, \\ \phi_2(\mathbf{z}) &= 4 + (2, 1) \cdot (\mathbf{z} - (1, 2)) = 2z_1 + z_2.\end{aligned}$$

Formula (9) therefore becomes

$$u(\mathbf{z}) = \min\{z_1 + 2z_2, 2z_1 + z_2\}.$$

At the observations, one affine function attains the assigned value and the other lies above it:

$$u(\mathbf{x}^1) = \min\{4, 5\} = 4 = U_1, \quad u(\mathbf{x}^2) = \min\{5, 4\} = 4 = U_2.$$

In budget t , the corresponding linear term is at most 4, so no feasible bundle has higher utility. The upper contour is the intersection

$$\{\mathbf{z} : u(\mathbf{z}) \geq 4\} = \{\mathbf{z} \geq 0 : z_1 + 2z_2 \geq 4, 2z_1 + z_2 \geq 4\}.$$

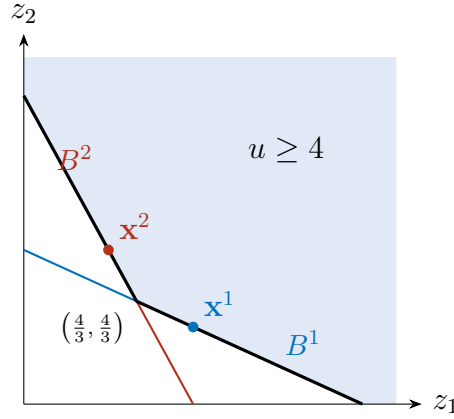


Figure 31: The Afriat construction in Example 10.8. The two budget lines support the shaded upper contour. The lower envelope of the affine utility functions yields a kinked indifference curve and allows multiple maximizers.

10.5 Multipliers, supporting planes and computation

For the constructed utility, observation t supplies the global inequality

$$u(\mathbf{z}) \leq u(\mathbf{x}^t) + \lambda_t \mathbf{p}^t \cdot (\mathbf{z} - \mathbf{x}^t).$$

Thus $\lambda_t \mathbf{p}^t$ is a *supergradient* of the concave utility at $\mathbf{x}^t$: a vector defining a supporting affine upper bound. This statement does not require differentiability.

Summing the Afriat inequalities along any cycle $t_1, \dots, t_m, t_{m+1} = t_1$ cancels the utility differences:

$$0 = \sum_{i=1}^m (U_{t_{i+1}} - U_{t_i}) \leq \sum_{i=1}^m \lambda_{t_i} \mathbf{p}^{t_i} \cdot (\mathbf{x}^{t_{i+1}} - \mathbf{x}^{t_i}).$$

These are the *cyclic monotonicity* inequalities with the sign convention for concave supergradients. They involve every cycle, not only cycles of affordable comparisons. Conversely, if positive multipliers make every such sum nonnegative, GARP must hold: a cycle with all $d_{t_i t_{i+1}} \leq 0$ and at least one strict inequality would make the sum negative. GARP is therefore equivalent to the existence of positive rescalings of observed prices that satisfy these cyclic inequalities (Chambers and Echenique, 2016, p. 41).

If the constructed utility is differentiable at an interior observation, the supporting vector equals the gradient: $\nabla u(\mathbf{x}^t) = \lambda_t \mathbf{p}^t$. This is the familiar FOC, and concavity gives the Afriat inequality by evaluating the supporting plane at $\mathbf{x}^s$. Under the usual envelope conditions, λ_t is marginal utility of wealth. At a boundary optimum, the KKT conditions instead give $u_k(\mathbf{x}^t) \leq \lambda_t p_k^t$, with equality when $x_k^t > 0$. The finite-data proof needs neither interior choices nor smooth utility.

The inequalities form a linear feasibility problem in $2T$ unknowns. Strict positivity can be replaced by $\lambda_t \geq 1$: any finite positive solution can be multiplied by $1/\min_t \lambda_t$. A common shift then sets $U_1 = 0$. Hence an ordinary linear program can find Afriat numbers:

$$U_s - U_t - \lambda_t d_{ts} \leq 0, \quad \lambda_t \geq 1, \quad U_1 = 0.$$

Solving these inequalities both tests GARP and produces a rationalizing utility, though not a uniquely identified preference.

Remark 10.9 (A smaller feasibility system). To test whether Afriat numbers exist, it is enough to impose (8) for pairs with $d_{ts} \leq 0$, retaining $\lambda_t > 0$ ([Chambers and Echenique, 2016](#), Remark 3.6). Any solution to this reduced system rules out a GARP violation: summing its inequalities around an affordable cycle with a strict edge would give $0 \leq \sum_i \lambda_{t_i} d_{t_i t_{i+1}} < 0$. GARP then guarantees a solution to the full system by the theorem. This is equivalence of *feasibility*; a particular solution to the reduced system need not satisfy the omitted inequalities.

12 Lecture 12. Choice under risk: expected utility

12.1 Prizes, lotteries and their probabilities

Under risk, probabilities are given. Following the course slides, the finite, nonempty set of prizes and the set of lotteries are

$$X = \{x_1, \dots, x_N\}, \quad P = \Delta(X) = \left\{ p : X \longrightarrow [0, 1] : \sum_{x \in X} p(x) = 1 \right\}.$$

The decision maker chooses a *lottery*, not a prize. In this lecture, the prize set is denoted by capital X , replacing the prize notation used earlier in the notes.

What do the probability functions mean?

For any two lotteries,

$$p, q \in P, \quad \begin{aligned} p(x) &= \text{probability that lottery } p \text{ delivers prize } x, \\ q(x) &= \text{probability that lottery } q \text{ delivers prize } x. \end{aligned}$$

Thus a lottery is a function, while its value at a particular prize is a number. For example,

x	0	10	20
$p(x)$	1/4	1/2	1/4
$q(x)$	0	1	0

$$\sum_{x \in X} p(x) = \sum_{x \in X} q(x) = 1.$$

The *degenerate lottery* at a prize gives that prize with certainty:

$$\delta_x(y) = \begin{cases} 1, & y = x, \\ 0, & y \neq x. \end{cases} \quad p = \sum_{x \in X} p(x) \delta_x.$$

The *support* consists of prizes that receive positive probability:

$$\text{supp}(p) = \{x \in X : p(x) > 0\}.$$

Mixtures and reduction of compound lotteries

First choose between two lotteries using an objective random device, then draw a prize from the selected lottery. Reduction identifies this compound lottery with its final distribution:

$$p, q \in P, \quad \alpha \in [0, 1],$$

$$[\alpha p + (1 - \alpha)q](x) = \alpha p(x) + (1 - \alpha)q(x) \quad \text{for every } x \in X.$$

Only final probabilities and prizes matter; the randomizing device and the order of its stages do not.

12.2 Strict preferences and the three axioms

We take strict preference as primitive and recover weak preference and indifference by

$$p \succsim q \iff \neg(q \succ p), \quad p \sim q \iff [\neg(p \succ q) \text{ and } \neg(q \succ p)].$$

A1. A strict weak order

Strict preference is asymmetric and negatively transitive:

$$\begin{aligned} p \succ q &\implies \neg(q \succ p), \\ p \succ q &\implies [p \succ r \text{ or } r \succ q] \quad \text{for every } r \in P. \end{aligned}$$

Proposition 12.1. *A1 is equivalent to the induced weak preference being complete and transitive, with the original relation as its strict part.*

Proof. Asymmetry gives completeness and identifies the strict part. To prove transitivity, suppose

$$p \succsim q, \quad q \succsim r, \quad r \succ p.$$

Negative transitivity would imply

$$r \succ q \quad \text{or} \quad q \succ p,$$

contradicting one of the two weak comparisons. Conversely, if weak preference is complete and transitive, failure of negative transitivity would give

$$p \succ q, \quad q \succsim r \succsim p,$$

a contradiction. Asymmetry follows from the definition of the strict part. $\square$

Strict preference is consequently irreflexive, transitive and acyclic. A1 alone does not guarantee a utility representation on an infinite choice set; see Example 3.8. Even with finitely many prizes, the set of lotteries is generally infinite.

A2. Independence

Mixing two strictly ranked lotteries with the same third lottery preserves the ranking whenever the original comparison receives positive weight:

$$p, q, r \in P, \quad 0 < \alpha \leq 1, \\ p \succ q \implies \alpha p + (1 - \alpha)r \succ \alpha q + (1 - \alpha)r.$$

Zero weight is excluded: both mixtures would then be the same lottery. The indifference version of substitution will be *proved* in L3.

A3. Archimedean continuity

For any three strictly ranked lotteries, suitable mixtures of the best and worst lie on either side of the middle lottery:

$$p \succ q \succ r \implies \text{there exist } \alpha, \beta \in (0, 1) \text{ such that} \\ \alpha p + (1 - \alpha)r \succ q \succ \beta p + (1 - \beta)r.$$

This rules out an infinitely important advantage that no sufficiently small change in probability could offset.

Allais: why independence is restrictive

The slide example uses the following lotteries, with prizes in thousands of dollars. Here the four lottery labels follow the slides; the monetary prizes are the column headings:

	25	24	0
x	.33	.66	.01
y	0	1	0
z	.33	0	.67
w	0	.34	.66

Under expected utility, the value of a lottery is

$$U(p) = \sum_{x \in X} p(x)u(x).$$

This representation cannot accommodate both of the commonly observed choices

$$y \succ x, \quad z \succ w,$$

because

$$\begin{aligned} U(y) > U(x) &\iff .34u(24) > .33u(25) + .01u(0) \\ &\iff U(w) > U(z). \end{aligned}$$

12.3 The theorem and the proof roadmap

Theorem 12.2 (von Neumann–Morgenstern). *On the lottery set over a finite, nonempty prize set, A1–A3 hold if and only if there is a Bernoulli utility index*

$$u : X \longrightarrow \mathbb{R}$$

such that

$$p \succ q \iff \sum_{x \in X} p(x)u(x) > \sum_{x \in X} q(x)u(x).$$

Any two such prize indices differ by a positive affine transformation:

$$u'(x) = cu(x) + d \quad \text{for every } x \in X, \quad c > 0.$$

The proof follows the slides: establish L1–L4, calibrate each lottery against the best and worst prizes, prove that calibration is affine, and then use induction on support size to obtain expected utility.

Keep the two utility objects separate

The calibration function evaluates an entire lottery; the Bernoulli index evaluates a single prize:

$$f : P \longrightarrow [0, 1], \quad u : X \longrightarrow [0, 1], \quad u(x) = f(\delta_x).$$

The central task is to prove, rather than assume, the identity

$$f(p) = \sum_{x \in X} p(x)u(x).$$

12.4 The four preparatory lemmas

L1. More weight on the preferred lottery is better

Lemma 12.3. *Under A1–A2,*

$$p \succ q, \quad 0 \leq \alpha < \beta \leq 1 \quad \implies \quad \beta p + (1 - \beta)q \succ \alpha p + (1 - \alpha)q.$$

Proof. Define the lottery with the larger weight on the preferred component:

$$r = \beta p + (1 - \beta)q.$$

Independence first gives

$$r \succ \beta q + (1 - \beta)q = q.$$

If the smaller weight is zero, this already proves the claim. Otherwise,

$$0 < \frac{\alpha}{\beta} < 1.$$

Apply independence again, now to the strict comparison just obtained:

$$\begin{aligned} r &= \left(1 - \frac{\alpha}{\beta}\right)r + \frac{\alpha}{\beta}r \\ &\succ \left(1 - \frac{\alpha}{\beta}\right)q + \frac{\alpha}{\beta}r \\ &= \alpha p + (1 - \alpha)q. \end{aligned}$$

□

L2. Calibration: a unique equivalent binary lottery

Lemma 12.4. *Under A1–A3,*

$$p \succsim q \succsim r, \quad p \succ r \implies \text{there is a unique } \alpha^* \in [0, 1]$$

such that

$$q \sim \alpha^* p + (1 - \alpha^*) r.$$

Proof. **Endpoints and uniqueness**

If the middle lottery is indifferent to an endpoint, use that endpoint:

$$q \sim p \implies \alpha^* = 1, \quad q \sim r \implies \alpha^* = 0.$$

L1 implies uniqueness, since two distinct mixture weights yield strictly ranked lotteries. It remains to establish existence in the strict case:

$$p \succ q \succ r.$$

Construct the candidate as a supremum

As in the slides, define the candidate directly:

$$\alpha^* = \sup\{\alpha \in [0, 1] : q \succsim \alpha p + (1 - \alpha)r\}.$$

The set contains zero and is bounded above by one. A3 supplies weights strictly inside the interval such that

$$\beta p + (1 - \beta)r \prec q \prec \alpha p + (1 - \alpha)r, \quad 0 < \beta, \alpha < 1.$$

By L1, no weight at or above the upper comparison belongs to the set. Consequently,

$$0 < \beta \leq \alpha^* \leq \alpha < 1.$$

Case 1: the candidate mixture is strictly better

Suppose

$$\alpha^*p + (1 - \alpha^*)r \succ q \succ r.$$

By A3 there is a weight strictly below one for which

$$\begin{aligned} \beta &\in (0, 1), \\ \beta[\alpha^*p + (1 - \alpha^*)r] + (1 - \beta)r &= \beta\alpha^*p + (1 - \beta\alpha^*)r \succ q. \end{aligned}$$

L1 implies that every weight at least as large as this new weight also gives a lottery strictly above the middle lottery. Therefore

$$\sup\{\alpha \in [0, 1] : q \succsim \alpha p + (1 - \alpha)r\} \leq \beta\alpha^* < \alpha^*,$$

contradicting the definition of the supremum.

Case 2: the candidate mixture is strictly worse

Suppose instead

$$p \succ q \succ \alpha^*p + (1 - \alpha^*)r.$$

By A3, for some positive weight,

$$\begin{aligned} \beta &\in (0, 1), \\ q &\succ \beta p + (1 - \beta)[\alpha^*p + (1 - \alpha^*)r] \\ &= [\alpha^* + \beta(1 - \alpha^*)]p + [1 - \alpha^* - \beta(1 - \alpha^*)]r. \end{aligned}$$

Thus the defining set contains a weight strictly above its supremum:

$$\begin{aligned} \alpha^* + \beta(1 - \alpha^*) &\in \{\alpha \in [0, 1] : q \succsim \alpha p + (1 - \alpha)r\}, \\ \alpha^* + \beta(1 - \alpha^*) &> \alpha^*. \end{aligned}$$

Both strict comparisons are impossible. Completeness leaves precisely the desired indifference. □

L3. Substituting indifferent lotteries

Lemma 12.5. *Under A1–A3,*

$$p \sim q, \quad r \in P, \quad \alpha \in [0, 1] \implies \alpha p + (1 - \alpha)r \sim \alpha q + (1 - \alpha)r.$$

Proof. The endpoint weights are immediate. For an interior weight, suppose, toward a contradiction, that

$$0 < \alpha < 1, \\ \alpha p + (1 - \alpha)r \succ \alpha q + (1 - \alpha)r.$$

The slide argument: perturb toward a better lottery

As in the slides, first suppose there is a lottery satisfying

$$s \succ p \sim q.$$

For every interior mixing weight, A2 implies

$$\beta \in (0, 1), \\ \beta s + (1 - \beta)q \succ \beta p + (1 - \beta)q = q \sim p.$$

Applying A2 again and using the assumed strict comparison gives

$$\begin{aligned} & \alpha[\beta s + (1 - \beta)q] + (1 - \alpha)r \\ & \succ \alpha p + (1 - \alpha)r \\ & \succ \alpha q + (1 - \alpha)r. \end{aligned}$$

By A3 there is a weight, depending on the perturbation, such that

$$\begin{aligned} & \alpha^*(\beta) \in (0, 1), \\ & \alpha p + (1 - \alpha)r \\ & \succ \alpha^*(\beta) \left(\alpha[\beta s + (1 - \beta)q] + (1 - \alpha)r \right) \\ & \quad + [1 - \alpha^*(\beta)] \left(\alpha q + (1 - \alpha)r \right). \end{aligned}$$

Following the slides, fix

$$\beta = \frac{1}{2}, \quad \alpha^* = \alpha^*\left(\frac{1}{2}\right).$$

The lottery on the right of the strict comparison reduces to

$$\begin{aligned} & \alpha^* \left(\alpha \left[\frac{1}{2}s + \frac{1}{2}q \right] + (1 - \alpha)r \right) + (1 - \alpha^*) \left(\alpha q + (1 - \alpha)r \right) \\ & = \frac{\alpha^* \alpha}{2} s + \left[\frac{\alpha^* \alpha}{2} + (1 - \alpha^*) \alpha \right] q + (1 - \alpha)r \\ & = \alpha \left[\frac{\alpha^*}{2} s + \left(1 - \frac{\alpha^*}{2} \right) q \right] + (1 - \alpha)r. \end{aligned}$$

But its inner mixture is strictly preferred to the original indifferent pair:

$$\frac{\alpha^*}{2}s + \left(1 - \frac{\alpha^*}{2}\right)q \succ q \sim p.$$

A2 therefore gives exactly the opposite strict comparison:

$$\alpha \left[\frac{\alpha^*}{2}s + \left(1 - \frac{\alpha^*}{2}\right)q \right] + (1 - \alpha)r \succ \alpha p + (1 - \alpha)r.$$

This is the contradiction. Interchanging the original indifferent pair rules out the reverse strict ranking of the two mixtures as well.

Boundary case

If no strictly better lottery exists but a strictly worse one does, apply the same argument to the reversed preference relation: reversing every comparison preserves A1–A3 and leaves indifference unchanged. If neither exists, A1 makes every lottery indifferent, and the result is immediate. $\square$

L4. Best and worst prizes bound every lottery

Lemma 12.6. *There exist a best prize and a worst prize, denoted by*

$$\bar{x}, \underline{x} \in X,$$

whose degenerate lotteries bound every lottery:

$$\delta_{\bar{x}} \succeq p \succeq \delta_{\underline{x}} \quad \text{for every } p \in P.$$

Proof. Because the prize set is finite and the induced weak order is complete and transitive, best and worst degenerate lotteries exist:

$$\delta_{\bar{x}} \succeq \delta_x \succeq \delta_{\underline{x}} \quad \text{for every } x \in X.$$

They need not be unique. A2 and L3 together permit weakly preferred components to replace original components in any mixture:

$$p \succeq q \implies \alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r, \quad \alpha \in [0, 1].$$

Apply this property successively to the finite degenerate-lottery decomposition. Replacing each prize by the best prize gives the upper bound; replacing each prize by the worst gives the lower bound:

$$\begin{aligned}\delta_{\bar{x}} &= \sum_{x \in X} p(x) \delta_{\bar{x}} \\ &\succeq \sum_{x \in X} p(x) \delta_x = p \\ &\succeq \sum_{x \in X} p(x) \delta_{\underline{x}} = \delta_{\underline{x}}.\end{aligned}$$

Formally, successive replacement is induction on the number of positive-probability components. □

12.5 Constructing expected utility: the main proof

We now prove the existence direction of Theorem 12.2. The order of the construction is the one used in the slides.

Step 1. Separate the case of universal indifference

If the best and worst degenerate lotteries are indifferent, L4 gives

$$\delta_{\bar{x}} \sim \delta_{\underline{x}} \implies p \sim q \quad \text{for every } p, q \in P.$$

A constant prize index represents this preference:

$$u(x) = 0 \quad \text{for every } x \in X.$$

For the rest of the construction, assume

$$\delta_{\bar{x}} \succ \delta_{\underline{x}}.$$

Step 2. Define the calibration function

By L4 every lottery lies between these endpoints. By L2 there is exactly one probability of the best prize that makes the binary lottery indifferent to it. As in the slides, define

$$f(p) = \alpha \quad \text{where} \quad \alpha \delta_{\bar{x}} + (1 - \alpha) \delta_{\underline{x}} \sim p.$$

The calibration probability lies in the unit interval:

$$\alpha \in [0, 1].$$

In particular,

$$f(\delta_{\bar{x}}) = 1, \quad f(\delta_{\underline{x}}) = 0.$$

Calibration is a probability of receiving the best prize in an *equivalent* binary lottery. It is generally different from the original lottery's probability of that prize.

Step 3. Show that calibration represents preference

Substitute the equivalent binary lotteries, then apply L1:

$$\begin{aligned}
 p &\succ q \\
 &\Downarrow \\
 f(p)\delta_{\bar{x}} + [1 - f(p)]\delta_{\underline{x}} &\succ f(q)\delta_{\bar{x}} + [1 - f(q)]\delta_{\underline{x}} \\
 &\Downarrow \\
 f(p) &> f(q).
 \end{aligned}$$

Similarly,

$$p \sim q \iff f(p) = f(q).$$

This is already an ordinal representation on lotteries. We must still prove its expected utility form.

Step 4. Prove affinity using L3 twice

Fix two lotteries and a mixture weight:

$$p, q \in P, \quad \alpha \in [0, 1].$$

Replace the first component by its equivalent binary lottery, then the second component:

$$\begin{aligned}
 &\alpha p + (1 - \alpha)q \\
 &\quad \sim \\
 &\alpha[f(p)\delta_{\bar{x}} + [1 - f(p)]\delta_{\underline{x}}] + (1 - \alpha)q \\
 &\quad \sim \\
 &\alpha[f(p)\delta_{\bar{x}} + [1 - f(p)]\delta_{\underline{x}}] \\
 &\quad + (1 - \alpha)[f(q)\delta_{\bar{x}} + [1 - f(q)]\delta_{\underline{x}}].
 \end{aligned}$$

Reduction collects the probabilities of the best and worst prizes:

$$\begin{aligned}
 &[\alpha f(p) + (1 - \alpha)f(q)]\delta_{\bar{x}} \\
 &+ [1 - \alpha f(p) - (1 - \alpha)f(q)]\delta_{\underline{x}}.
 \end{aligned}$$

Uniqueness of the calibration weight therefore yields

$$\boxed{f(\alpha p + (1 - \alpha)q) = \alpha f(p) + (1 - \alpha)f(q).}$$

This is the key bridge from preference axioms to expected utility.

Step 5. Define utility on prizes

Evaluate each prize through the lottery that delivers it for sure:

$$u(x) := f(\delta_x) \quad \text{for every } x \in X.$$

The resulting normalization is

$$u(\bar{x}) = 1, \quad u(\underline{x}) = 0, \quad 0 \leq u(x) \leq 1.$$

Equivalently, the utility of a prize is its calibration probability:

$$\delta_x \sim u(x)\delta_{\bar{x}} + [1 - u(x)]\delta_{\underline{x}}.$$

Step 6. Induct on the size of the support

The target identity is

$$f(p) = \sum_{x \in X} p(x)u(x).$$

Base case: a single prize in the support

Such a lottery is degenerate, so the identity follows from the definition:

$$p = \delta_{x'} \implies f(p) = f(\delta_{x'}) = u(x') = \sum_{x \in X} p(x)u(x).$$

Induction hypothesis and the residual lottery

Assume the identity holds for every lottery with support of size at most one less than the current size. Take

$$|\text{supp}(p)| = n > 1, \quad x' \in \text{supp}(p), \quad 0 < p(x') < 1.$$

Remove the probability assigned to the selected prize and renormalize the probabilities of all remaining prizes:

$$q(x) = \begin{cases} 0, & x = x', \\ \frac{p(x)}{1 - p(x')}, & x \neq x'. \end{cases}$$

Here the residual lottery is constructed from the original lottery; it is not an additional arbitrary choice. It is a valid probability distribution because

$$q(x) \geq 0, \quad \sum_{x \in X} q(x) = \frac{\sum_{x \neq x'} p(x)}{1 - p(x')} = 1.$$

Its support is smaller, and it reconstructs the original lottery:

$$|\text{supp}(q)| = n - 1, \quad p = p(x')\delta_{x'} + [1 - p(x')]q.$$

The reconstruction holds prize by prize:

$$[p(x')\delta_{x'} + (1 - p(x'))q](x) = \begin{cases} p(x'), & x = x', \\ [1 - p(x')] \frac{p(x)}{1 - p(x')} = p(x), & x \neq x'. \end{cases}$$

Apply affinity, then the induction hypothesis

Each equality now has a specific justification:

$$\begin{aligned} f(p) &= p(x')f(\delta_{x'}) + [1 - p(x')]f(q) \\ &= p(x')u(x') + [1 - p(x')] \sum_{x \neq x'} q(x)u(x) \\ &= p(x')u(x') + [1 - p(x')] \sum_{x \neq x'} \frac{p(x)}{1 - p(x')} u(x) \\ &= p(x')u(x') + \sum_{x \neq x'} p(x)u(x) \\ &= \sum_{x \in X} p(x)u(x). \end{aligned}$$

The first line uses Step 4; the second uses Step 5 and the induction hypothesis; the third substitutes the residual probabilities. Finiteness of the prize set completes the induction.

Combine representation and the expectation identity

Define lottery utility by

$$U(p) := \sum_{x \in X} p(x)u(x) = f(p).$$

Step 3 now gives exactly the desired representation:

$$p \succ q \iff f(p) > f(q) \iff U(p) > U(q).$$

12.6 Uniqueness and the converse

Positive affine uniqueness

Start with the normalized prize index constructed above and let another prize index represent the same lottery preferences by expected utility. Every prize has the calibration indifference

$$\delta_x \sim u(x)\delta_{\bar{x}} + [1 - u(x)]\delta_{\underline{x}}.$$

Evaluating both sides with the second index forces

$$\begin{aligned} u'(x) &= u(x)u'(\bar{x}) + [1 - u(x)]u'(\underline{x}) \\ &= [u'(\bar{x}) - u'(\underline{x})]u(x) + u'(\underline{x}). \end{aligned}$$

Thus

$$u'(x) = cu(x) + d, \quad c = u'(\bar{x}) - u'(\underline{x}) > 0, \quad d = u'(\underline{x}).$$

The slope is positive because the best prize is strictly preferred to the worst. Since every representing index is a positive affine transformation of the normalized one, any two representing indices are positive affine transformations of each other.

Conversely, such a transformation preserves every lottery comparison:

$$\begin{aligned} \sum_{x \in X} p(x)u'(x) &= \sum_{x \in X} p(x)[cu(x) + d] \\ &= c \sum_{x \in X} p(x)u(x) + d, \quad c > 0. \end{aligned}$$

In the universal-indifference case, every representing prize index must be constant. Any two constant indices are also related by a positive affine transformation.

Expected utility implies A1–A3

A1 follows from the order on real numbers. For A2, compute

$$\begin{aligned} &U(\alpha p + (1 - \alpha)r) - U(\alpha q + (1 - \alpha)r) \\ &= \alpha[U(p) - U(q)] > 0 \quad \text{if } U(p) > U(q), \quad \alpha > 0. \end{aligned}$$

For A3, take a strictly ranked triple and locate the middle utility between the endpoints:

$$U(p) > U(q) > U(r), \quad c = \frac{U(q) - U(r)}{U(p) - U(r)} \in (0, 1).$$

Choose weights on either side of that number:

$$0 < \beta < c < \alpha < 1.$$

Affinity of expected utility then gives

$$U(\alpha p + (1 - \alpha)r) > U(q) > U(\beta p + (1 - \beta)r).$$

This proves the converse and completes the theorem.

What utility differences tell us

A strictly increasing transformation of *lottery utility* preserves its ranking. To retain an expected utility representation through a *prize index*, the admissible transformations are positive affine. For example,

$$\begin{aligned} \delta_x &\succ \delta_{x'} \succ \delta_{x''}, \\ u(x) - u(x'') &= 2[u(x') - u(x'')] \\ \iff \frac{1}{2}\delta_x + \frac{1}{2}\delta_{x''} &\sim \delta_{x'}. \end{aligned}$$

The ratio of utility differences records an observable lottery indifference, not a claim that one prize is twice as good as another.

12.7 Beyond expected utility: betweenness

Betweenness requires an interior mixture to lie strictly between two strictly ranked lotteries:

$$\begin{aligned} 0 < \alpha < 1, \\ p \succ q \implies p \succ \alpha p + (1 - \alpha)q \succ q. \end{aligned}$$

Independence implies this property by L1. With A1, betweenness also preserves indifference along mixtures of indifferent endpoints. To see this, suppose

$$p \sim q, \quad m = \alpha p + (1 - \alpha)q \succ p, \quad 0 < \alpha < 1,$$

and define

$$s = \frac{1}{2}p + \frac{1}{2}m.$$

Betweenness and indifference imply

$$m \succ s \succ p \sim q.$$

But the original mixture is itself an interior mixture of the new lottery and the second endpoint:

$$m = \frac{2\alpha}{1+\alpha}s + \frac{1-\alpha}{1+\alpha}q.$$

Betweenness would therefore imply the reverse strict comparison:

$$s \succ m \succ q.$$

The case of a mixture strictly below the indifferent endpoints is excluded by reversing all comparisons.

A model satisfying betweenness but not independence

Give each prize a strictly positive weight and define

$$w : X \longrightarrow (0, \infty), \quad D(p) = \sum_{x \in X} p(x)w(x),$$

$$V(p) = \frac{\sum_{x \in X} p(x)w(x)u(x)}{D(p)}.$$

The utility of a mixture is a weighted average of the component utilities:

$$V(\alpha p + (1-\alpha)q) = \theta V(p) + (1-\theta)V(q),$$

$$\theta = \frac{\alpha D(p)}{\alpha D(p) + (1-\alpha)D(q)} \in (0, 1).$$

This proves betweenness. Indifference sets are intersections of the simplex with hyperplanes:

$$V(p) = k \iff \sum_{x \in X} p(x)w(x)[u(x) - k] = 0.$$

With three prizes the indifference lines are straight, but they need not be parallel.

For an explicit failure of substitution, take

$$X = \{x_1, x_2, x_3\},$$

$$(u(x_1), u(x_2), u(x_3)) = (0, 1, 2),$$

$$(w(x_1), w(x_2), w(x_3)) = (1, 2, 1),$$

$$p = \delta_{x_2}, \quad q = \frac{1}{2}\delta_{x_1} + \frac{1}{2}\delta_{x_3}, \quad r = \delta_{x_1}.$$

Then

$$V(p) = V(q) = 1, \quad V\left(\frac{1}{2}p + \frac{1}{2}r\right) = \frac{2}{3} > \frac{1}{2} = V\left(\frac{1}{2}q + \frac{1}{2}r\right).$$

These preferences are continuous and satisfy betweenness, but they violate L3 and hence admit no expected utility representation.

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A.1 The question and the anomalies

Dozens of non-EU models exist. Starmer asks whether any is a serious *descriptive* replacement for EU, and sorts the evidence into two classes.

Class A: independence fails, but preferences exist. The Allais effects of §1.4. Reshaping the indifference curves can absorb them.

Class B: the ranking itself is unstable. Subjects choose the P-bet (large chance of a small prize) over the \$-bet, yet price the \$-bet higher: $P \succ \$$ and $M(\$) > M(P)$. Either the ranking depends on how the question is put, or, if prices are certainty equivalents, preferences cycle, $P \succ \$ \sim M(\$) \succ M(P) \sim P$. Framing does the same damage: in the “Asian disease” problem 72% take the sure option when outcomes are described as lives saved and 22% when the same options are described as lives lost (Tversky and Kahneman, 1981). Class B attacks the primitive rather than the axioms, so no conventional repair reaches it.

A.2 Conventional theories

Call a theory *conventional* if the agent ranks probability distributions over outcomes by a complete and transitive preference and chooses by maximizing it, with independence dropped. The data ask for indifference curves that are steep near the left edge and flat near the bottom-right corner.

Fanning out (Machina, 1982). For smooth V , the *local utility function* $U(\cdot; q)$, defined by $U(x_i; q) = \partial V(q) / \partial p_i$, makes the agent behave near q like an EU maximizer with index $U(\cdot; q)$. If that index becomes more concave as q improves in the FOSD order, the curves fan out to the north-west.

Betweenness says a mixture always ranks between the two prospects mixed: if $q \succ r$ then $q \succ (q, p; r, 1 - p) \succ r$ for every $p < 1$. In the triangle this is the same as straight indifference curves, because $q \sim r$ puts every mixture of q and r on the segment joining them, and betweenness puts that whole segment on the curve.

EU asks for straight *and* parallel lines; betweenness gives up parallel. The leading case is *weighted utility* (Chew and MacCrimmon, 1979), whose lines all pass through one point outside the triangle, a device rather than a lottery. Placed to the south-west it makes the lines steeper towards the north-west, which is the fanning out the Allais evidence asks for (Figure 32).

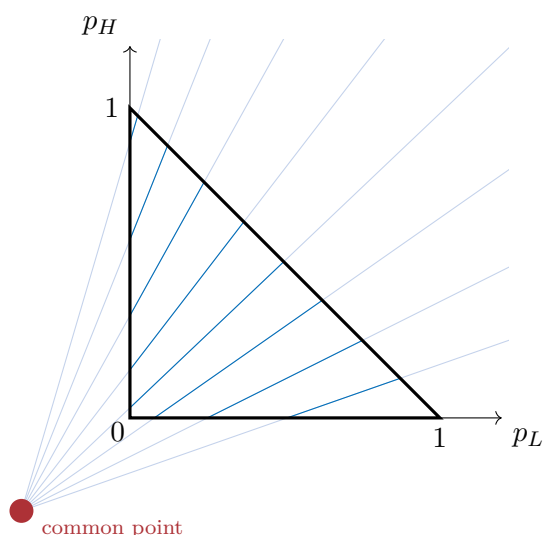


Figure 32: Weighted utility. The indifference lines are straight, as betweenness requires, but not parallel: they all pass through one point outside the triangle, drawn here to the south-west. Moving north-west the lines get steeper, so the agent becomes more risk averse in that direction. EU is the case in which the common point is sent to infinity and the lines become parallel.

Implicit EU (Dekel, 1986) and disappointment aversion (Gul, 1991) keep the lines straight and let the fanning go one way in one part of the triangle and the other way elsewhere.

Drop betweenness and the curves bend, in a direction that has meaning. Mix two prospects the agent is indifferent between: if she likes the mixture *less*, she dislikes randomizing, V is quasi-convex and the curves are concave; if *more*, V is quasi-concave and the curves are convex (Figure 33). This matters outside decision theory, since existence of Nash equilibrium needs quasi-concavity: a player who dislikes randomizing will not play a mixed strategy.

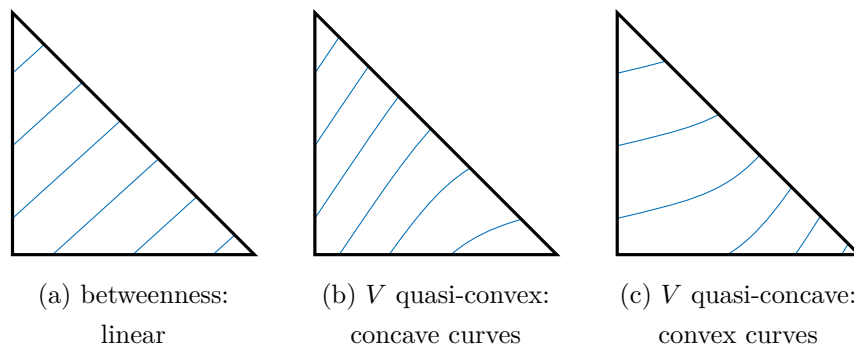


Figure 33: Betweenness and its failures (Starmer, Fig. 5). Under betweenness the probability mixtures of two indifferent prospects lie on the same indifference curve, so curves are straight. Quasi-convexity means the agent dislikes randomizing between equally-valued prospects; quasi-concavity means she likes it. Nash equilibrium existence needs quasi-concavity.

A.3 Decision weights and prospect theory

The other strategy keeps the utility u on prizes and changes the probabilities instead: the agent maximizes $V(q) = \sum_i w_i u(x_i)$, where w_i is the *weight* she puts on prize x_i rather than its true probability p_i .

The obvious version fails. Take $w_i = \pi(p_i)$ with π increasing, $\pi(0) = 0$, $\pi(1) = 1$. A convex π gives $\pi(p) + \pi(1 - p) < 1$ for $0 < p < 1$, so the weights add up to less than one. Compare x for sure with the gamble paying x with probability p and $x + \varepsilon$ with probability $1 - p$: the gamble is better whatever $\varepsilon > 0$ is, yet its value $\pi(p)u(x) + \pi(1 - p)u(x + \varepsilon)$ falls below $u(x)$ for small ε , so the agent turns down a gamble that can only help her. Fishburn (1978) showed the only escape here is $\pi(p) = p$, which is EU again.

Quiggin (1982) weights *cumulative* probabilities instead. Order the prizes from worst x_1 to best x_n and set

$$w_i = \pi(p_i + \dots + p_n) - \pi(p_{i+1} + \dots + p_n), \quad w_n = \pi(p_n).$$

The weights now add to one and the problem disappears, at the cost that a prize's weight depends on its *position* in the ranking: change what it pays without changing its place and its weight does not move. Estimates give π an inverse-S shape, small probabilities counting for more than they should and large ones for less (Figure 34).

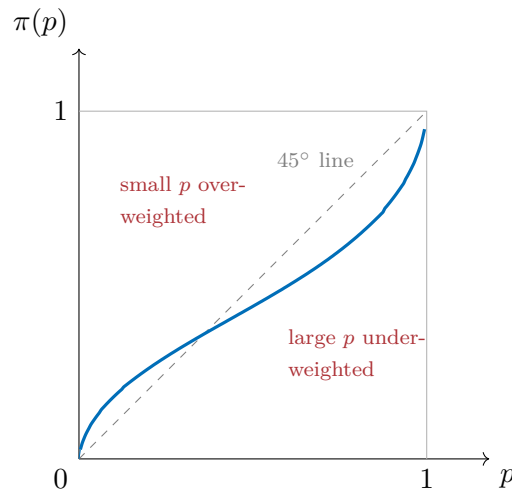


Figure 34: The inverse-S probability weighting function, drawn with the [Tversky and Kahneman \(1992\)](#) form $\pi(p) = p^\delta / (p^\delta + (1 - p)^\delta)^{1/\delta}$, $\delta = 0.65$. It crosses the 45° line once. Diminishing sensitivity with respect to the natural reference points 0 and 1 makes π steep near the endpoints and flat in the middle.

Prospect theory ([Kahneman and Tversky, 1979](#)) changes two further things. First, the agent *edits* the problem before evaluating it, rounding numbers, merging outcomes that pay the same, cancelling what the options share. This is how the theory explains framing and also its weak point, since nothing fixes the order of the steps. Second, she values *gains and losses* from a reference point rather than final wealth, with a value function kinked there: concave for gains, convex for losses, steeper for losses, so losing \$100 hurts more than gaining \$100 pleases (Figure 35). One consequence is choice cycles, which [Starmer \(1999\)](#) looked for and found.

Cumulative prospect theory ([Tversky and Kahneman, 1992](#)) applies Quiggin’s weights to gains and losses separately and drops the editing stage: consistency returns, the explanation of framing is lost. Regret theory ([Loomes and Sugden, 1982](#)) compares the two options outcome by outcome, asking how the agent would feel having taken one rather than the other, so an option’s value depends on what it is compared with and choices can cycle.

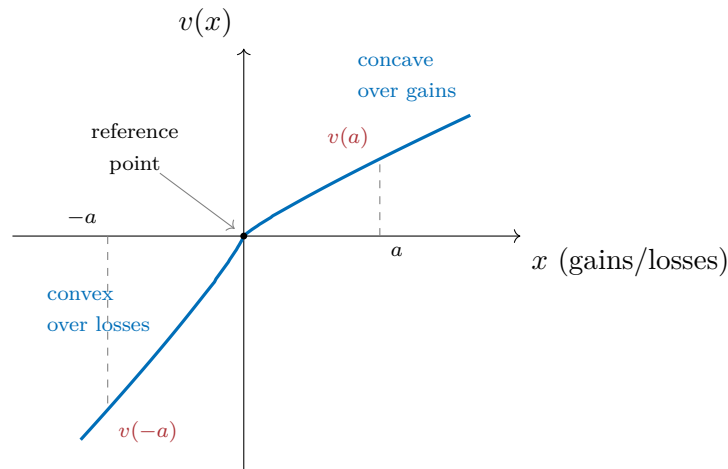


Figure 35: The prospect theory value function: kinked at the reference point, concave for gains, convex for losses, and steeper for losses, so $|v(-a)| > v(a)$ for every $a > 0$ (loss aversion). Here $v(x) = x^{0.88}$ for $x \geq 0$ and $v(x) = -2.25(-x)^{0.88}$ for $x < 0$ (Tversky–Kahneman 1992 estimates).

A.4 What the evidence selects

The experiments give three results.

1. The curves do not all fan out. In one part of the triangle they fan out, in another they fan the opposite way. Any theory that forces them to fan out everywhere is wrong.
2. The curves are not straight, so betweenness fails (Camerer and Ho, 1994).
3. Behaviour is close to EU in the middle of the triangle and far from it near the edges.

Only the decision-weight models can do all three at once. And when these models are fitted to data, most of the gain in fit comes from adding one inverse-S weighting function to EU; the rest of the machinery adds little (Harless and Camerer, 1994; Hey and Orme, 1994).

Three other results cannot be explained by any theory of the conventional kind.

- People sometimes choose an option that pays less in every case, when the problem is presented in a way that hides this. A conventional theory assumes they never do.
- Describe one event paying a prize as two events paying the same prize. The total probability is unchanged, but people act as if the prize were more likely.
- Choice cycles are common, while transitivity rules them out.

Evidence outside the laboratory says the same. With a smooth u , EU makes an agent almost risk neutral over small stakes, which is why it cannot explain how much insurance people buy and why someone who turns down small favourable gambles at every level of wealth would have to turn down enormous ones too ([Rabin, 2000](#)). Weighting probabilities avoids this, since the distortion does not vanish as the stake shrinks. The price of giving up independence is that a plan optimal at the start of a sequence of choices need not be followed later ([Machina, 1989](#)), although most comparative statics survive through local utility functions. Starmer's conclusion is that there is no reason to expect the best *descriptive* model to be built on principles of rational choice, however attractive those principles are as a normative standard.

B Utility representation and Debreu's theorem

Lecture 3 showed that completeness and transitivity are necessary for a utility representation (Proposition 3.2) and sufficient on a countable set (Theorems 3.6 and 3.7). On an uncountable set they are not sufficient: lexicographic preferences have no representation (Example 3.8). Section 4.2 repaired this for continuous, monotone preferences on $\mathbb{R}_+^K$ by measuring bundles against the diagonal (Lemma 4.8). This appendix drops monotonicity. It first isolates what countability was doing, and then shows that continuity supplies it: on convex subsets of $\mathbb{R}^n$, and then on any connected and separable space (Debreu, 1954). See also Chambers and Echenique (2016, Theorem 1.1), Kreps (2012, Proposition 1.12), Mas-Colell et al. (1995, Chapters 1 and 3) and Rubinstein (2012).

Throughout, $\succsim$ is a complete and transitive relation on a nonempty set X . The following lemma is used repeatedly.

Lemma B.1 (Mixing weak and strict). *If $x \succsim y \succ z$ then $x \succ z$. If $x \succ y \succsim z$ then $x \succ z$.*

Proof. Assume $x \succsim y \succ z$. Transitivity gives $x \succsim z$. If also $z \succsim x$, then $z \succsim x \succsim y$ gives $z \succsim y$, contradicting $y \succ z$. Hence $x \succ z$. The second statement is symmetric. $\square$

B.1 Order-dense sets and Cantor's criterion

Countability of X was used in Theorem 3.7 only to produce a countable “ruler” against which every alternative could be measured. What matters is that the ruler be countable and fine enough, not that X itself be countable.

Definition B.2 (Order-dense set). $Z \subseteq X$ is *order-dense* for $\succsim$ if for all $x, y \in X$ with $x \succ y$ there is $z \in Z$ with $x \succsim z \succsim y$.

Example B.3. $\mathbb{Q}$ is order-dense in $(\mathbb{R}, \geq)$: between any two reals there is a rational. For X countable, $Z = X$ is order-dense, which is why Theorem 3.7 needed no extra hypothesis.

Theorem B.4 (Cantor). *The following are equivalent.*

- (i) *There exists $u : X \rightarrow \mathbb{R}$ representing $\succsim$.*
- (ii) *There exists an at most countable order-dense $Z \subseteq X$.*

Proof. (ii) $\Rightarrow$ (i). Enumerate $Z = \{z_1, z_2, \dots\}$, put $w_k = 2^{-k}$ and

$$K(x) = \{k : x \succsim z_k\}, \quad S(x) = \{k : x \succ z_k\}, \quad u(x) = \sum_{k \in K(x)} w_k + \sum_{k \in S(x)} w_k \in [0, 2].$$

If $x \succsim y$ then $K(y) \subseteq K(x)$ by transitivity and $S(y) \subseteq S(x)$ by Lemma B.1, so $u(x) \geq u(y)$.

Let $x \succ y$. By (ii) there is z_k with $x \succsim z_k \succsim y$; in particular $k \in K(x)$. Two cases. If $y \succsim z_k$ fails, then $k \notin K(y)$, so the first sum is strictly larger for x while the second is at least as large. If $y \succsim z_k$, then $z_k \sim y$, so $x \succ y \sim z_k$ and Lemma B.1 give $x \succ z_k$, i.e. $k \in S(x)$; and $k \notin S(y)$, since $y \succ z_k$ would contradict $z_k \succsim y$. Now the second sum is strictly larger for x and the first is at least as large. Either way $u(x) > u(y)$. The converse implication follows from completeness as before.

(i) $\Rightarrow$ (ii). Let u represent $\succsim$ and put $A = u(X)$. Let $\{I_n\}_{n \in \mathbb{N}}$ enumerate the closed intervals with rational endpoints, $I_n = [q_n, q'_n]$ with $q_n < q'_n$; there are countably many because $\mathbb{Q} \times \mathbb{Q}$ is countable. For each n define x_n by:

(R1) if $A \cap I_n \neq \emptyset$, choose $x_n \in X$ with $u(x_n) \in I_n$;

(R2) if $A \cap I_n = \emptyset$ and the set $\{r \in A : r > q'_n\}$ is nonempty with infimum $\rho_n \in A$, choose x_n with $u(x_n) = \rho_n$;

(R3) otherwise x_n is undefined.

Let $Z = \{x_n : x_n \text{ defined}\}$, countable since at most one element arises per n .

Let $x \succ y$, so $u(x) > u(y)$. Pick a rational q with $u(y) < q < u(x)$ and set $\rho = \inf\{r \in A : r > q\}$, which exists because the set is nonempty (it contains $u(x)$) and bounded below by q ; thus $q \leq \rho \leq u(x)$.

If $u(x) > \rho$, choose a rational q' with $\rho < q' < u(x)$ and let n be the index with $I_n = [q, q']$. By definition of the infimum there is $r \in A$ with $q < r < q'$, so $A \cap I_n \neq \emptyset$ and (R1) applies: $u(y) < q \leq u(x_n) \leq q' < u(x)$, giving $x \succ x_n \succ y$.

If $u(x) = \rho$, choose a rational q' with $u(y) < q' < q$ and let n be the index with $I_n = [q', q]$. If $A \cap I_n \neq \emptyset$ then (R1) gives $u(y) < q' \leq u(x_n) \leq q < u(x)$ and again $x \succ x_n \succ y$. If $A \cap I_n = \emptyset$, the right endpoint is q , so $\rho_n = \rho = u(x) \in A$ and (R2) gives $u(x_n) = u(x)$, hence $x \sim x_n$ and $x_n \succ y$.

In every case some $z \in Z$ satisfies $x \succsim z \succsim y$. $\square$

By Theorem B.4, the lexicographic order of Example 3.8 is a complete and transitive relation with no countable order-dense subset.

B.2 Topological preliminaries

From here on X carries a topology. Only the notions used below are recorded, each with an example and a non-example.

Definition B.5 (Continuity of $\succsim$). $\succsim$ is *continuous* if for every $x \in X$ the contour sets

$$U(x) = \{y \in X : y \succsim x\}, \quad L(x) = \{y \in X : x \succsim y\}$$

are closed. By completeness their complements $\{y : x \succ y\}$ and $\{y : y \succ x\}$ are then open. Equivalently: if $x_\lambda \succsim y_\lambda$ for all λ , $x_\lambda \rightarrow x$ and $y_\lambda \rightarrow y$, then $x \succsim y$, where (x_λ) and (y_λ) are nets; in a metric space it is enough to check this for sequences (Proposition 3.11).

Example B.6. Leontief preferences on $\mathbb{R}_+^2$, $x \succsim y \iff \min\{x_1, x_2\} \geq \min\{y_1, y_2\}$, are continuous: $\min$ is a continuous function and the contour sets are preimages of closed rays.

Example B.7 (Not continuous). On $\mathbb{R}_+^2$ let $x \succsim y \iff \lfloor x_1 \rfloor \lfloor x_2 \rfloor \geq \lfloor y_1 \rfloor \lfloor y_2 \rfloor$. Then $y_m = (2 - \frac{1}{m}, 2 - \frac{1}{m})$ satisfies $y_m \sim (1, 1)$, so $y_m \in L(1, 1)$, while $y_m \rightarrow (2, 2)$ and $\lfloor 2 \rfloor \lfloor 2 \rfloor = 4 > 1$, so the limit is not in $L(1, 1)$: the lower contour set is not closed. Lexicographic preferences are not continuous either (Example 3.13).

Definition B.8 (Dense subset, separability). $D \subseteq X$ is *dense* if $\overline{D} = X$, equivalently if D meets every nonempty open set. X is *separable* if it has a countable dense subset.

Example B.9. $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$, so $\mathbb{R}^n$ is separable. Every subset of $\mathbb{R}^n$ with the subspace topology is separable as well, but this is not automatic: separability does not pass to subspaces in general, and the honest route is through second countability (Example B.11 below). An uncountable set with the discrete topology is not separable: every singleton is open, so the only dense set is the whole space.

Definition B.10 (Second countability). X is *second countable* if its topology has a countable base, i.e. a countable family $\{B_n\}$ of open sets such that every open set is a union of some of them.

Example B.11. $\mathbb{R}^n$ is second countable: the balls $B(z, \rho)$ with $z \in \mathbb{Q}^n$ and $\rho \in \mathbb{Q}_{>0}$ form a countable base. Given U open and $x \in U$, pick ε with $B(x, \varepsilon) \subseteq U$, then $z \in \mathbb{Q}^n$ with $\|z - x\| < \varepsilon/3$ and a rational $\rho \in (\varepsilon/3, 2\varepsilon/3)$; then $x \in B(z, \rho) \subseteq B(x, \varepsilon) \subseteq U$. Every second countable space is separable, by choosing one point in each nonempty basic set; the converse fails in general. Unlike separability, second countability *is* inherited by every subspace: intersect the members of a countable base with the subspace. Hence every $X \subseteq \mathbb{R}^n$ is second countable and therefore separable, which settles the claim of Example B.9 and is the form in which separability enters Theorem B.17.

Definition B.12 (Connectedness). X is *connected* if it cannot be written as $A \cup B$ with A, B nonempty, open and disjoint.

Example B.13. Intervals of $\mathbb{R}$ are connected, and so is every convex $X \subseteq \mathbb{R}^n$: convex sets are path connected, since $t \mapsto (1-t)x + ty$ is a continuous map from the connected interval $[0, 1]$ into X , and the continuous image of a connected set is connected. By contrast $[0, 1] \cup [2, 3]$ is not connected, and neither is $\mathbb{Q}$ with the topology induced from $\mathbb{R}$.

Definition B.14 (Dense linear order, endpoints). A linear order $(P, \prec)$ is *dense* if $p \prec p'$ implies $p \prec p'' \prec p'$ for some p'' ; it is *without endpoints* if it has neither a maximum nor a minimum.

Example B.15. $\mathbb{Z}$ is not dense: nothing lies between 3 and 4. $\mathbb{Q}$ is dense and without endpoints. $\mathbb{Q} \cap [0, 1)$ is dense but has a minimum, and $\mathbb{Q} \cap (0, 1)$ is dense and without endpoints. The two properties are independent, and both are needed in Lemma B.20: denseness makes every interior interval of the construction split eventually, and the absence of endpoints does the same for the two boundary intervals.

B.3 The monotone case

Lemma 4.8 needs no countable ruler: it measures every bundle against the diagonal, and $u(x) = t(x)$ is the unique t with $t\mathbf{1} \sim x$. Monotonicity does real work there, in three places: it makes $\{t : t\mathbf{1} \precsim x\}$ nonempty, it gives $x \precsim 0$, and it makes the crossing point unique. Only its strict part, $y \gg x \Rightarrow y \succ x$, is needed. For x on the boundary of $\mathbb{R}_+^K$, $x + \varepsilon\mathbf{1} \gg 0$ gives $x + \varepsilon\mathbf{1} \succ 0$ for every $\varepsilon > 0$, and closedness of $U(0)$ gives $x \precsim 0$. This is the construction behind Proposition 3.C.1 of Mas-Colell et al. (1995). Without monotonicity the diagonal need not meet every indifference set, and the next two subsections dispense with it at the cost of a longer construction.

B.4 Convex $X \subseteq \mathbb{R}^n$: the general case

The idea is to build a representation on a countable dense subset of X , where Theorem 3.7 applies, and then extend it to X by a supremum. The natural candidate $\mathbb{Q}^n \cap X$ does *not* work in general: density of $\mathbb{Q}^n$ in $\mathbb{R}^n$ does not pass to subsets, and $\mathbb{Q}^n \cap X$ can even be empty, as for the convex set $X = \{\sqrt{2}\} \times [0, 1]$. What does work is separability of X , which is automatic by second countability (Example B.11). Connectedness enters through the following lemma, which says that a continuous preference has no jumps.

Lemma B.16 (No jumps). *Let $\precsim$ be continuous on a connected X . If $x \succ y$, there is $z \in X$ with $x \succ z \succ y$.*

Proof. Suppose not, so $\{z : x \succ z\} \cap \{z : z \succ y\} = \emptyset$. Both sets are open by continuity and nonempty, since y belongs to the first and x to the second. They also cover X : given w , completeness gives $w \succsim x$ or $x \succ w$; in the first case $w \succsim x \succ y$ gives $w \succ y$ by Lemma B.1. So X is the union of two disjoint nonempty open sets, contradicting connectedness. $\square$

Theorem B.17 (Convex domain). *Let $X \subseteq \mathbb{R}^n$ be convex and let $\succsim$ be a continuous preference relation on X . Then $\succsim$ admits a continuous utility representation.*

Proof of existence. Let Y be a countable dense subset of X , which exists because X , as a subspace of $\mathbb{R}^n$, is second countable and hence separable (Example B.11). By Theorem 3.7 there is $v : Y \rightarrow (0, 1]$ representing the restriction of $\succsim$ to Y . For $x \in X$ define

$$u(x) = \sup A_x, \quad A_x = \{v(y) : y \in Y, x \succ y\},$$

with $\sup \emptyset = 0$. Since v is bounded, u is well defined.

If $x \sim z$ then $x \succ y \iff z \succ y$ for every y , so $A_x = A_z$ and $u(x) = u(z)$.

Let $x \succ z$. By Lemma B.16 there is w with $x \succ w \succ z$, and by continuity the set $\{w' : x \succ w' \succ z\}$ is open and nonempty; as Y is dense it contains some $y_1 \in Y$, so $x \succ y_1 \succ z$. Applying the lemma and density again inside $\{w' : y_1 \succ w' \succ z\}$ gives $y_2 \in Y$ with $y_1 \succ y_2 \succ z$. Now

$$u(x) \geq v(y_1) > v(y_2) \geq u(z),$$

where the first inequality holds because $y_1 \in A_x$, the second because v represents $\succsim$ on Y , and the third because every $y \in Y$ with $z \succ y$ satisfies $y_2 \succ y$ and hence $v(y) < v(y_2)$. Therefore $u(x) > u(z)$.

Combining the two displays, $x \succsim z$ implies $u(x) \geq u(z)$; and if $u(x) \geq u(z)$ while $x \not\succsim z$ failed, completeness would give $z \succ x$ and the previous paragraph $u(z) > u(x)$. Hence u represents $\succsim$. $\square$

Remark B.18 (Why continuity needs a better v). The u just built need not be continuous. The culprit is the image of v : the openness of $\{u < a\}$ in Step (e) of Theorem B.21 below rests on the values of the auxiliary function being *dense* in an interval, and the v of Theorem 3.7 takes values among sums of powers 2^{-k} , which are not dense. Concretely, on $X = [0, 1]$ with $x \succsim y \iff x \geq y$ and $Y = \mathbb{Q} \cap [0, 1]$ enumerated as $\{y_1, y_2, \dots\}$, the resulting u jumps at every rational: at $y_0 = y_k$ one has $\lim_{x \uparrow y_0} u(x) = u(y_0) = v(y_0) - 2^{-k}$, while $u(x) \geq v(y_0)$ for every $x > y_0$, a jump of at least 2^{-k} . The representation survives; continuity does not. Steps (a) and (b) of Theorem B.21 repair exactly this, replacing v by the midpoint embedding of Lemma B.20, whose image is dense in $(0, 1)$, after

which Step (e) proves continuity. Since a convex X is connected (Example B.13) and separable (Example B.11), Theorem B.17 in full, continuity included, is a special case of Theorem B.21.

B.5 Connected and separable X : Debreu's theorem

Nothing in the previous subsection used the linear structure of $\mathbb{R}^n$ beyond connectedness and the existence of a countable dense set. Debreu's theorem keeps only those two properties. Throughout this subsection we assume that $x_0 \succ y_0$ for some x_0, y_0 ; otherwise $u \equiv 0$ works.

Lemma B.19 (Order density of a dense set). *Let $\succsim$ be continuous on a connected X and let $D \subseteq X$ be dense. If $x \succ y$, there is $d \in D$ with $x \succ d \succ y$.*

Proof. The set $\{z : x \succ z \succ y\}$ is the intersection of two open sets, and it is nonempty by Lemma B.16. A dense set meets every nonempty open set. $\square$

Lemma B.20 (Midpoint embedding). *Let $(P, \prec)$ be a countable, nonempty, dense linear order without endpoints, enumerated without repetitions as $P = \{c_1, c_2, \dots\}$. Then there is a strictly increasing $q : P \rightarrow (0, 1)$ whose image is dense in $(0, 1)$.*

Proof. Construction. Set $q(c_1) = 1/2$. For $n \geq 2$, the points $0, 1, q(c_1), \dots, q(c_{n-1})$ partition $[0, 1]$ into n closed intervals. Since $\prec$ is linear, c_n occupies exactly one position relative to $c_1, \dots, c_{n-1}$: either $c_i \prec c_n \prec c_j$ for two consecutive placed elements, or c_n lies below all of them, or above all of them. Let (ℓ, r) be the corresponding interval, with $\ell = q(c_i)$ or $\ell = 0$, and $r = q(c_j)$ or $r = 1$, and set $q(c_n) = (\ell + r)/2$.

Monotonicity. By induction: if q preserves the order on $\{c_1, \dots, c_{n-1}\}$, then $q(c_n) \in (\ell, r)$ sits exactly where it must relative to all previous images. So q is strictly increasing, in particular injective.

Density. Every interval arising in the construction is dyadic, of the form $[k2^{-m}, (k+1)2^{-m}]$. First, every such interval is eventually split. An interval $(q(c_i), q(c_j))$ present at stage n corresponds to consecutive placed elements $c_i \prec c_j$; by density of P there is $c \in P$ with $c_i \prec c \prec c_j$, say $c = c_N$ with $N > n$, and placing c_N splits that interval. The two boundary intervals $(0, q(c_i))$ and $(q(c_j), 1)$ are split because P has no minimum and no maximum. Second, for every m there is a stage N_m after which all intervals have length at most 2^{-m} : at stage N_m there are finitely many intervals of length exactly 2^{-m} , each is split at some stage, and N_{m+1} is the largest of those stages.

Now let $(a, b) \subseteq (0, 1)$ and choose m with $2^{-m} < b - a$. At stage N_m no interval of the partition contains (a, b) , so some partition point lies in (a, b) ; that point is neither 0 nor 1, hence equals $q(c_i)$ for some i . Therefore $q(P) \cap (a, b) \neq \emptyset$. $\square$

Theorem B.21 (Debreu 1954). *Let (X, τ) be a connected and separable topological space and let $\succsim$ be a continuous preference relation on X . Then there is a continuous $u : X \rightarrow [0, 1]$ representing $\succsim$.*

Proof. Step (a): a countable order-dense set without endpoints. Let D be countable and dense. On the quotient $D/\sim$ the relation $[d] \succ [d'] \iff d \succ d'$ is a well defined linear order, and it is dense by Lemma B.19. Let D° be D with the maximal and minimal classes removed, whenever they exist. Then $D^\circ/\sim$ is nonempty, indeed infinite, since applying Lemma B.19 repeatedly to $x_0 \succ y_0$ produces $d_1 \succ d_2 \succ \dots$ in D , all but at most two of which lie in D° . It has no endpoints: if $[d^*]$ were maximal in D and $[d]$ maximal in $D \setminus [d^*]$, no element of D would lie strictly between d and d^* , contradicting density; the minimum is symmetric. And it is dense, being a dense order with its endpoints removed. Finally, if $x \succ d \succ y$ with $d \in D$, then $d \in D^\circ$, since Lemma B.19 would otherwise produce an element of D strictly above d and below x . So Lemma B.19 holds with D° .

Step (b): an auxiliary map. Apply Lemma B.20 to $P = D^\circ/\sim$ and set $q(d) = q([d])$. Then $q(d) > q(d') \iff d \succ d'$ and $q(d) = q(d') \iff d \sim d'$, and $S := q(D^\circ)$ is dense in $(0, 1)$.

Step (c): the candidate. For $x \in X$ put

$$u(x) = \sup\{q(d) : d \in D^\circ, x \succ d\}, \quad \sup \emptyset = 0,$$

so that $u(X) \subseteq [0, 1]$.

Step (d): u represents $\succsim$. If $x \succsim y$ then $\{d : y \succ d\} \subseteq \{d : x \succ d\}$ and $u(x) \geq u(y)$. If $x \succ y$, Step (a) gives $d, d' \in D^\circ$ with $x \succ d \succ d' \succ y$, so $u(x) \geq q(d) > q(d')$; and every $d'' \in D^\circ$ with $y \succ d''$ satisfies $d' \succ d''$, hence $q(d'') < q(d')$ and $u(y) \leq q(d')$. Therefore $u(x) > u(y)$. Completeness gives the converse.

Step (e): u is continuous. It suffices that $\{u > a\}$ and $\{u < a\}$ be open for every $a \in \mathbb{R}$. For the first, if $a < 0$ the set is all of X , so let $a \geq 0$; then

$$\{u > a\} = \bigcup_{d \in D^\circ, q(d) > a} \{x : x \succ d\},$$

a union of open sets, because for $a \geq 0$ the inequality $u(x) > a$ holds exactly when some $d \in D^\circ$ has $x \succ d$ and $q(d) > a$. For the second, the set is empty if $a \leq 0$ and all of X if

$a > 1$, so let $0 < a \leq 1$; we claim

$$\{u < a\} = \bigcup_{d \in D^\circ, q(d) < a} \{x : d \succ x\}.$$

($\supseteq$) If $d \succ x$ and $q(d) < a$, every d'' with $x \succ d''$ satisfies $d \succ d''$, so $q(d'') < q(d)$ and $u(x) \leq q(d) < a$. ($\subseteq$) Let $u(x) < a$. Since S is dense in $(0, 1)$ there is $r \in S$ with $u(x) < r < a$; write $r = q(d)$. We claim $d \succ x$. Otherwise $x \lesssim d$, and for every $s \in S$ with $s < r$, writing $s = q(d')$ with $d' \prec d \lesssim x$, we get $x \succ d'$ and $u(x) \geq s$; since S is dense and $r > 0$, $\sup\{s \in S : s < r\} = r$, so $u(x) \geq r$, a contradiction. Hence x lies in the union, which is open. $\square$

B.6 Where each hypothesis is used

- **Completeness and transitivity** are necessary (Proposition 3.2) and, on a countable X , sufficient (Theorem 3.7).
- **Countability** of a ruler, not of X , is what really matters: Theorem B.4 says a representation exists if and only if a countable order-dense set does. Example 3.8 is a weak order with none.
- **Connectedness** is used only to rule out jumps (Lemma B.16); **separability** only to make a dense set order-dense (Lemma B.19); **continuity** of $\lesssim$ in the openness of the strict contour sets and in Step (e).
- **Monotonicity**, when available, replaces the whole construction by the diagonal crossing of Lemma 4.8.
- **Convexity** of $X \subseteq \mathbb{R}^n$ is never used for its own sake: it delivers connectedness, and separability is automatic for subsets of $\mathbb{R}^n$ by second countability (Example B.11).

Remark B.22. In Theorem I of Debreu (1954) the hypothesis “connected and separable” is replaced by “second countable”. Without connectedness jumps may exist, the image $q(D^\circ)$ cannot be dense in an interval, and the proof needs an extra ingredient, the open gap lemma of Debreu (1964): every $S \subseteq \mathbb{R}$ admits a strictly increasing g such that every gap of $g(S)$ is an open interval. Composing with g turns a representation into a continuous one.

C Differentiable and smooth preferences

Definition 5.20 still needs directions, small steps and an inner product. Richter et al. (2025) propose a notion that needs none of these, so it makes sense on any abstract set, a grid of indivisible bundles included.

Let X be any nonempty set and let Λ be a set of *primitive orderings*: complete, transitive and reflexive relations $\succeq$ on X , each one a consideration by which alternatives can be ranked. Following Richter and Rubinstein (2019), an ordering $\succeq \in \Lambda$ is *critical* at x if $y \succ x$ implies $y \triangleright x$, and $\succsim$ is Λ -*convex* if at every x some ordering in Λ is critical: improving on x requires advancing in that consideration. The differentiable version strengthens critical to a *gradient*: $\succeq \in \Lambda$ is a gradient of $\succsim$ at x if

$$(i) \quad y \succ x \Rightarrow y \triangleright x, \quad (ii) \quad x \triangleright y \Rightarrow x \succ y.$$

Advancing in the gradient ordering is necessary for improvement, and falling in it is sufficient for strict deterioration. The relation $\succsim$ is Λ -*convex-differentiable* (Λ -CD) if at every non-maximal x there is exactly one gradient, written $\nabla_{\succsim}(x)$. Uniqueness is the differentiability: a single consideration governs the preference at each point, and nothing in the wording mentions ε -moves.

On $X = \mathbb{R}^K$ the natural primitives are the *algebraic linear orderings* $\Psi = \{\succeq_v : v \in \mathbb{R}^K \setminus \{0\}\}$, where $y \succeq_v z$ if $v \cdot y \geq v \cdot z$, and then the abstract gradient is the usual one.

Proposition C.1 (Richter et al. 2025). *If $\succsim$ on $\mathbb{R}^K$ is monotone, convex and represented by a differentiable u with $\nabla u(x) \neq 0$ everywhere, then $\succsim$ is Ψ -CD and $\nabla_{\succsim}(x) = \succeq_{\nabla u(x)}$.*

Proof. Fix x and write $g = \nabla u(x)$. Since $u(x + \varepsilon g) = u(x) + \varepsilon \|g\|^2 + o(\varepsilon)$, no point is $\succsim$ -maximal, so a gradient is required at every x .

$\succeq_g$ is a gradient. Convexity gives $x + \varepsilon(z - x) \succsim x$ for $z \succsim x$ and $\varepsilon \in (0, 1)$; dividing $u(x + \varepsilon(z - x)) \geq u(x)$ by ε and letting $\varepsilon \downarrow 0$ yields the supporting inequality

$$z \succsim x \implies g \cdot (z - x) \geq 0. \quad (*)$$

Property (ii) is its contrapositive: $x \triangleright_g y$ means $g \cdot (y - x) < 0$, so $y \not\succsim x$ and completeness gives $x \succ y$. For (i), let $y \succ x$. Then y is interior to the upper contour set, so $g \cdot (y - x) = 0$ would leave $y - \delta g$ in that set while violating (*); hence $g \cdot (y - x) > 0$.

Uniqueness. Let v not be a positive multiple of g and write $v = \lambda g + r$ with $r \perp g$. Take $d = g$ if $r = 0$, in which case $\lambda < 0$, and $d = \varepsilon g - r$ with $\varepsilon > 0$ small otherwise. Either way $g \cdot d > 0 > v \cdot d$, so $x + td \succ x$ for small $t > 0$ while $v \cdot (td) < 0$, which contradicts (i) for $\succeq_v$. $\square$

Remark C.2. The extension is strict: ranking $\mathbb{R}^2$ by x_1 when $x_1 \leq 0$ and by e^{x_2} when $x_1 > 0$ is monotone, convex and Ψ -CD but not continuous. At boundary points the gradient need not be unique; requiring only a *tight* gradient, one with the smallest upper contour set, repairs this (Λ -TCD). With two agents, finitely many indivisible goods and strict Λ -TCD preferences over bundles, every Pareto-efficient allocation is a linear-price equilibrium outcome (Richter et al., 2025).

Definition C.3 (Smooth preferences). Let $X \subseteq \mathbb{R}^K$ be open and let $\succsim$ be continuous and monotone. Then $\succsim$ is of class C^r , $r \geq 1$, if the indifference relation

$$I = \{(x, y) \in X \times X : x \sim y\}$$

is a C^r submanifold of $X \times X$ of codimension one.²

Theorem C.4 (Debreu 1972). A continuous and monotone $\succsim$ on an open X is of class C^r if and only if it admits a representation u of class C^r with $\nabla u(x) \neq 0$ for every x . Any two such representations differ by composition with a C^r function of one variable with strictly positive derivative.

The second sentence is what makes C^r ordinal after all: requiring a nonvanishing gradient shrinks the admissible changes of representation from all strictly increasing g to the increasing C^r diffeomorphisms of $\mathbb{R}$, and smoothness survives that group.

Corollary C.5 (Gauss map). If $\succsim$ is of class C^r , the map

$$g : X \rightarrow S^{K-1}, \quad g(x) = \frac{\nabla u(x)}{\|\nabla u(x)\|},$$

is of class C^{r-1} and does not depend on the choice of C^r representation.

Proof. If $\tilde{u} = h \circ u$ with $h' > 0$, then $\nabla \tilde{u}(x) = h'(u(x))\nabla u(x)$ is a positive multiple of $\nabla u(x)$, so the normalization is the same. $\square$

So $g(x)$ is the $v(x)$ of Definition 5.20, normalized and with regularity attached. Class C^1 implies differentiability, because $\nabla u(x) \cdot d > 0$ makes u increase along d . The converse is false.

²A set $M \subseteq \mathbb{R}^m$ is a C^r submanifold of codimension one if every point of M has an open neighbourhood V on which there is a C^r function $F : V \rightarrow \mathbb{R}$ with $\nabla F \neq 0$ and $M \cap V = F^{-1}(0)$. Locally M is the solution set of one non-degenerate equation, so it is a surface of dimension $m - 1$ with a well defined tangent space and normal direction at each of its points. Here $m = 2K$ and $M = I$.

Example C.6 (Differentiable but not of class C^1). Near the origin let $u(x) = 2x_1 + h(x_1) + x_2$, where $h(t) = t^2 \sin(1/t)$ and $h(0) = 0$. Then $h'(t) = 2t \sin(1/t) - \cos(1/t)$ exists everywhere, so $\nabla u = (2 + h'(x_1), 1) \neq 0$ and the preference is differentiable at every point. But h' has no limit at 0, so the slope $-(2 + h'(x_1))$ of the indifference curves does not converge, and by Corollary C.5 no C^1 representation exists.

Theorem C.7 (Debreu 1972). *Let $\succsim$ be of class C^r with $r \geq 2$, strictly convex, and let the indifference hypersurfaces have Gaussian curvature different from zero everywhere, equivalently a nonsingular bordered Hessian. Then the demand function $f(p, w)$ is of class C^{r-1} .*

The curvature condition is strict convexity in its differentiable form. Perfect substitutes have flat curves, curvature zero, and demand is not even a function: at one price ratio the whole budget line is optimal and on either side of it the consumer spends everything on one good. Outside the class the condition is not necessary: Leontief is nowhere differentiable, yet $f(p, w) = \frac{w}{p_1 + p_2}(1, 1)$ is smooth.

What smoothness is ultimately for is not the consumer but the counting of equilibria. With C^1 demand, Walrasian equilibrium is the zero set of an excess demand function $Z(p, \omega)$, and Sard's theorem applied to the projection $(p, \omega) \mapsto \omega$ gives Debreu (1970): for almost every endowment the equilibria are finite and each is locally a C^1 function of the endowment. Such economies are called *regular*. Balasko (1988) maps the global structure of the equilibrium set, Smale (1976) recasts optimality and equilibrium as transversality statements, Mas-Colell (1985) is the unified treatment, and the index theorem of Dierker (1972) gives uniqueness whenever every equilibrium has index +1.

D The envelope theorem

Unconstrained. Let $f : X \times T \rightarrow \mathbb{R}$ with $X \subseteq \mathbb{R}^n$ and $T \subseteq \mathbb{R}$ open, let $x(t) \in \arg \max_{x \in X} f(x, t)$ and $M(t) = f(x(t), t)$. If f is C^1 and $x(\cdot)$ is differentiable at t , then

$$M'(t) = \frac{\partial f}{\partial t}(x(t), t).$$

Lagrange. Let $M(t) = \max\{f(x, t) : g_i(x, t) = 0, i = 1, \dots, m\}$ with f, g_i of class C^1 , let $x(t)$ be a solution at which the gradients $\nabla_x g_i$ are linearly independent, and let $\lambda(t)$ be the multipliers of $\mathcal{L}(x, \lambda, t) = f(x, t) + \sum_i \lambda_i g_i(x, t)$. Then

$$M'(t) = \frac{\partial \mathcal{L}}{\partial t}(x(t), \lambda(t), t) = \frac{\partial f}{\partial t} + \sum_i \lambda_i(t) \frac{\partial g_i}{\partial t},$$

everything evaluated at $(x(t), t)$.

Karush-Kuhn-Tucker. The same formula holds with inequality constraints $g_i(x, t) \leq 0$ and multipliers $\lambda_i(t) \geq 0$ satisfying complementary slackness, under a constraint qualification. A slack constraint has $\lambda_i = 0$, so only the binding constraints contribute.

Arbitrary choice sets (Milgrom and Segal, 2002). Let $V(t) = \sup_{x \in X} f(x, t)$ with X an arbitrary set and $T = [0, 1]$, and let $X^*(t) = \arg \max_{x \in X} f(x, t)$.

- (i) If $f(x, \cdot)$ is differentiable for every x , and V is differentiable at t with $x^* \in X^*(t)$, then $V'(t) = f_t(x^*, t)$.
- (ii) If $f(x, \cdot)$ is absolutely continuous for every x , if $|f_t(x, t)| \leq b(t)$ for an integrable b , and if $X^*(t) \neq \emptyset$ for almost every t , then V is absolutely continuous and

$$V(t) = V(0) + \int_0^t f_t(x^*(s), s) ds$$

for any selection $x^*(s) \in X^*(s)$.

E Temptation and self-control

[Gul and Pesendorfer \(2001\)](#) ask why someone may prefer fewer options, even when she would resist the excluded temptation. At date 1 the agent chooses a *menu* A ; at date 2 she chooses a lottery $x \in A$. Commitment restricts the menu beforehand; self-control resists temptation within it. The primitive is a preference $\succsim$ over menus, including singletons $\{x\}$ that commit the agent to x .

Representation (Theorem 1). Let prizes form a compact metric space and menus be nonempty compact sets of lotteries, with the Hausdorff topology induced by weak convergence. For menu mixtures, define

$$\alpha A + (1 - \alpha)B = \{\alpha x + (1 - \alpha)y : x \in A, y \in B\}.$$

Completeness, transitivity, continuity, mixture independence and *set betweenness*,

$$A \succsim B \implies A \succsim A \cup B \succsim B,$$

hold if and only if continuous affine lottery indices u, v exist such that

$$\boxed{U(A) = \max_{x \in A} \{u(x) + v(x)\} - \max_{y \in A} v(y)} \quad (10)$$

represents $\succsim$. Independence means that mixing both menus with a common menu preserves their ranking. Set betweenness allows additional options to hurt, but requires their union to lie between the original menus.

Interpretation. Singletons reveal commitment utility: $U(\{x\}) = u(x)$. The index v ranks temptation. Choosing x from A incurs the nonnegative self-control cost

$$c(x, A) = \max_{y \in A} v(y) - v(x), \quad U(A) = \max_{x \in A} \{u(x) - c(x, A)\}.$$

In the paper’s associated second-period model, choice maximizes $u + v$: the agent trades commitment utility against the cost of resisting temptation. This choice interpretation uses the paper’s additional axioms on preferences over menu–choice pairs (A, x) ; the representation theorem itself concerns date-1 menu preferences.

The proof first identifies u from singletons and v from how binary-menu utility changes when the tempting option changes. Set betweenness extends the binary representation to finite menus; continuity extends it to compact menus ([Gul and Pesendorfer, 2001](#), pp. 1409–1410).

A numerical example. Let h be healthy and t tempting, with $(u(h), v(h)) = (2, 0)$ and $(u(t), v(t)) = (0, 1)$. Then

$$U(\{h\}) = 2 > U(\{h, t\}) = 1 > U(\{t\}) = 0.$$

The agent chooses h from $\{h, t\}$ because $2 > 1$ under $u + v$, but pays a self-control cost of one. Removing the unchosen t raises utility without changing consumption. If instead $v(t) = 3$, she chooses t and $U(\{h, t\}) = U(\{t\}) = 0$: she succumbs. Commitment is valuable in both cases; successful self-control occurs only in the first.

Overwhelming temptation. As the weight on v becomes arbitrarily large, the limiting menu value is

$$U_\infty(A) = \max_{x \in \arg \max_{y \in A} v(y)} u(x).$$

The agent maximizes temptation utility, breaking ties with commitment utility. Theorem 3 accommodates this case by weakening continuity; the limit need not satisfy the strong continuity used in Theorem 1.

What the model adds. Final choices alone miss the cost of resisting rejected options. The example therefore explains a demand for commitment even when it does not change consumption. Preferences can remain dynamically consistent over (A, x) : the same evaluation $u(x) - c(x, A)$ governs both anticipated and actual choice. A change in the menu changes the full consequence, so commitment need not reflect a conflict between changing “selves.”

The paper also separates greater preference for commitment from greater self-control: the former concerns disagreement between u and v ; the latter, between the actual choice ranking $u + v$ and v . They need not move together (Gul and Pesendorfer, 2001, Theorems 8–9).

Scope and restrictions. Only the most tempting option determines the cost of a fixed choice. Adding options that change neither maximum in (10) has no effect. Thus the model excludes cumulative temptation and many forms of random temptation or processing costs. Menu independence also imposes strong restrictions on mixing and the timing of uncertainty resolution. The contribution is an axiomatic explanation, not an empirical test: within this model, removing an unchosen temptation can improve welfare, but removing arbitrary options need not.